

WHO recommendations on
maternal and newborn care for a positive postnatal experience

Web Supplement. Evidence base



WHO recommendations on maternal and newborn care for a positive postnatal experience. Web Supplement. Evidence base

ISBN 978-92-4-004617-7 (electronic version)

© World Health Organization 2022

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; <https://creativecommons.org/licenses/by-nc-sa/3.0/igo>).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: “This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition”.

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization (<http://www.wipo.int/amc/en/mediation/rules/>).

Suggested citation. Web Supplement. Evidence base. In: WHO recommendations on maternal and newborn care for a positive postnatal experience. Geneva: World Health Organization; 2022. Licence: [CC BY-NC-SA 3.0 IGO](https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

Cataloguing-in-Publication (CIP) data. CIP data are available at <http://apps.who.int/iris>.

Sales, rights and licensing. To purchase WHO publications, see <http://apps.who.int/bookorders>. To submit requests for commercial use and queries on rights and licensing, see <https://www.who.int/copyright>.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers’ products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

This publication forms part of the WHO guideline entitled *WHO recommendations on maternal and newborn care for a positive postnatal experience*. It is being made publicly available for transparency purposes and information, in accordance with the *WHO handbook for guideline development*, 2nd edition (2014).

Contents

Acronyms and abbreviations	vii
A. MATERNAL CARE	1
A.2 Interventions for common physiological signs and symptoms.....	1
EB table A.2.1: Local cooling for perineal pain relief.....	1
Comparison 1: Perineal local cooling compared with no pain relief or usual care	1
Comparison 2: Perineal local cooling compared with other forms of non-pharmacological interventions	5
EB table A.2.2: Oral analgesia for perineal pain relief.....	12
Comparison 1: Single-dose oral analgesic (any dose) compared with placebo	12
Comparison 2: Single-dose oral analgesic compared with a higher single dose of the same analgesic	29
Comparison 3: Single-dose oral analgesic compared with a single dose of an alternative oral analgesic	41
EB table A.2.3a: Pharmacological relief of pain due to uterine cramping/involution (pharmacological interventions compared with placebo).....	49
Comparison 1: Paracetamol (oral, single-dose) compared with placebo	49
Comparison 2: NSAIDs compared with placebo	50
Comparison 3: Opioids compared with placebo	56
EB table A.2.3b: Pharmacological relief of pain due to uterine cramping/involution (pharmacological interventions compared with other pharmacological interventions)	58
Comparison 1: Lower dose of an oral analgesic compared with a higher dose of the same analgesic.....	58
Comparison 2: An oral analgesic compared with an alternative oral analgesic of the same class	61
Comparison 3: An oral analgesic compared with an alternative oral analgesic from a different class	66
EB table A.2.4: Postnatal pelvic floor muscle training (PFMT) for pelvic floor strengthening	74
Comparison 1: Postnatal PFMT compared with no intervention or usual care for (mixed) prevention or treatment of incontinence.....	74
Comparison 2: Postnatal PFMT compared with no intervention or usual care for treatment of incontinence	76

EB table A.2.5: Non-pharmacological interventions to treat postpartum breast engorgement	78
Comparison 1: Cabbage leaf extract cream compared with placebo	78
Comparison 2: Cold cabbage leaves applied directly to the breast compared with usual care	79
Comparison 3: Cold gel packs applied directly to the breast compared with usual care	80
Comparison 4: Warm herbal compresses compared with usual care (including warm compresses without herbs).....	81
EB table A.2.6: Pharmacological interventions to treat postpartum breast engorgement	82
Comparison 1: Subcutaneous oxytocin compared with placebo	82
Comparison 2: Proteolytic enzymes compared with placebo	83
A.3 Preventive measures	85
EB table A.3.1: Non-pharmacological interventions to prevent postpartum mastitis	85
Comparison 1: Probiotics compared with placebo	85
Comparison 2: Hydrothermally processed cereal with anti-secretory factor-inducing properties compared with standard cereal (serving as a placebo) ..	86
Comparison 3: Specialist breastfeeding education compared with usual care	87
Comparison 4: Acupoint massage compared with usual care	89
EB table A.3.2: Pharmacological interventions to prevent postpartum mastitis.....	90
Comparison 1: Oral prophylactic antibiotics compared with placebo or usual care	90
Comparison 2: Topical prophylactic antibiotics compared with usual care (breastfeeding advice).....	92
EB table A.3.3: Prevention of postpartum constipation	93
Comparison: Laxatives compared with placebo.....	93
A.4 Mental health interventions	95
EB table A.4.1: Screening for postpartum depression and anxiety.....	95
Comparison: Screening for common mental disorders (CMDs: depression, anxiety) in the postpartum period compared with no screening or usual care	95
EB table A.4.2: Prevention of postpartum depression and anxiety	97

Comparison: Interventions to prevent common mental disorders (CMDs: depression, anxiety) in the postpartum period, delivered at any time, compared with no intervention or usual care.....	97
B. NEWBORN CARE.....	101
B.1 Newborn assessment	101
EB table B.1.2: Universal screening for abnormalities of the eye	101
Comparison: Universal newborn screening for abnormalities of the eye compared with no screening	101
EB table B.1.3: Universal screening for hearing impairment	103
Comparison: Universal newborn hearing screening (UNHS) compared with no screening or selective screening.....	103
EB table B.1.4a: Universal screening for neonatal hyperbilirubinaemia (TcB).....	105
Comparison: Universal screening for identification of neonatal hyperbilirubinaemia by TcB at discharge compared with clinical screening (visual inspection and/or assessment of risk factors), followed by TcB or TSB if required.....	105
EB table B.1.4b: Universal screening for neonatal hyperbilirubinaemia (TSB)	107
Comparison: Universal screening of TSB before discharge compared with clinical screening (visual inspection and/or risk factor assessment)	107
B.2 Preventive measures	108
EB table B.2.1: Timing of first bath to prevent hypothermia and its sequelae	108
Comparison 1: Delayed first bath (after 24 hours) compared with early first bath (at or before 24 hours)	108
Comparison 2: Delayed first bath (after 6 hours; i.e. at or after 9, 12 or 24 hours) compared with early first bath (at or before 6 hours)	109
EB table B.2.2: Use of emollients for the prevention of skin conditions	110
Comparison: Topical emollients compared with no intervention or skin care without emollients.....	110
EB table B.2.3: Application of chlorhexidine to the umbilical cord stump for the prevention of neonatal infection	112
Comparison: Routine application of chlorhexidine to the umbilical cord stump compared with dry cord care or usual care	112
EB table B.2.4: Sleeping position for the prevention of sudden infant death syndrome	115
Comparison: Supine (back) sleep position compared with non-supine (prone or side) sleep position	115
B.3 Nutritional interventions.....	117

EB table B.3.1: Neonatal vitamin A supplementation	117
Comparison: Neonatal vitamin A supplementation compared with placebo or no vitamin A supplementation	117
EB table B.3.2: Vitamin D supplementation for breastfed, term infants	118
Comparison: Vitamin D supplementation for breastfed, term infants compared with placebo or no supplementation	118
B.4 Infant growth and development	125
EB table B.4.1: Whole-body massage	125
Comparison: Whole-body massage compared with no massage	125
C. HEALTH SYSTEMS AND HEALTH PROMOTION INTERVENTIONS	129
EB table C.1: Schedules for postnatal care contacts	129
Comparison 1: Schedules involving four postnatal home visits (3, 7, 28 and 42 days after birth) compared with one postnatal home visit (at about 42 days after birth)	129
Comparison 2: Schedules involving two postnatal visits (3–5 and 10–14 days after birth) compared with one outpatient visit (10–14 days after birth)	131
EB table C.2: Length of stay in health facilities after birth	132
Comparison 1: Early discharge following vaginal birth compared with usual discharge	132
Comparison 2: Early discharge following caesarean birth compared with usual discharge	134
Ad-hoc analyses by time of discharge and mode of birth	136
EB table C.4: Approaches to strengthen preparation for discharge from the facility to home after birth	142
Comparison 1: Written education booklets for women compared with control leaflets	142
Comparison 2: Discharge education by a designated nurse compared with usual care	143
EB table C.5a: Home visits for postnatal care contacts compared with usual care	144
Comparison: Home visits for postnatal care contacts compared with usual care (evidence source 1)	144
Comparison: Home visits for postnatal care contacts compared with usual care (evidence source 2)	147
EB table C.5b: Home visits for postnatal care contacts compared with routine outpatient postnatal care	148
Comparison: Home visits for postnatal care contacts compared with routine outpatient postnatal care	148

EB table C.9: Involvement of men in postnatal care and maternal and newborn health.....	151
Comparison 1: Couples education compared with no intervention or usual care	151
Comparison 2: Couples education compared with women’s education alone.....	157
Comparison 3. Men’s education compared with no intervention or usual care	158
Comparison 4. Father as a labour companion compared with no companion.....	160
Comparison 5. Father as a labour companion compared with a female friend as a labour companion.....	161
Comparison 6. Multi-component interventions compared with no intervention or usual care	162

Note: The labelling convention in this document (e.g. “EB table A.2.1”) aligns with the evidence and recommendations of the guideline. GRADE tables are shown only where applicable for new and/or updated recommendations (not integrated recommendations); therefore, numbers may not be presented in consecutive order in this document.

The full guideline document is available at <https://www.who.int/publications/i/item/9789240045989>

Acronyms and abbreviations

25(OH)D	25-hydroxyvitamin D (vitamin D)
AF	antiseecretory factor
ALTE	apparently life-threatening event
CI	confidence interval
CMDs	common mental disorders
DID	difference-in-difference
EB	evidence base
GRADE	Grading of Recommendations Assessment, Development and Evaluation
HR	hazard ratio
ITT	intention-to-treat
IU	international units
MD	mean difference
NSAID	nonsteroidal anti-inflammatory drug
OR	odds ratio
PBHL	permanent bilateral hearing loss
PFMT	pelvic floor muscle training
PSBI	possible serious bacterial infection
RCT	randomized controlled trial

RR	risk ratio
SMD	standardized mean difference
SIDS	sudden infant death syndrome
SUDI	sudden unexpected death in infancy
TcB	transcutaneous bilirubinometry
TSB	total serum bilirubin
UNHS	universal newborn hearing screening
VAS	visual analogue scale

A. MATERNAL CARE

A.2 Interventions for common physiological signs and symptoms

EB table A.2.1: Local cooling for perineal pain relief

Comparison 1: Perineal local cooling compared with no pain relief or usual care

Source: East CE, Dorward EDF, Whale RE, Liu J. Local cooling for relieving pain from perineal trauma sustained during childbirth. Cochrane Database Syst Rev. 2020;(10):CD006304.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack or cold gel pad)	No pain relief or usual care	Relative (95% CI)	Absolute (95% CI)		
Perineal pain within 4–6 hours of birth												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^c	none	50	50	-	MD 4.46 lower (5.07 lower to 3.85 lower)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain within 24 hours of birth – moderate + severe pain												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	114/215 (53.0%)	52/101 (51.5%)	RR 1.03 (0.82 to 1.29)	15 more per 1000 (from 93 fewer to 149 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain within 24 hours of birth												
3	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	83	83	-	MD 0.41 lower (1.78 lower to 0.95 higher)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain 24–48 hours after birth – moderate + severe pain												
1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	84/215 (39.1%)	54/101 (53.5%)	RR 0.73 (0.57 to 0.94)	144 fewer per 1000 (from 230 fewer to 32 fewer)	⊕⊕○○ LOW	IMPORTANT

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack or cold gel pad)	No pain relief or usual care	Relative (95% CI)	Absolute (95% CI)		

Perineal pain 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	35	36	-	MD 0.53 lower (1.45 lower to 0.39 higher)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----	----	---	---	------------------	-----------

Perineal oedema within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	156/215 (72.6%)	73/101 (72.3%)	RR 1.00 (0.87 to 1.16)	0 fewer per 1000 (from 94 fewer to 116 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-------------	------	--------------------	-------------------	----------------------------------	--	-------------	-----------

Perineal oedema 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^e	none	121/215 (56.3%)	69/101 (68.3%)	RR 0.82 (0.69 to 0.98)	123 fewer per 1000 (from 212 fewer to 14 fewer)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	--------------------	-------------------	----------------------------------	---	------------------	-----------

Perineal bruising within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	127/215 (59.1%)	61/101 (60.4%)	RR 0.98 (0.81 to 1.19)	12 fewer per 1000 (from 115 fewer to 115 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-------------	------	--------------------	-------------------	----------------------------------	--	-------------	-----------

Perineal bruising 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^e	none	164/215 (76.3%)	68/101 (67.3%)	RR 1.13 (0.97 to 1.32)	88 more per 1000 (from 20 fewer to 215 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	--------------------	-------------------	----------------------------------	--	------------------	-----------

Perineal redness, oedema, bruising, discharge, wound gaping within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	35	36	-	MD 0.38 lower (1.14 lower to 0.38 higher)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----	----	---	---	------------------	-----------

Perineal redness, oedema, bruising, discharge, wound gaping 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^c	none	35	36	-	MD 1.19 lower (2.07 lower to 0.31 lower)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----	----	---	--	------------------	-----------

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack or cold gel pad)	No pain relief or usual care	Relative (95% CI)	Absolute (95% CI)		

Additional analgesia for relief of perineal pain within 24 hours of birth – non-prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	59/215 (27.4%)	32/101 (31.7%)	RR 0.87 (0.60 to 1.24)	41 fewer per 1000 (from 127 fewer to 76 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	---	------------------	-----------

Additional analgesia for relief of perineal pain within 24 hours of birth – prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	61/215 (28.4%)	23/101 (22.8%)	RR 1.25 (0.82 to 1.89)	57 more per 1000 (from 41 fewer to 203 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	--	------------------	-----------

Additional analgesia for relief of perineal pain 24–48 hours after birth – non-prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	51/215 (23.7%)	28/101 (27.7%)	RR 0.86 (0.58 to 1.27)	39 fewer per 1000 (from 116 fewer to 75 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	---	------------------	-----------

Additional analgesia for relief of perineal pain 24–48 hours after birth – prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	60/215 (27.9%)	21/101 (20.8%)	RR 1.34 (0.87 to 2.08)	71 more per 1000 (from 27 fewer to 225 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	--	------------------	-----------

Pain associated with activities of daily living (sitting) within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	206/212 (97.2%)	94/100 (94.0%)	RR 1.03 (0.98 to 1.09)	28 more per 1000 (from 19 fewer to 85 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-------------	------	-----------------	----------------	----------------------------------	---	-------------	-----------

Pain associated with activities of daily living (sitting) 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	203/212 (95.8%)	96/100 (96.0%)	RR 1.00 (0.95 to 1.05)	0 fewer per 1000 (from 48 fewer to 48 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-------------	------	-----------------	----------------	----------------------------------	---	-------------	-----------

Pain associated with activities of daily living (walking) within 24 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	196/212 (92.5%)	92/100 (92.0%)	RR 1.00 (0.94 to 1.08)	0 fewer per 1000 (from 55 fewer to 74 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-------------	------	-----------------	----------------	----------------------------------	---	-------------	-----------

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack or cold gel pad)	No pain relief or usual care	Relative (95% CI)	Absolute (95% CI)		

Pain associated with activities of daily living (walking) 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	190/212 (89.6%)	89/100 (89.0%)	RR 1.01 (0.93 to 1.09)	9 more per 1000 (from 62 fewer to 80 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-------------	------	-----------------	----------------	----------------------------------	--	-------------	-----------

Pain associated with activities of daily living (feeding baby) within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	88/212 (41.5%)	36/99 (36.4%)	RR 1.14 (0.84 to 1.55)	51 more per 1000 (from 58 fewer to 200 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	---------------	----------------------------------	--	------------------	-----------

Pain associated with activities of daily living (feeding baby) 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	66/211 (31.3%)	36/100 (36.0%)	RR 0.87 (0.63 to 1.21)	47 fewer per 1000 (from 133 fewer to 76 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	---	------------------	-----------

Maternal views and experiences of treatment at day 10 – satisfaction with overall perineal care (good + very good + excellent)

1	randomized trials	serious ^a	not serious	serious ^b	not serious	none	187/208 (89.9%)	84/100 (84.0%)	RR 1.07 (0.97 to 1.18)	59 more per 1000 (from 25 fewer to 151 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-------------	------	-----------------	----------------	----------------------------------	--	-------------	-----------

Women providing any breastmilk to baby 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	122/215 (56.7%)	64/100 (64.0%)	RR 0.89 (0.73 to 1.07)	70 fewer per 1000 (from 173 fewer to 45 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	-----------------	----------------	----------------------------------	---	------------------	-----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

b. Intervention was mainly focused on prevention and not relief of pain.

c. Less than 400 participants.

d. Wide confidence interval crossing the line of no effect.

e. Less than 300 participants.

Comparison 2: Perineal local cooling compared with other forms of non-pharmacological interventions

Comparison 2a: Perineal cooling and compression compared with uncooled gel pads and compression after vaginal birth in women with non-severe perineal trauma

Source: East CE, Dorward EDF, Whale RE, Liu J. Local cooling for relieving pain from perineal trauma sustained during childbirth. Cochrane Database Syst Rev. 2020;(10):CD006304.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (cold gel pad) + compression	Uncooled gel pad + compression	Relative (95% CI)	Absolute (95% CI)		
Perineal pain within 4–6 hours of birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{b,c}	none	125	125	-	MD 0.32 lower (0.78 lower to 0.14 higher)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain within 24–48 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	serious ^c	none	125	125	-	MD 0.43 lower (0.73 lower to 0.13 lower)	⊕⊕○○ LOW	IMPORTANT
Perineal oedema 24–48 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	serious ^c	none	125	125	-	MD 0.15 lower (0.28 lower to 0.03 lower)	⊕⊕○○ LOW	IMPORTANT
Perineal bruising 24–48 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	125	125	-	MD 0 (0 to 0)	⊕○○○ VERY LOW	IMPORTANT
Satisfaction with perineal care												
1	randomized trials	not serious	not serious	serious ^a	serious ^c	none	125	125	-	MD 0.88 higher (0.38 higher to 1.38 higher)	⊕⊕○○ LOW	IMPORTANT

CI: confidence interval; MD: mean difference.

a. Intervention was mainly focused on prevention and not relief of pain.

b. Wide confidence interval crossing the line of no effect.

c. Less than 400 participants.

d. No events.

Comparison 2b: Perineal cooling (ice packs) compared with room temperature water packs after vaginal birth in women with non-severe perineal trauma

Source: East CE, Dorward EDF, Whale RE, Liu J. Local cooling for relieving pain from perineal trauma sustained during childbirth. Cochrane Database Syst Rev. 2020;(10):CD006304.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack)	Room temperature water pack	Relative (95% CI)	Absolute (95% CI)		
Perineal pain within 4–6 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{b,c}	none	0/28 (0.0%)	0/35 (0.0%)	not estimable	-	⊕○○○ VERY LOW	IMPORTANT
Perineal pain within 24 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{b,c}	none	0/28 (0.0%)	0/35 (0.0%)	not estimable	-	⊕○○○ VERY LOW	IMPORTANT
Perineal oedema within 4–6 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	10/28 (35.7%)	13/35 (37.1%)	RR 0.96 (0.50 to 1.86)	15 fewer per 1000 (from 186 fewer to 319 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal oedema within 24 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	2/28 (7.1%)	7/35 (20.0%)	RR 0.36 (0.08 to 1.59)	128 fewer per 1000 (from 184 fewer to 118 more)	⊕○○○ VERY LOW	IMPORTANT
Additional analgesia for relief of perineal pain within 24 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	10/28 (35.7%)	20/35 (57.1%)	RR 0.63 (0.35 to 1.11)	211 fewer per 1000 (from 371 fewer to 63 more)	⊕○○○ VERY LOW	IMPORTANT
Maternal exhaustion within 4–6 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{b,c}	none	0/28 (0.0%)	0/35 (0.0%)	not estimable	-	⊕○○○ VERY LOW	IMPORTANT
Maternal exhaustion within 24 hours after birth												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{b,c}	none	0/28 (0.0%)	0/35 (0.0%)	not estimable	-	⊕○○○ VERY LOW	IMPORTANT

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack)	Room temperature water pack	Relative (95% CI)	Absolute (95% CI)		

Maternal views and experiences with treatment – satisfied with treatment

1	randomized trials	not serious	not serious	serious ^a	serious ^c	none	24/28 (85.7%)	33/35 (94.3%)	RR 0.91 (0.77 to 1.08)	85 fewer per 1000 (from 217 fewer to 75 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	-------------	-------------	----------------------	----------------------	------	---------------	---------------	----------------------------------	---	-------------	-----------

Maternal views and experiences with treatment – would repeat treatment in future childbirth

1	randomized trials	not serious	not serious	serious ^a	serious ^c	none	24/28 (85.7%)	34/35 (97.1%)	RR 0.88 (0.75 to 1.04)	117 fewer per 1000 (from 243 fewer to 39 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	-------------	-------------	----------------------	----------------------	------	---------------	---------------	----------------------------------	--	-------------	-----------

Maternal views and experiences with treatment – would recommend treatment

1	randomized trials	not serious	not serious	serious ^a	serious ^c	none	25/28 (89.3%)	35/35 (100.0%)	RR 0.89 (0.77 to 1.03)	110 fewer per 1000 (from 230 fewer to 30 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	-------------	-------------	----------------------	----------------------	------	---------------	----------------	----------------------------------	--	-------------	-----------

Women providing any breastmilk to the baby 48 hours after birth

1	randomized trials	not serious	not serious	serious ^a	serious ^c	none	28/28 (100.0%)	35/35 (100.0%)	RR 1.00 (0.94 to 1.06)	0 fewer per 1000 (from 60 fewer to 60 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	-------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	---	-------------	-----------

CI: confidence interval; RR: risk ratio.

a. Intervention was mainly focused on prevention and not relief of pain.

b. No events.

c. Less than 300 participants.

d. Wide confidence interval crossing the line of no effect.

e. Less than 30 events.

Comparison 2c: Perineal cooling (ice packs) compared with cold gel pads after vaginal birth in women with non-severe perineal trauma

Source: East CE, Dorward EDF, Whale RE, Liu J. Local cooling for relieving pain from perineal trauma sustained during childbirth. Cochrane Database Syst Rev. 2020;(10):CD006304.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack)	Cooling treatment (cold gel pad)	Relative (95% CI)	Absolute (95% CI)		
Perineal pain within 4–6 hours after birth – moderate + severe pain												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	6/22 (27.3%)	13/27 (48.1%)	RR 0.57 (0.26 to 1.24)	207 fewer per 1000 (from 356 fewer to 116 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain within 24 hours of birth – moderate + severe pain												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	68/129 (52.7%)	73/135 (54.1%)	RR 0.98 (0.78 to 1.22)	11 fewer per 1000 (from 119 fewer to 119 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain within 24 hours of birth												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	35	39	-	MD 0.58 higher (0.44 lower to 1.6 higher)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain 24–48 hours after birth – moderate + severe pain												
2	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	54/129 (41.9%)	46/134 (34.3%)	RR 1.21 (0.89 to 1.65)	72 more per 1000 (from 38 fewer to 223 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal pain 24–48 hours after birth												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	35	39	-	MD 0.86 higher (0.1 lower to 1.82 higher)	⊕○○○ VERY LOW	IMPORTANT
Perineal oedema within 4–6 hours after birth												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	17/22 (77.3%)	15/27 (55.6%)	RR 1.39 (0.93 to 2.09)	217 more per 1000 (from 39 fewer to 606 more)	⊕○○○ VERY LOW	IMPORTANT

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack)	Cooling treatment (cold gel pad)	Relative (95% CI)	Absolute (95% CI)		
Perineal oedema within 24 hours after birth												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	92/129 (71.3%)	99/135 (73.3%)	RR 0.97 (0.84 to 1.13)	22 fewer per 1000 (from 117 fewer to 95 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal oedema 24–48 hours after birth												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	79/129 (61.2%)	53/135 (39.3%)	RR 1.69 (1.03 to 2.77)	271 more per 1000 (from 12 more to 695 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal bruising within 4–6 hours of birth												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	7/22 (31.8%)	7/27 (25.9%)	RR 1.23 (0.51 to 2.97)	60 more per 1000 (from 127 fewer to 511 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal bruising within 24 hours of birth												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	78/129 (60.5%)	87/135 (64.4%)	RR 0.95 (0.79 to 1.14)	32 fewer per 1000 (from 135 fewer to 90 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal bruising 24–48 hours of birth												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	96/129 (74.4%)	94/135 (69.6%)	RR 1.07 (0.92 to 1.25)	49 more per 1000 (from 56 fewer to 174 more)	⊕○○○ VERY LOW	IMPORTANT
Perineal redness, oedema, bruising, discharge, wound gaping within 24 hours of birth												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^e	none	35	39	-	MD 0.13 lower (0.85 lower to 0.59 higher)	⊕○○○ VERY LOW	IMPORTANT
Perineal redness, oedema, bruising, discharge, wound gaping 24–48 hours after birth												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	35	39	-	MD 0.2 higher (0.33 lower to 0.73 higher)	⊕○○○ VERY LOW	IMPORTANT

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack)	Cooling treatment (cold gel pad)	Relative (95% CI)	Absolute (95% CI)		

Additional analgesia for relief of perineal pain: within 24 hours of birth – non-prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	29/107 (27.1%)	30/108 (27.8%)	RR 0.98 (0.63 to 1.51)	6 fewer per 1000 (from 103 fewer to 142 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	----------------	----------------------------------	---	------------------	-----------

Additional analgesia for relief of perineal pain: within 24 hours of birth – prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	28/107 (26.2%)	33/108 (30.6%)	RR 0.86 (0.56 to 1.31)	43 fewer per 1000 (from 134 fewer to 95 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	----------------	----------------------------------	---	------------------	-----------

Additional analgesia for relief of perineal pain: 24–48 hours after birth – non-prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	27/107 (25.2%)	24/108 (22.2%)	RR 1.14 (0.70 to 1.84)	31 more per 1000 (from 67 fewer to 187 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	----------------	----------------------------------	--	------------------	-----------

Additional analgesia for relief of perineal pain: 24–48 hours after birth – prescription analgesia

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	24/107 (22.4%)	36/108 (33.3%)	RR 0.67 (0.43 to 1.05)	110 fewer per 1000 (from 190 fewer to 17 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	----------------	----------------------------------	--	------------------	-----------

Pain associated with activities of daily living (sitting) within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	101/105 (96.2%)	105/107 (98.1%)	RR 0.98 (0.94 to 1.03)	20 fewer per 1000 (from 59 fewer to 29 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	-----------------	-----------------	----------------------------------	--	------------------	-----------

Pain associated with activities of daily living (sitting) 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	99/105 (94.3%)	104/107 (97.2%)	RR 0.97 (0.92 to 1.03)	29 fewer per 1000 (from 78 fewer to 29 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	-----------------	----------------------------------	--	------------------	-----------

Pain associated with activities of daily living (walking) within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	95/105 (90.5%)	101/107 (94.4%)	RR 0.96 (0.89 to 1.04)	38 fewer per 1000 (from 104 fewer to 38 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	-----------------	----------------------------------	---	------------------	-----------

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cooling treatment (ice pack)	Cooling treatment (cold gel pad)	Relative (95% CI)	Absolute (95% CI)		

Pain associated with activities of daily living (walking) 24–48 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	91/105 (86.7%)	99/107 (92.5%)	RR 0.94 (0.85 to 1.03)	56 fewer per 1000 (from 139 fewer to 28 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	---	------------------	-----------

Pain associated with activities of daily living (feeding baby) within 24 hours of birth

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	42/105 (40.0%)	46/107 (43.0%)	RR 0.93 (0.68 to 1.28)	30 fewer per 1000 (from 138 fewer to 120 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	----------------	----------------------------------	--	------------------	-----------

Pain associated with activities of daily living (feeding baby) 24–48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	34/104 (32.7%)	32/107 (29.9%)	RR 1.09 (0.73 to 1.63)	27 more per 1000 (from 81 fewer to 188 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	----------------	----------------------------------	--	------------------	-----------

Maternal views and experiences with treatment at day 5 – opinion on treatment effects (good + very good + excellent)

1	randomized trials	serious ^a	not serious	serious ^b	serious ^f	none	6/22 (27.3%)	22/27 (81.5%)	RR 0.33 (0.17 to 0.68)	546 fewer per 1000 (from 676 fewer to 261 fewer)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	--------------	---------------	----------------------------------	--	------------------	-----------

Maternal views and experiences with treatment at day 10 satisfaction with overall perineal care (good + very good + excellent)

1	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	78/102 (76.5%)	99/106 (93.4%)	RR 0.82 (0.73 to 0.92)	168 fewer per 1000 (from 252 fewer to 75 fewer)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	----------------------	------	----------------	----------------	----------------------------------	---	------------------	-----------

Women providing any breastmilk to the baby 48 hours after birth

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	62/105 (59.0%)	60/107 (56.1%)	RR 1.05 (0.84 to 1.33)	28 more per 1000 (from 90 fewer to 185 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	----------------	----------------------------------	--	------------------	-----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Intervention is mainly focused on prevention and not relief of pain.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 participants.

e. Less than 400 participants.

f. Less than 300 women and less than 30 events.

EB table A.2.2: Oral analgesia for perineal pain relief

Comparison 1: Single-dose oral analgesic (any dose) compared with placebo

Comparison 1a: Single-dose paracetamol compared with placebo

Source: Abalos E, Gyte GML, Sguassero Y. Paracetamol/acetaminophen (single administration) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2021;(1):CD008407.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Paracetamol (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
10	randomized trials	very serious ^a	serious ^b	not serious	not serious	none	422/700 (60.3%)	157/579 (27.1%)	RR 2.14 (1.59 to 2.89)	309 more per 1000 (from 160 more to 512 more)	⊕○○○ VERY LOW	IMPORTANT
Adequate pain relief as reported by the woman – paracetamol 500–650 mg												
5	randomized trials	very serious ^a	serious ^b	not serious	not serious	none	146/275 (53.1%)	56/207 (27.1%)	RR 1.86 (1.20 to 2.87)	233 more per 1000 (from 54 more to 506 more)	⊕○○○ VERY LOW	IMPORTANT
Adequate pain relief as reported by the woman – paracetamol 1000 mg												
6	randomized trials	serious ^c	serious ^b	not serious	not serious	none	276/425 (64.9%)	101/372 (27.2%)	RR 2.42 (1.53 to 3.81)	386 more per 1000 (from 144 more to 763 more)	⊕⊕○○ LOW	IMPORTANT
Additional pain relief												
8	randomized trials	serious ^c	serious ^b	not serious	not serious	none	65/620 (10.5%)	156/512 (30.5%)	RR 0.34 (0.21 to 0.55)	201 fewer per 1000 (from 241 fewer to 137 fewer)	⊕⊕○○ LOW	IMPORTANT
Additional pain relief – paracetamol 500–650 mg												
3	randomized trials	serious ^c	not serious	not serious	serious ^d	none	15/193 (7.8%)	33/124 (26.6%)	RR 0.30 (0.17 to 0.53)	186 fewer per 1000 (from 221 fewer to 125 fewer)	⊕⊕○○ LOW	IMPORTANT

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Paracetamol (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		

Additional pain relief – paracetamol 1000 mg

6	randomized trials	serious ^c	serious ^b	not serious	not serious	none	50/427 (11.7%)	123/388 (31.7%)	RR 0.36 (0.19 to 0.67)	203 fewer per 1000 (from 257 fewer to 105 fewer)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	----------------------	----------------------	-------------	-------------	------	----------------	-----------------	----------------------------------	--	-------------	-----------

Maternal nausea – paracetamol 1000 mg

1	randomized trials	not serious	not serious	not serious	very serious ^{d,e}	none	0/123 (0.0%)	2/109 (1.8%)	RR 0.18 (0.01 to 3.66)	15 fewer per 1000 (from 18 fewer to 49 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	-------------	-------------	-------------	-----------------------------	------	--------------	--------------	----------------------------------	--	-------------	-----------

Maternal sleepiness – paracetamol 1000 mg

1	randomized trials	not serious	not serious	not serious	very serious ^{d,e}	none	3/123 (2.4%)	3/109 (2.8%)	RR 0.89 (0.18 to 4.30)	3 fewer per 1000 (from 23 fewer to 91 more)	⊕⊕○○ LOW	IMPORTANT
---	-------------------	-------------	-------------	-------------	-----------------------------	------	--------------	--------------	----------------------------------	---	-------------	-----------

Maternal bowel movements (not pre-specified)

1	randomized trials	serious ^c	not serious	not serious	very serious ^{d,e}	none	26/175 (14.9%)	13/88 (14.8%)	RR 1.01 (0.54 to 1.86)	1 more per 1000 (from 68 fewer to 127 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	----------------	---------------	----------------------------------	---	------------------	-----------

Maternal bowel movements (not pre-specified) – paracetamol 500–650 mg

1	randomized trials	serious ^c	not serious	not serious	very serious ^{d,e}	none	13/88 (14.8%)	6/44 (13.6%)	RR 1.08 (0.44 to 2.66)	11 more per 1000 (from 76 fewer to 226 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	---------------	--------------	----------------------------------	--	------------------	-----------

Maternal bowel movements (not pre-specified) – paracetamol 1000 mg

1	randomized trials	serious ^c	not serious	not serious	very serious ^{d,e}	none	13/87 (14.9%)	7/44 (15.9%)	RR 0.94 (0.40 to 2.18)	10 fewer per 1000 (from 95 fewer to 188 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	---------------	--------------	----------------------------------	---	------------------	-----------

Maternal gastric discomfort (not pre-specified) – paracetamol 1000 mg

1	randomized trials	serious ^c	not serious	not serious	very serious ^{d,e}	none	13/75 (17.3%)	11/75 (14.7%)	RR 1.18 (0.57 to 2.47)	26 more per 1000 (from 63 fewer to 216 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	---------------	---------------	----------------------------------	--	------------------	-----------

CI: confidence interval; RR: risk ratio.

- a. Most of the pooled effect provided by studies “B” or “C” but with a substantial proportion (i.e. > 50%) from studies “C”.
- b. Severe, unexplained, heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 < 0.05$)
- c. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.
- d. Small sample size and or few events.
- e. Wide confidence interval crossing the line of no effect.

Comparison 1b: Single-dose aspirin compared with placebo

Source: Shepherd E, Grivell RM. Aspirin (single dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2020;(7):CD012129.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Aspirin (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
13	randomized trials	serious ^a	not serious	serious ^b	not serious	none	258/550 (46.9%)	114/451 (25.3%)	RR 2.03 (1.69 to 2.42)	260 more per 1000 (from 174 more to 359 more)	⊕⊕○○ LOW	IMPORTANT
Adequate pain relief as reported by the woman – aspirin 300 mg												
1	randomized trials	very serious ^c	not serious	not serious	very serious ^{d,e}	none	8/40 (20.0%)	1/13 (7.7%)	RR 2.60 (0.36 to 18.88)	123 more per 1000 (from 49 fewer to 1000 more)	⊕○○○ VERY LOW	IMPORTANT
Adequate pain relief as reported by the woman – aspirin 500–650 mg												
11	randomized trials	serious ^a	not serious	serious ^b	not serious	none	209/417 (50.1%)	101/383 (26.4%)	RR 1.98 (1.64 to 2.39)	258 more per 1000 (from 169 more to 367 more)	⊕⊕○○ LOW	IMPORTANT
Adequate pain relief as reported by the woman – aspirin 900 mg												
1	randomized trials	very serious ^c	not serious	serious ^b	very serious ^{d,e}	none	11/20 (55.0%)	6/20 (30.0%)	RR 1.83 (0.84 to 3.99)	249 more per 1000 (from 48 fewer to 897 more)	⊕○○○ VERY LOW	IMPORTANT
Adequate pain relief as reported by the woman – aspirin 1200 mg												
3	randomized trials	very serious ^c	not serious	serious ^b	very serious ^{d,f}	none	30/73 (41.1%)	6/35 (17.1%)	RR 2.75 (1.25 to 6.06)	300 more per 1000 (from 43 more to 867 more)	⊕○○○ VERY LOW	IMPORTANT

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Aspirin (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Need for additional pain relief 4–8 hours after drug administration												
10	randomized trials	serious ^a	not serious	serious ^b	not serious	none	26/422 (6.2%)	86/322 (26.7%)	RR 0.25 (0.17 to 0.37)	200 fewer per 1000 (from 222 fewer to 168 fewer)	⊕⊕○○ LOW	IMPORTANT
Need for additional pain relief – aspirin 300 mg												
1	randomized trials	very serious ^c	not serious	not serious	serious ^e	none	2/40 (5.0%)	4/13 (30.8%)	RR 0.16 (0.03 to 0.79)	258 fewer per 1000 (from 298 fewer to 65 fewer)	⊕○○○ VERY LOW	IMPORTANT
Need for additional pain relief – aspirin 500–650 mg												
9	randomized trials	very serious ^c	not serious	serious ^b	not serious	none	21/302 (7.0%)	73/267 (27.3%)	RR 0.27 (0.17 to 0.41)	200 fewer per 1000 (from 227 fewer to 161 fewer)	⊕○○○ VERY LOW	IMPORTANT
Need for additional pain relief – aspirin 900 mg												
1	randomized trials	very serious ^g	not serious	serious ^b	very serious ^{d,e}	none	0/20 (0.0%)	3/20 (15.0%)	RR 0.14 (0.01 to 2.60)	129 fewer per 1000 (from 149 fewer to 240 more)	⊕○○○ VERY LOW	IMPORTANT
Need for additional pain relief – aspirin 1200 mg												
2	randomized trials	very serious ^c	not serious	serious ^b	serious ^e	none	3/60 (5.0%)	6/22 (27.3%)	RR 0.20 (0.06 to 0.70)	218 fewer per 1000 (from 256 fewer to 82 fewer)	⊕○○○ VERY LOW	IMPORTANT
Maternal adverse effects												
14	randomized trials	very serious ^c	not serious	serious ^b	very serious ^{d,h}	none	16/583 (2.7%)	13/484 (2.7%)	RR 1.08 (0.57 to 2.06)	2 more per 1000 (from 12 fewer to 28 more)	⊕○○○ VERY LOW	IMPORTANT

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Aspirin (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		

Maternal adverse effects – aspirin 300 mg

1	randomized trials	very serious ^g	not serious	not serious	very serious ⁱ	none	0/40 (0.0%)	0/13 (0.0%)	not estimable		⊕○○○ VERY LOW	IMPORTANT
---	-------------------	---------------------------	-------------	-------------	---------------------------	------	-------------	-------------	---------------	--	------------------	-----------

Maternal adverse effects – aspirin 500–650 mg

13	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,h}	none	11/463 (2.4%)	9/429 (2.1%)	RR 1.13 (0.51 to 2.53)	3 more per 1000 (from 10 fewer to 32 more)	⊕○○○ VERY LOW	IMPORTANT
----	-------------------	----------------------	-------------	----------------------	-----------------------------	------	---------------	--------------	---------------------------	---	------------------	-----------

Maternal adverse effects – aspirin 900 mg

1	randomized trials	very serious ^g	not serious	serious ^b	very serious ^{d,e}	none	5/20 (25.0%)	2/20 (10.0%)	RR 2.50 (0.55 to 11.41)	150 more per 1000 (from 45 fewer to 1000 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	---------------------------	-------------	----------------------	-----------------------------	------	--------------	--------------	----------------------------	---	------------------	-----------

Maternal adverse effects – aspirin 1200 mg

2	randomized trials	very serious ^g	not serious	serious ^b	very serious ^{d,e}	none	0/60 (0.0%)	2/22 (9.1%)	RR 0.10 (0.01 to 1.80)	82 fewer per 1000 (from 90 fewer to 73 more)	⊕○○○ VERY LOW	IMPORTANT
---	-------------------	---------------------------	-------------	----------------------	-----------------------------	------	-------------	-------------	---------------------------	---	------------------	-----------

CI: confidence interval; RR: risk ratio.

- a. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.
- b. Many studies excluded breastfeeding women – the evidence cannot be extrapolated to all women during the postpartum period.
- c. Most of the pooled effect provided by studies “B” or “C” but with a substantial proportion (i.e. > 50%) from studies “C”.
- d. Wide confidence interval crossing the line of no effect.
- e. Less than 300 women and few events.
- f. Less than 300 women.
- g. All of the pooled effect provided by study “C”.
- h. Less than 30 events.
- i. No events.

Comparison 1c: Single-dose NSAID compared with placebo

Source: Wuytack F, Smith V, Cleary BJ. Oral non-steroidal anti-inflammatory drugs (single dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2021;(1):CD011352.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration)												
10	randomized trials	serious ^a	not serious	serious ^b	not serious	publication bias strongly suspected ^c	597/1105 (54.0%)	133/468 (28.4%)	RR 1.91 (1.64 to 2.23)	259 more per 1000 (from 182 more to 350 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – ibuprofen 300–400 mg												
3	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	64/146 (43.8%)	16/94 (17.0%)	RR 2.64 (1.62 to 4.30)	279 more per 1000 (from 106 more to 562 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – ibuprofen 800 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e}	none	25/80 (31.3%)	7/41 (17.1%)	RR 1.83 (0.87 to 3.87)	142 more per 1000 (from 22 fewer to 490 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diclofenac 25 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e}	none	32/52 (61.5%)	4/13 (30.8%)	RR 2.00 (0.86 to 4.65)	308 more per 1000 (from 43 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diclofenac 50 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e}	none	34/50 (68.0%)	4/13 (30.8%)	RR 2.21 (0.96 to 5.11)	372 more per 1000 (from 12 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diclofenac 100 mg												
1	randomized trials	not serious	not serious	serious ^b	serious ^d	none	37/51 (72.5%)	4/13 (30.8%)	RR 2.36 (1.03 to 5.42)	418 more per 1000 (from 9 more to 1000 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration) – ketoprofen 25 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	serious ^{d,g}	none	20/28 (71.4%)	3/14 (21.4%)	RR 3.33 (1.19 to 9.34)	499 more per 1000 (from 41 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diflunisal 125 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	12/33 (36.4%)	1/8 (12.5%)	RR 2.91 (0.44 to 19.22)	239 more per 1000 (from 70 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – meclofenamate sodium 100 mg												
3	randomized trials	serious ^a	not serious	not serious	serious ^d	none	110/173 (63.6%)	39/87 (44.8%)	RR 1.42 (1.10 to 1.82)	188 more per 1000 (from 45 more to 368 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (4 hours after administration) – meclofenamate sodium 200 mg												
3	randomized trials	serious ^a	not serious	not serious	serious ^d	none	112/175 (64.0%)	39/87 (44.8%)	RR 1.42 (1.10 to 1.83)	188 more per 1000 (from 45 more to 372 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (4 hours after administration) – ketoprofen 50 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	very serious ^{d,g}	none	18/26 (69.2%)	3/14 (21.4%)	RR 3.23 (1.15 to 9.10)	478 more per 1000 (from 32 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diflunisal 250 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	9/30 (30.0%)	1/8 (12.5%)	RR 2.40 (0.35 to 16.26)	175 more per 1000 (from 81 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diflunisal 500 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	14/30 (46.7%)	1/8 (12.5%)	RR 3.73 (0.57 to 24.29)	341 more per 1000 (from 54 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration) – flurbiprofen 25 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	11/32 (34.4%)	1/8 (12.5%)	RR 2.75 (0.41 to 18.29)	219 more per 1000 (from 74 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – flurbiprofen 50 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	15/29 (51.7%)	1/8 (12.5%)	RR 4.14 (0.64 to 26.76)	392 more per 1000 (from 45 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – flurbiprofen 100 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	15/31 (48.4%)	1/8 (12.5%)	RR 3.87 (0.60 to 25.09)	359 more per 1000 (from 50 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration)												
17	randomized trials	serious ^a	not serious	serious ^b	not serious	none	870/1455 (59.8%)	200/624 (32.1%)	RR 1.92 (1.69 to 2.17)	295 more per 1000 (from 221 more to 375 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (6 hours after administration) – ibuprofen 300–400 mg												
2	randomized trials	not serious	not serious	serious ^b	serious ^d	none	44/69 (63.8%)	16/55 (29.1%)	RR 2.08 (1.30 to 3.32)	314 more per 1000 (from 87 more to 675 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (6 hours after administration) – ibuprofen 900 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	17/20 (85.0%)	2/7 (28.6%)	RR 2.98 (0.91 to 9.74)	566 more per 1000 (from 26 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – ketoprofen 25 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	serious ^{d,g}	none	18/28 (64.3%)	3/14 (21.4%)	RR 3.00 (1.06 to 8.49)	429 more per 1000 (from 13 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – ketoprofen 50 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	serious ^{d,g}	none	17/26 (65.4%)	3/14 (21.4%)	RR 3.05 (1.08 to 8.64)	439 more per 1000 (from 17 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – meclofenamate sodium 100 mg												
3	randomized trials	serious ^a	not serious	not serious	serious ^d	none	103/173 (59.5%)	38/87 (43.7%)	RR 1.36 (1.05 to 1.76)	157 more per 1000 (from 22 more to 332 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (6 hours after administration) – meclofenamate sodium 200 mg												
3	randomized trials	serious ^a	not serious	not serious	serious ^d	none	105/175 (60.0%)	37/87 (42.5%)	RR 1.40 (1.07 to 1.83)	170 more per 1000 (from 30 more to 353 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (6 hours after administration) – diflunisal 125 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	13/33 (39.4%)	1/8 (12.5%)	RR 3.15 (0.48 to 20.69)	269 more per 1000 (from 65 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – diflunisal 250 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	10/30 (33.3%)	1/8 (12.5%)	RR 2.67 (0.40 to 17.86)	209 more per 1000 (from 75 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – diflunisal 500 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	16/30 (53.3%)	1/8 (12.5%)	RR 4.27 (0.66 to 27.51)	409 more per 1000 (from 42 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – dipyron 500 mg												
1	randomized trials	serious ^a	not serious	not serious	serious ^d	none	67/89 (75.3%)	15/44 (34.1%)	RR 2.21 (1.44 to 3.39)	413 more per 1000 (from 150 more to 815 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – aceclofenac 50 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	14/18 (77.8%)	2/4 (50.0%)	RR 1.56 (0.57 to 4.27)	280 more per 1000 (from 215 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aceclofenac 100 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	20/24 (83.3%)	2/4 (50.0%)	RR 1.67 (0.62 to 4.51)	335 more per 1000 (from 190 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aceclofenac 150 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	19/21 (90.5%)	2/4 (50.0%)	RR 1.81 (0.67 to 4.87)	405 more per 1000 (from 165 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – etodolac 25 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e,g}	none	11/40 (27.5%)	4/13 (30.8%)	RR 0.89 (0.34 to 2.33)	34 fewer per 1000 (from 203 fewer to 409 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – etodolac 100 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e,g}	none	15/40 (37.5%)	4/13 (30.8%)	RR 1.22 (0.49 to 3.02)	68 more per 1000 (from 157 fewer to 622 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – antrafenine 300 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^{d,g}	none	16/29 (55.2%)	3/29 (10.3%)	RR 5.33 (1.74 to 16.36)	448 more per 1000 (from 77 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – flurbiprofen 25 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	13/32 (40.6%)	1/8 (12.5%)	RR 3.25 (0.50 to 21.31)	281 more per 1000 (from 63 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – flurbiprofen 50 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	18/29 (62.1%)	1/8 (12.5%)	RR 4.97 (0.78 to 31.75)	496 more per 1000 (from 27 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – flurbiprofen 100 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	19/31 (61.3%)	1/8 (12.5%)	RR 4.90 (0.77 to 31.33)	488 more per 1000 (from 29 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 12.5 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	very serious ^{d,e,g}	none	10/24 (41.7%)	1/5 (20.0%)	RR 2.08 (0.34 to 12.80)	216 more per 1000 (from 132 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 25 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	very serious ^{d,e,g}	none	11/23 (47.8%)	1/5 (20.0%)	RR 2.39 (0.39 to 14.53)	278 more per 1000 (from 122 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 50 mg												
2	randomized trials	very serious ^f	not serious	serious ^b	very serious ^{d,e}	none	28/50 (56.0%)	2/12 (16.7%)	RR 3.38 (0.93 to 12.26)	397 more per 1000 (from 12 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 100 mg												
2	randomized trials	very serious ^f	not serious	serious ^b	serious ^d	none	33/50 (66.0%)	2/12 (16.7%)	RR 3.95 (1.10 to 14.19)	492 more per 1000 (from 17 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 200 mg												
2	randomized trials	very serious ^f	not serious	serious ^b	serious ^d	none	32/49 (65.3%)	2/12 (16.7%)	RR 3.95 (1.10 to 14.19)	492 more per 1000 (from 17 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – fenoprofen 300 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	very serious ^{d,e,g}	none	19/27 (70.4%)	1/7 (14.3%)	RR 4.93 (0.79 to 30.74)	561 more per 1000 (from 30 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (4 hours after administration)												
4	randomized trials	serious ^a	not serious	serious ^b	not serious	none	30/296 (10.1%)	58/190 (30.5%)	RR 0.39 (0.26 to 0.58)	186 fewer per 1000 (from 226 fewer to 128 fewer)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (4 hours after administration) – ibuprofen 300–400 mg												
3	randomized trials	not serious	not serious	serious ^b	serious ^d	none	12/146 (8.2%)	32/94 (34.0%)	RR 0.32 (0.18 to 0.56)	231 fewer per 1000 (from 279 fewer to 150 fewer)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (4 hours after administration) – ibuprofen 800 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	7/80 (8.8%)	7/41 (17.1%)	RR 0.51 (0.19 to 1.36)	84 fewer per 1000 (from 138 fewer to 61 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration)												
10	randomized trials	serious ^a	not serious	serious ^b	not serious	publication bias strongly suspected ^c	81/628 (12.9%)	168/384 (43.8%)	RR 0.32 (0.26 to 0.40)	298 fewer per 1000 (from 324 fewer to 263 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – ibuprofen 300–400 mg												
3	randomized trials	not serious	not serious	serious ^b	serious ^d	none	16/100 (16.0%)	46/86 (53.5%)	RR 0.33 (0.20 to 0.54)	358 fewer per 1000 (from 428 fewer to 246 fewer)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – ibuprofen 900 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	0/20 (0.0%)	1/7 (14.3%)	RR 0.13 (0.01 to 2.81)	124 fewer per 1000 (from 141 fewer to 259 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Need for additional analgesia (6 hours after administration) – meclofenamate sodium 100 mg												
3	randomized trials	serious ^a	not serious	not serious	serious ^d	none	23/173 (13.3%)	37/126 (29.4%)	RR 0.34 (0.21 to 0.53)	194 fewer per 1000 (from 232 fewer to 138 fewer)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – meclofenamate sodium 200 mg												
2	randomized trials	serious ^a	not serious	not serious	serious ^d	none	24/95 (25.3%)	26/47 (55.3%)	RR 0.45 (0.29 to 0.70)	304 fewer per 1000 (from 393 fewer to 166 fewer)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – antrafenine 300 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^{d,g}	none	5/29 (17.2%)	16/29 (55.2%)	RR 0.31 (0.13 to 0.74)	381 fewer per 1000 (from 480 fewer to 143 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – flurbiprofen 25 mg												
1	randomized trials	not serious	not serious	serious ^b	serious ^{d,g}	none	1/32 (3.1%)	4/8 (50.0%)	RR 0.06 (0.01 to 0.49)	470 fewer per 1000 (from 495 fewer to 255 fewer)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – flurbiprofen 50 mg												
1	randomized trials	not serious	not serious	serious ^b	serious ^{d,g}	none	0/29 (0.0%)	4/8 (50.0%)	RR 0.03 (0.00 to 0.56)	485 fewer per 1000 (from 220 fewer to --)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – flurbiprofen 100 mg												
1	randomized trials	not serious	not serious	serious ^b	serious ^{d,g}	none	0/31 (0.0%)	4/8 (50.0%)	RR 0.03 (0.00 to 0.53)	485 fewer per 1000 (from 235 fewer to --)	⊕⊕○○ LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration)												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,h}	none	0/60 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Maternal drug adverse effects (4 hours after administration) – aspirin 500–650 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,h}	none	0/30 (0.0%)	0/15 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – ibuprofen 300–400 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,h}	none	0/30 (0.0%)	0/15 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration)												
13	randomized trials	not serious	not serious	serious ^b	serious ^e	none	24/897 (2.7%)	11/491 (2.2%)	RR 1.38 (0.71 to 2.70)	9 more per 1000 (from 6 fewer to 38 more)	⊕⊕○○ LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – ibuprofen 300–400 mg												
3	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	5/100 (5.0%)	3/86 (3.5%)	RR 1.01 (0.27 to 3.85)	0 fewer per 1000 (from 25 fewer to 99 more)	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – ibuprofen 900 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,e,g}	none	3/20 (15.0%)	1/7 (14.3%)	RR 1.05 (0.13 to 8.52)	7 more per 1000 (from 124 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – ketoprofen 25 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	very serious ^{d,h}	none	0/28 (0.0%)	0/14 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – ketoprofen 50 mg												
1	randomized trials	very serious ^f	not serious	serious ^b	very serious ^{d,h}	none	0/26 (0.0%)	0/14 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aceclofenac 50 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e,g}	none	1/18 (5.6%)	0/4 (0.0%)	RR 0.79 (0.04 to 16.59)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Maternal drug adverse effects (6 hours after administration) – aceclofenac 100 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,h}	none	0/24 (0.0%)	0/4 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aceclofenac 150 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,h}	none	0/21 (0.0%)	0/4 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – diflunisal 125 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,h}	none	0/33 (0.0%)	0/8 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – diflunisal 250 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,h}	none	0/30 (0.0%)	0/8 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – diflunisal 500 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,h}	none	0/30 (0.0%)	0/8 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – dipyronе 500 mg												
2	randomized trials	not serious	not serious	not serious	very serious ^{e,g}	none	5/190 (2.6%)	2/145 (1.4%)	RR 2.48 (0.49 to 12.46)	20 more per 1000 (from 7 fewer to 158 more)	⊕⊕○○ LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – antrafenine 300 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,h}	none	0/29 (0.0%)	0/29 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – flurbiprofen 25 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,h}	none	0/32 (0.0%)	0/8 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – flurbiprofen 50 mg												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,h}	none	0/29 (0.0%)	0/8 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Placebo	Relative (95% CI)	Absolute (95% CI)		

Maternal drug adverse effects (6 hours after administration) – flurbiprofen 100 mg

1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,h}	none	0/31 (0.0%)	0/8 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
---	-------------------	-------------	-------------	----------------------	-----------------------------	------	-------------	------------	---------------	---	------------------	----------

CI: confidence interval; RR: risk ratio.

a. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

b. Many studies excluded breastfeeding women – the evidence cannot be extrapolated to all women during the postpartum period.

c. Evident asymmetry in funnel plot with at least 10 studies.

d. Less than 300 participants.

e. Wide confidence interval crossing the line of no effect.

f. Most of the pooled effect provided by studies “B” or “C” but with a substantial proportion (i.e. > 50%) from studies “C”.

g. Few events.

h. No events.

Comparison 2: Single-dose oral analgesic compared with a higher single dose of the same analgesic

Comparison 2a(i): Single-dose aspirin compared with a higher single dose of aspirin (300 mg versus 600 mg)

Source: Shepherd E, Grivell RM. Aspirin (single dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2020;(7):CD012129.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	300 mg aspirin single dose	600 mg aspirin single dose	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	8/40 (20.0%)	10/41 (24.4%)	RR 0.82 (0.36 to 1.86)	44 fewer per 1000 (from 156 fewer to 210 more)	⊕○○○ VERY LOW	IMPORTANT
Need for additional pain relief												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	2/40 (5.0%)	3/41 (7.3%)	RR 0.68 (0.12 to 3.88)	23 fewer per 1000 (from 64 fewer to 211 more)	⊕○○○ VERY LOW	IMPORTANT
Maternal adverse effects												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{d,e}	none	0/40 (0.0%)	0/41 (0.0%)	not estimable	-	⊕○○○ VERY LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. All of the pooled effect provided by study "C".

b. Wide confidence interval crossing the line of no effect.

c. Less than 300 women and less than 30 events.

d. No events.

e. Less than 300 women.

Comparison 2a(ii): Single-dose aspirin compared with a higher single dose of aspirin (600 mg versus 1200 mg)

Source: Shepherd E, Grivell RM. Aspirin (single dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2020;(7):CD012129.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	600 mg aspirin single dose	1200 mg aspirin single dose	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
2	randomized trials	very serious ^a	not serious	serious ^b	very serious ^{c,d}	none	19/61 (31.1%)	22/60 (36.7%)	RR 0.85 (0.52 to 1.39)	55 fewer per 1000 (from 176 fewer to 143 more)	⊕○○○ VERY LOW	IMPORTANT
Need for additional pain relief												
2	randomized trials	very serious ^a	not serious	serious ^b	very serious ^{c,d,e}	none	4/61 (6.6%)	3/60 (5.0%)	RR 1.32 (0.30 to 5.68)	16 more per 1000 (from 35 fewer to 234 more)	⊕○○○ VERY LOW	IMPORTANT
Maternal adverse effects												
2	randomized trials	very serious ^a	not serious	serious ^b	very serious ^{c,d,e}	none	1/61 (1.6%)	0/60 (0.0%)	RR 3.00 (0.13 to 69.52)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. All of the pooled effect provided by studies "C".

b. One of the studies reporting this outcome excluded breastfeeding women – thus the data cannot be extrapolated to all women during postnatal period.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women.

e. Few events.

Comparison 2a(iii): Single-dose aspirin compared with a higher single dose of aspirin (300 mg versus 1200 mg)

Source: Shepherd E, Grivell RM. Aspirin (single dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2020;(7):CD012129.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	300 mg aspirin single dose	1200 mg aspirin single dose	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	8/40 (20.0%)	13/40 (32.5%)	RR 0.62 (0.29 to 1.32)	124 fewer per 1000 (from 231 fewer to 104 more)	⊕○○○ VERY LOW	IMPORTANT
Need for additional pain relief												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	2/40 (5.0%)	1/40 (2.5%)	RR 2.00 (0.19 to 21.18)	25 more per 1000 (from 20 fewer to 505 more)	⊕○○○ VERY LOW	IMPORTANT
Maternal adverse effects												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^d		0/40 (0.0%)	0/40 (0.0%)	not estimable	-	⊕○○○ VERY LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. All of the pooled effect provided by study "C".

b. Wide confidence interval crossing the line of no effect.

c. Less than 300 women and few events.

d. No events.

Comparison 2b: Single-dose NSAID compared with a higher single dose of the same NSAID

Source: Wuytack F, Smith V, Cleary BJ. Oral non-steroidal anti-inflammatory drugs (single-dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2021;(1):CD011352.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration) – ibuprofen 300–400 mg (A) vs ibuprofen 800 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	25/80 (31.3%)	25/80 (31.3%)	RR 1.00 (0.63 to 1.58)	0 fewer per 1000 (from 116 fewer to 181 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diflunisal 125 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	12/33 (36.4%)	9/30 (30.0%)	RR 1.21 (0.60 to 2.46)	63 more per 1000 (from 120 fewer to 438 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diflunisal 125 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	12/33 (36.4%)	14/30 (46.7%)	RR 0.78 (0.43 to 1.41)	103 fewer per 1000 (from 266 fewer to 191 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – diflunisal 250 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	9/30 (30.0%)	14/30 (46.7%)	RR 0.64 (0.33 to 1.25)	168 fewer per 1000 (from 313 fewer to 117 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – meclofenamate sodium 100 mg (A) vs meclofenamate sodium 200 mg (B)												
3	randomized trials	serious ^a	not serious	not serious	not serious	none	110/173 (63.6%)	112/175 (64.0%)	RR 1.00 (0.85 to 1.17)	0 fewer per 1000 (from 96 fewer to 109 more)	⊕⊕⊕○ MODERATE	CRITICAL
Adequate pain relief (4 hours after administration) – diclofenac 25 mg (A) vs diclofenac 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c}	none	32/52 (61.5%)	34/50 (68.0%)	RR 0.90 (0.68 to 1.21)	68 fewer per 1000 (from 218 fewer to 143 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration) – diclofenac 25 mg (A) vs diclofenac 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c}	none	32/52 (61.5%)	37/51 (72.5%)	RR 0.85 (0.65 to 1.11)	109 fewer per 1000 (from 254 fewer to 80 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – ketoprofen 25 mg (A) vs ketoprofen 50 mg (B)												
1	randomized trials	very serious ^f	not serious	serious ^e	very serious ^{b,c}	none	20/28 (71.4%)	18/26 (69.2%)	RR 1.03 (0.73 to 1.46)	21 more per 1000 (from 187 fewer to 318 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aceclofenac 50 mg (A) vs aceclofenac 100 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	12/18 (66.7%)	16/24 (66.7%)	RR 1.00 (0.65 to 1.54)	0 fewer per 1000 (from 233 fewer to 360 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aceclofenac 50 mg (A) vs aceclofenac 150 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	12/18 (66.7%)	17/21 (81.0%)	RR 0.82 (0.56 to 1.21)	146 fewer per 1000 (from 356 fewer to 170 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aceclofenac 100 mg (A) vs aceclofenac 150 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	16/24 (66.7%)	17/21 (81.0%)	RR 0.82 (0.58 to 1.17)	146 fewer per 1000 (from 340 fewer to 138 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c,d}	none	11/32 (34.4%)	15/29 (51.7%)	RR 0.66 (0.37 to 1.20)	176 fewer per 1000 (from 326 fewer to 103 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c,d}	none	11/32 (34.4%)	15/31 (48.4%)	RR 0.71 (0.39 to 1.30)	140 fewer per 1000 (from 295 fewer to 145 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 after administration) – flurbiprofen 50 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c}	none	15/29 (51.7%)	15/31 (48.4%)	RR 1.07 (0.64 to 1.77)	34 more per 1000 (from 174 fewer to 373 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – ibuprofen 300–400 mg (A) vs ibuprofen 900 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c}	none	17/20 (85.0%)	17/20 (85.0%)	RR 1.00 (0.77 to 1.30)	0 fewer per 1000 (from 195 fewer to 255 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – diflunisal 125 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	13/33 (39.4%)	10/30 (33.3%)	RR 1.18 (0.61 to 2.29)	60 more per 1000 (from 130 fewer to 430 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – diflunisal 125 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	13/33 (39.4%)	16/30 (53.3%)	RR 0.74 (0.43 to 1.27)	139 fewer per 1000 (from 304 fewer to 144 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – diflunisal 250 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	10/30 (33.3%)	16/30 (53.3%)	RR 0.63 (0.34 to 1.15)	197 fewer per 1000 (from 352 fewer to 80 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – meclofenamate sodium 100 mg (A) vs meclofenamate sodium 200 mg (B)												
3	randomized trials	serious ^a	not serious	not serious	not serious	none	103/173 (59.5%)	105/175 (60.0%)	RR 1.00 (0.84 to 1.18)	0 fewer per 1000 (from 96 fewer to 108 more)	⊕⊕⊕○ MODERATE	CRITICAL
Adequate pain relief (6 hours after administration) – ketoprofen 25 mg (A) vs ketoprofen 50 mg (B)												
1	randomized trials	very serious ^f	not serious	serious ^e	very serious ^{b,c}	none	18/28 (64.3%)	17/26 (65.4%)	RR 0.98 (0.66 to 1.46)	13 fewer per 1000 (from 222 fewer to 301 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – aceclofenac 50 mg (A) vs aceclofenac 100 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	14/18 (77.8%)	20/24 (83.3%)	RR 0.93 (0.69 to 1.27)	58 fewer per 1000 (from 258 fewer to 225 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aceclofenac 50 mg (A) vs aceclofenac 150 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	14/18 (77.8%)	19/21 (90.5%)	RR 0.86 (0.65 to 1.14)	127 fewer per 1000 (from 317 fewer to 127 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aceclofenac 100 mg (A) vs aceclofenac 150 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	20/24 (83.3%)	19/21 (90.5%)	RR 0.92 (0.73 to 1.16)	72 fewer per 1000 (from 244 fewer to 145 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – etodolac 25 mg (A) vs etodolac 100 mg (B)												
1	randomized trials	serious ^a	not serious	serious ^e	very serious ^{b,c,d}	none	11/40 (27.5%)	15/40 (37.5%)	RR 0.73 (0.39 to 1.39)	101 fewer per 1000 (from 229 fewer to 146 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c}	none	13/32 (40.6%)	18/29 (62.1%)	RR 0.65 (0.39 to 1.09)	217 fewer per 1000 (from 379 fewer to 56 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c}	none	13/32 (40.6%)	19/31 (61.3%)	RR 0.66 (0.40 to 1.10)	208 fewer per 1000 (from 368 fewer to 61 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – flurbiprofen 50 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c}	none	18/29 (62.1%)	19/31 (61.3%)	RR 1.01 (0.68 to 1.51)	6 more per 1000 (from 196 fewer to 313 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – fenoprofen 50 mg (A) vs fenoprofen 100 mg (B)												
2	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c}	none	28/50 (56.0%)	33/50 (66.0%)	RR 0.85 (0.62 to 1.16)	99 fewer per 1000 (from 251 fewer to 106 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 50 mg (A) vs fenoprofen 200 mg (B)												
2	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c}	none	28/50 (56.0%)	32/49 (65.3%)	RR 0.86 (0.62 to 1.17)	91 fewer per 1000 (from 248 fewer to 111 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 50 mg (A) vs fenoprofen 300 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c}	none	17/27 (63.0%)	19/27 (70.4%)	RR 0.89 (0.61 to 1.31)	77 fewer per 1000 (from 274 fewer to 218 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 100 mg (A) vs fenoprofen 200 mg (B)												
2	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c}	none	33/50 (66.0%)	32/49 (65.3%)	RR 1.01 (0.76 to 1.34)	7 more per 1000 (from 157 fewer to 222 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 100 mg (A) vs fenoprofen 300 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c}	none	19/26 (73.1%)	19/27 (70.4%)	RR 1.04 (0.74 to 1.46)	28 more per 1000 (from 183 fewer to 324 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 200 mg (A) vs fenoprofen 300 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c}	none	19/26 (73.1%)	19/27 (70.4%)	RR 1.04 (0.74 to 1.46)	28 more per 1000 (from 183 fewer to 324 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 12.5 mg (A) vs fenoprofen 25 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c,d}	none	10/24 (41.7%)	11/23 (47.8%)	RR 0.87 (0.46 to 1.65)	62 fewer per 1000 (from 258 fewer to 311 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – fenoprofen 12.5 mg (A) vs fenoprofen 50 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c,d}	none	10/24 (41.7%)	11/23 (47.8%)	RR 0.87 (0.46 to 1.65)	62 fewer per 1000 (from 258 fewer to 311 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 12.5 mg (A) vs fenoprofen 100 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c,d}	none	10/24 (41.7%)	15/23 (65.2%)	RR 0.64 (0.37 to 1.12)	235 fewer per 1000 (from 411 fewer to 78 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 12.5 mg (A) vs fenoprofen 200 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c,d}	none	10/24 (41.7%)	13/23 (56.5%)	RR 0.74 (0.41 to 1.33)	147 fewer per 1000 (from 333 fewer to 187 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 25 mg (A) vs fenoprofen 50 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c,d}	none	11/23 (47.8%)	11/23 (47.8%)	RR 1.00 (0.55 to 1.83)	0 fewer per 1000 (from 215 fewer to 397 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 25 mg (A) vs fenoprofen 100 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c,d}	none	11/23 (47.8%)	15/23 (65.2%)	RR 0.73 (0.44 to 1.23)	176 fewer per 1000 (from 365 fewer to 150 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – fenoprofen 25 mg (A) vs fenoprofen 200 mg (B)												
1	randomized trials	very serious _f	not serious	serious ^e	very serious ^{b,c,d}	none	11/23 (47.8%)	13/23 (56.5%)	RR 0.85 (0.48 to 1.48)	85 fewer per 1000 (from 294 fewer to 271 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (4 hours after administration) – ibuprofen 300–400 mg (A) vs ibuprofen 800 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	4/80 (5.0%)	7/80 (8.8%)	RR 0.57 (0.17 to 1.88)	38 fewer per 1000 (from 73 fewer to 77 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Need for additional analgesia (6 hours after administration) – ibuprofen 300–400 mg (A) vs ibuprofen 900 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c,d}	none	1/20 (5.0%)	0/20 (0.0%)	RR 3.00 (0.13 to 69.52)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – meclofenamate sodium 100 mg (A) vs meclofenamate sodium 200 mg (B)												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	22/96 (22.9%)	24/95 (25.3%)	RR 0.91 (0.55 to 1.50)	23 fewer per 1000 (from 114 fewer to 126 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c,d}	none	1/32 (3.1%)	0/29 (0.0%)	RR 2.73 (0.12 to 64.42)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c,d}	none	1/32 (3.1%)	0/31 (0.0%)	RR 2.91 (0.12 to 68.81)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – flurbiprofen 50 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{c,g}	none	0/29 (0.0%)	0/31 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – diflunisal 125 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/33 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – diflunisal 125 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/33 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – diflunisal 250 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/30 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		
Maternal drug adverse effects (4 hours after administration) – ibuprofen 300–400 mg (A) vs ibuprofen 800 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/80 (0.0%)	0/80 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – ibuprofen 300 mg (A) vs ibuprofen 900 mg (B)												
1	randomized trials	not serious	not serious	serious ^e	very serious ^{b,c,d}	none	3/20 (15.0%)	3/20 (15.0%)	RR 1.00 (0.23 to 4.37)	0 fewer per 1000 (from 115 fewer to 505 more)	⊕○○○ VERY LOW	-
Maternal drug adverse effects (6 hours after administration) – diflunisal 125 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/33 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – diflunisal 125 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/33 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – diflunisal 250 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/30 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – ketoprofen 25 mg (A) vs ketoprofen 50 mg (B)												
1	randomized trials	very serious ^f	not serious	serious ^e	very serious ^{c,g}	none	0/28 (0.0%)	0/26 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aceclofenac 50 mg (A) vs aceclofenac 100 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	1/18 (5.6%)	0/24 (0.0%)	RR 3.95 (0.17 to 91.61)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aceclofenac 50 mg (A) vs aceclofenac 150 mg (B)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	1/18 (5.6%)	0/21 (0.0%)	RR 3.47 (0.15 to 80.35)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Higher single dose of the same NSAID	Relative (95% CI)	Absolute (95% CI)		

Maternal drug adverse effects (6 hours after administration) – aceclofenac 100 mg (A) vs aceclofenac 150 mg (B)

1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,g}	none	0/24 (0.0%)	0/21 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	-------------	-------------	---------------	---	------------------	----------

Maternal drug adverse effects (6 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 100 mg (B)

1	randomized trials	not serious	not serious	serious ^e	very serious ^{c,g}	none	0/32 (0.0%)	0/29 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
---	-------------------	-------------	-------------	----------------------	-----------------------------	------	-------------	-------------	---------------	---	------------------	----------

Maternal drug adverse effects (6 hours after administration) – flurbiprofen 25 mg (A) vs flurbiprofen 100 mg (B)

1	randomized trials	not serious	not serious	serious ^e	very serious ^{c,g}	none	0/32 (0.0%)	0/31 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
---	-------------------	-------------	-------------	----------------------	-----------------------------	------	-------------	-------------	---------------	---	------------------	----------

Maternal drug adverse effects (6 hours after administration) – flurbiprofen 50 mg (A) vs flurbiprofen 100 mg (B)

1	randomized trials	not serious	not serious	serious ^e	very serious ^{c,g}	none	0/29 (0.0%)	0/31 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
---	-------------------	-------------	-------------	----------------------	-----------------------------	------	-------------	-------------	---------------	---	------------------	----------

CI: confidence interval; RR: risk ratio.

a. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

b. Wide confidence interval crossing the line of no effect.

c. Less than 300 participants.

d. Few events.

e. Some studies included in this outcome excluded breastfeeding women – the evidence cannot be extrapolated to all women during the postpartum period.

f. Most of the pooled effect provided by studies “B” or “C” but with a substantial proportion (i.e. > 50%) from studies “C”.

g. No events.

Comparison 3: Single-dose oral analgesic compared with a single dose of an alternative oral analgesic

Comparison 3a: Single-dose NSAID compared with single-dose paracetamol

Source: Wuytack F, Smith V, Cleary BJ. Oral non-steroidal anti-inflammatory drugs (single-dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2021;(1):CD011352.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Paracetamol	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration)												
3	randomized trials	serious ^d	not serious	not serious	not serious	none	54/171 (31.6%)	35/171 (20.5%)	RR 1.54 (1.07 to 2.22)	111 more per 1000 (from 14 more to 250 more)	⊕⊕⊕○ MODERATE	CRITICAL
Adequate pain relief (4 hours after administration) – ibuprofen 300–400 mg vs paracetamol 1000 mg												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b,c}	none	18/36 (50.0%)	11/37 (29.7%)	RR 1.68 (0.93 to 3.04)	202 more per 1000 (from 21 fewer to 606 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (4 hours after administration) – ibuprofen 300–400 mg vs paracetamol 500 mg												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	30/106 (28.3%)	21/104 (20.2%)	RR 1.40 (0.86 to 2.28)	81 more per 1000 (from 28 fewer to 258 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aceclofenac 100 mg vs paracetamol 650 mg												
1	randomized trials	serious ^d	not serious	not serious	very serious ^{a,b,c}	none	6/29 (20.7%)	3/30 (10.0%)	RR 2.07 (0.57 to 7.50)	107 more per 1000 (from 43 fewer to 650 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aceclofenac 100 mg vs paracetamol 650 mg												
2	randomized trials	serious ^d	not serious	not serious	very serious ^{a,b,c}	none	18/49 (36.7%)	10/50 (20.0%)	RR 1.82 (0.61 to 5.47)	164 more per 1000 (from 78 fewer to 894 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (4 hours after administration) – ibuprofen 300–400 mg vs paracetamol 1000 mg												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b,c}	none	8/36 (22.2%)	15/37 (40.5%)	RR 0.55 (0.27 to 1.13)	182 fewer per 1000 (from 296 fewer to 53 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID (single administration, any dose)	Paracetamol	Relative (95% CI)	Absolute (95% CI)		

Need for additional analgesia (6 hours after administration) – ibuprofen 300–400 mg vs paracetamol 1000 mg

1	randomized trials	serious ^d	not serious	not serious	serious ^{b,c}	none	5/31 (16.1%)	16/28 (57.1%)	RR 0.28 (0.12 to 0.67)	411 fewer per 1000 (from 503 fewer to 189 fewer)	⊕⊕○○ LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	------------------------	------	--------------	---------------	----------------------------------	--	-------------	----------

Maternal drug adverse effects (4 hours after administration) – ibuprofen 300–400 mg vs paracetamol 500 mg

1	randomized trials	serious ^d	not serious	not serious	very serious ^{b,e}	none	0/106 (0.0%)	0/104 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	--------------	--------------	---------------	---	------------------	----------

Maternal drug adverse effects (6 hours after administration)

3	randomized trials	not serious	not serious	not serious	very serious ^{a,c}	none	6/150 (4.0%)	8/150 (5.3%)	RR 0.74 (0.27 to 2.08)	14 fewer per 1000 (from 39 fewer to 58 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	-------------	-------------	-------------	-----------------------------	------	--------------	--------------	----------------------------------	--	-------------	----------

Maternal drug adverse effects (6 hours after administration) – dipyrone 500 mg vs paracetamol 500 mg

1	randomized trials	not serious	not serious	not serious	very serious ^{a,b,c}	none	5/101 (5.0%)	7/100 (7.0%)	RR 0.71 (0.23 to 2.15)	20 fewer per 1000 (from 54 fewer to 81 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	-------------	-------------	-------------	-------------------------------	------	--------------	--------------	----------------------------------	--	-------------	----------

Maternal drug adverse effects (6 hours after administration) – aceclofenac 100 mg vs paracetamol 650 mg

2	randomized trials	serious ^d	not serious	not serious	very serious ^{a,b,c}	none	1/49 (2.0%)	1/50 (2.0%)	RR 1.00 (0.07 to 14.90)	0 fewer per 1000 (from 19 fewer to 278 more)	⊕○○○ VERY LOW	-
---	-------------------	----------------------	-------------	-------------	-------------------------------	------	-------------	-------------	-----------------------------------	--	------------------	---

CI: confidence interval; RR: risk ratio.

a. Wide confidence interval crossing the line of no effect.

b. Less than 300 participants.

c. Few events.

d. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

e. No events.

Comparison 3b: Single-dose NSAID (aspirin) compared with a single dose of another NSAID

Source: Wuytack F, Smith V, Cleary BJ. Oral non-steroidal anti-inflammatory drugs (single-dose) for perineal pain in the early postpartum period. Cochrane Database Syst Rev. 2021;(1):CD011352.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID single dose	Alternative NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration)												
4	randomized trials	not serious	not serious	serious ^a	not serious	none	187/363 (51.5%)	200/368 (54.3%)	RR 0.95 (0.83 to 1.09)	27 fewer per 1000 (from 92 fewer to 49 more)	⊕⊕⊕○ MODERATE	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 125 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{c,d,e}	none	17/32 (53.1%)	12/33 (36.4%)	RR 1.46 (0.84 to 2.55)	167 more per 1000 (from 58 fewer to 564 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{c,d,e}	none	17/32 (53.1%)	9/30 (30.0%)	RR 1.77 (0.94 to 3.35)	231 more per 1000 (from 18 fewer to 705 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{c,d,e}	none	17/32 (53.1%)	14/30 (46.7%)	RR 1.14 (0.69 to 1.88)	65 more per 1000 (from 145 fewer to 411 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs ibuprofen 300–400 mg (B)												
1	randomized trials	serious ^b	not serious	serious ^a	very serious ^{c,d}	none	16/30 (53.3%)	21/30 (70.0%)	RR 0.76 (0.51 to 1.15)	168 fewer per 1000 (from 343 fewer to 105 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs diclofenac 25 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	27/50 (54.0%)	32/52 (61.5%)	RR 0.88 (0.63 to 1.23)	74 fewer per 1000 (from 228 fewer to 142 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID single dose	Alternative NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs diclofenac 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	27/50 (54.0%)	34/50 (68.0%)	RR 0.79 (0.58 to 1.09)	143 fewer per 1000 (from 286 fewer to 61 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs diclofenac 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	27/50 (54.0%)	37/51 (72.5%)	RR 0.74 (0.55 to 1.01)	189 fewer per 1000 (from 326 fewer to 7 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 25 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	13/29 (44.8%)	11/32 (34.4%)	RR 1.30 (0.70 to 2.44)	103 more per 1000 (from 103 fewer to 495 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	13/29 (44.8%)	15/29 (51.7%)	RR 0.87 (0.51 to 1.48)	67 fewer per 1000 (from 253 fewer to 248 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (4 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	13/29 (44.8%)	15/31 (48.4%)	RR 0.93 (0.54 to 1.60)	34 fewer per 1000 (from 223 fewer to 290 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 900 mg (A) vs ibuprofen 300–400 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	18/20 (90.0%)	17/20 (85.0%)	RR 1.06 (0.84 to 1.34)	51 more per 1000 (from 136 fewer to 289 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 900 mg (A) vs ibuprofen 900 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	18/20 (90.0%)	17/20 (85.0%)	RR 1.06 (0.84 to 1.34)	51 more per 1000 (from 136 fewer to 289 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID single dose	Alternative NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 125 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{c,d}	none	18/32 (56.3%)	13/33 (39.4%)	RR 1.43 (0.85 to 2.41)	169 more per 1000 (from 59 fewer to 555 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{c,d,e}	none	18/32 (56.3%)	10/30 (33.3%)	RR 1.69 (0.93 to 3.05)	230 more per 1000 (from 23 fewer to 683 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{c,d}	none	18/32 (56.3%)	16/30 (53.3%)	RR 1.05 (0.67 to 1.66)	27 more per 1000 (from 176 fewer to 352 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs etodolac 25 mg (B)												
1	randomized trials	serious ^b	not serious	serious ^a	very serious ^{c,d,e}	none	16/39 (41.0%)	11/40 (27.5%)	RR 1.49 (0.80 to 2.80)	135 more per 1000 (from 55 fewer to 495 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs etodolac 100 mg (B)												
1	randomized trials	serious ^b	not serious	serious ^a	very serious ^{c,d}	none	16/39 (41.0%)	15/40 (37.5%)	RR 1.09 (0.63 to 1.89)	34 more per 1000 (from 139 fewer to 334 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 25 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	14/29 (48.3%)	13/32 (40.6%)	RR 1.19 (0.68 to 2.09)	77 more per 1000 (from 130 fewer to 443 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	14/29 (48.3%)	18/29 (62.1%)	RR 0.78 (0.49 to 1.25)	137 fewer per 1000 (from 317 fewer to 155 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID single dose	Alternative NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d}	none	14/29 (48.3%)	19/31 (61.3%)	RR 0.79 (0.49 to 1.26)	129 fewer per 1000 (from 313 fewer to 159 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief (6 hours after administration) – aspirin 500–650 mg (A) vs dipyron 500 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	serious ^d	none	61/90 (67.8%)	67/89 (75.3%)	RR 0.90 (0.75 to 1.08)	75 fewer per 1000 (from 188 fewer to 60 more)	⊕⊕○○ LOW	CRITICAL
Need for additional analgesia (4 hours after administration) – aspirin 500–650 mg (A) vs ibuprofen 300–400 mg (B)												
1	randomized trials	serious ^b	not serious	serious ^a	very serious ^{c,d,e}	none	5/30 (16.7%)	0/30 (0.0%)	RR 11.00 (0.64 to 190.53)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – aspirin 900 mg (A) vs ibuprofen 300–400 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	1/20 (5.0%)	0/20 (0.0%)	RR 3.00 (0.13 to 69.52)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – aspirin 900 mg (A) vs ibuprofen 900 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{d,f}	none	0/20 (0.0%)	0/20 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 25 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	1/29 (3.4%)	1/32 (3.1%)	RR 1.10 (0.07 to 16.85)	3 more per 1000 (from 29 fewer to 495 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional analgesia (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	1/29 (3.4%)	0/29 (0.0%)	RR 3.00 (0.13 to 70.74)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID single dose	Alternative NSAID	Relative (95% CI)	Absolute (95% CI)		
Need for additional analgesia (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	1/29 (3.4%)	0/31 (0.0%)	RR 3.20 (0.14 to 75.55)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – aspirin 600 mg (A) vs diflunisal 125 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{d,f}	none	0/32 (0.0%)	0/33 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – aspirin 600 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{d,f}	none	0/32 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – aspirin 600 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{d,f}	none	0/32 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (4 hours after administration) – aspirin 600 mg (A) vs ibuprofen 400 mg (B)												
1	randomized trials	serious ^b	not serious	serious ^b	very serious ^{d,f}	none	0/30 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 900 mg (A) vs ibuprofen 300–400 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	5/20 (25.0%)	3/20 (15.0%)	RR 1.67 (0.46 to 6.06)	100 more per 1000 (from 81 fewer to 759 more)	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 900 mg (A) vs ibuprofen 900 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{c,d,e}	none	5/20 (25.0%)	3/20 (15.0%)	RR 1.67 (0.46 to 6.06)	100 more per 1000 (from 81 fewer to 759 more)	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 500–650 mg (A) vs dipyrrone 500 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{d,f}	none	0/89 (0.0%)	0/89 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID single dose	Alternative NSAID	Relative (95% CI)	Absolute (95% CI)		
Maternal drug adverse effects (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 25 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{d,f}	none	0/29 (0.0%)	0/32 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 50 mg (B)												
1	randomized trials	not serious	not serious	serious ^b	very serious ^{d,f}	none	0/29 (0.0%)	0/29 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 500–650 mg (A) vs flurbiprofen 100 mg (B)												
1	randomized trials	not serious	not serious	serious ^a	very serious ^{d,f}	none	0/29 (0.0%)	0/31 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 125 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{d,f}	none	0/32 (0.0%)	0/33 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 250 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{d,f}	none	0/32 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Maternal drug adverse effects (6 hours after administration) – aspirin 500–650 mg (A) vs diflunisal 500 mg (B)												
1	randomized trials	serious ^b	not serious	not serious	very serious ^{d,f}	none	0/32 (0.0%)	0/30 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. Some studies included in this outcome excluded breastfeeding women – the evidence cannot be extrapolated to all women during the postpartum period.

b. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 participants.

e. Few events.

f. No events.

EB table A.2.3a: Pharmacological relief of pain due to uterine cramping/involution (pharmacological interventions compared with placebo)

Comparison 1: Paracetamol (oral, single-dose) compared with placebo

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Paracetamol	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – paracetamol 650 mg vs placebo												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	15/22 (68.2%)	14/26 (53.8%)	RR 1.27 (0.80 to 2.00)	145 more per 1000 (from 108 fewer to 538 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – paracetamol 1000 mg vs placebo												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,d}	none	4/39 (10.3%)	5/36 (13.9%)	RR 0.74 (0.21 to 2.54)	36 fewer per 1000 (from 110 fewer to 214 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – paracetamol 650 mg vs placebo												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	10/22 (45.5%)	5/26 (19.2%)	RR 2.36 (0.95 to 5.88)	262 more per 1000 (from 10 fewer to 938 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – paracetamol 1000 mg vs placebo												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	2/39 (5.1%)	1/36 (2.8%)	RR 1.85 (0.17 to 19.50)	24 more per 1000 (from 23 fewer to 514 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women and less than 30 events.

Comparison 2: NSAIDs compared with placebo

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAIDs	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
11	randomized trials	serious ^a	not serious	serious ^b	not serious	none	466/649 (71.8%)	131/297 (44.1%)	RR 1.66 (1.45 to 1.91)	291 more per 1000 (from 198 more to 401 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief as reported by the woman – aspirin 650 mg												
6	randomized trials	serious ^a	not serious	serious ^b	not serious	none	123/168 (73.2%)	60/114 (52.6%)	RR 1.33 (1.09 to 1.61)	174 more per 1000 (from 47 more to 321 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief as reported by the woman – naproxen 275 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	27/30 (90.0%)	9/15 (60.0%)	RR 1.50 (0.98 to 2.31)	300 more per 1000 (from 12 fewer to 786 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – naproxen 300 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	27/35 (77.1%)	9/17 (52.9%)	RR 1.46 (0.90 to 2.36)	244 more per 1000 (from 53 fewer to 720 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – naproxen 550 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^f	none	23/30 (76.7%)	9/30 (30.0%)	RR 2.56 (1.43 to 4.57)	468 more per 1000 (from 129 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – naproxen 600 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	30/35 (85.7%)	10/18 (55.6%)	RR 1.54 (1.00 to 2.38)	300 more per 1000 (from 0 fewer to 767 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAIDs	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – flurbiprofen 50 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	26/30 (86.7%)	10/16 (62.5%)	RR 1.39 (0.93 to 2.08)	244 more per 1000 (from 44 fewer to 675 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – ketorolac 5 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	26/30 (86.7%)	5/10 (50.0%)	RR 1.73 (0.92 to 3.27)	365 more per 1000 (from 40 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – ketorolac 10 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	25/30 (83.3%)	5/10 (50.0%)	RR 1.67 (0.88 to 3.16)	335 more per 1000 (from 60 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 12.5 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	12/27 (44.4%)	1/5 (20.0%)	RR 2.22 (0.37 to 13.48)	244 more per 1000 (from 126 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 25 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	15/27 (55.6%)	1/5 (20.0%)	RR 2.78 (0.47 to 16.56)	356 more per 1000 (from 106 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 50 mg												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	33/54 (61.1%)	2/12 (16.7%)	RR 3.72 (1.03 to 13.39)	453 more per 1000 (from 5 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAIDs	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – fenoprofen 100 mg												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	37/56 (66.1%)	3/13 (23.1%)	RR 2.86 (1.04 to 7.89)	429 more per 1000 (from 9 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 200 mg												
3	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	42/68 (61.8%)	5/25 (20.0%)	RR 2.67 (1.15 to 6.23)	334 more per 1000 (from 30 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 300 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	20/29 (69.0%)	2/7 (28.6%)	RR 2.41 (0.73 to 7.99)	403 more per 1000 (from 77 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief												
4	randomized trials	serious ^a	not serious	serious ^b	serious ^g	none	5/250 (2.0%)	24/125 (19.2%)	RR 0.15 (0.07 to 0.33)	163 fewer per 1000 (from 179 fewer to 129 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – aspirin 650 mg												
2	randomized trials	serious ^a	not serious	serious ^b	serious ^f	none	1/60 (1.7%)	5/25 (20.0%)	RR 0.11 (0.02 to 0.63)	178 fewer per 1000 (from 196 fewer to 74 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – ketorolac 5 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	1/30 (3.3%)	2/10 (20.0%)	RR 0.17 (0.02 to 1.65)	166 fewer per 1000 (from 196 fewer to 130 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAIDs	Placebo	Relative (95% CI)	Absolute (95% CI)		
Need for additional pain relief – ketorolac 10 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	1/30 (3.3%)	2/10 (20.0%)	RR 0.17 (0.02 to 1.65)	166 fewer per 1000 (from 196 fewer to 130 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – naproxen 275 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	0/30 (0.0%)	2/15 (13.3%)	RR 0.10 (0.01 to 2.02)	120 fewer per 1000 (from 132 fewer to 136 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – naproxen 300 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	0/35 (0.0%)	2/17 (11.8%)	RR 0.10 (0.01 to 1.98)	106 fewer per 1000 (from 116 fewer to 115 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – naproxen 600 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	0/35 (0.0%)	2/18 (11.1%)	RR 0.11 (0.01 to 2.09)	99 fewer per 1000 (from 110 fewer to 121 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – naproxen 550 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^f	none	2/30 (6.7%)	9/30 (30.0%)	RR 0.22 (0.05 to 0.94)	234 fewer per 1000 (from 285 fewer to 18 fewer)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects												
8	randomized trials	serious ^a	not serious	serious ^b	serious ^c	none	92/372 (24.7%)	52/211 (24.6%)	RR 1.05 (0.78 to 1.41)	12 more per 1000 (from 54 fewer to 101 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAIDs	Placebo	Relative (95% CI)	Absolute (95% CI)		
Maternal adverse effects – aspirin 650 mg												
5	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	36/140 (25.7%)	24/83 (28.9%)	RR 0.93 (0.58 to 1.47)	20 fewer per 1000 (from 121 fewer to 136 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – flurbiprofen 50 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	6/30 (20.0%)	3/16 (18.8%)	RR 1.07 (0.31 to 3.71)	13 more per 1000 (from 129 fewer to 508 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – naproxen 275 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	5/30 (16.7%)	2/15 (13.3%)	RR 1.25 (0.27 to 5.70)	33 more per 1000 (from 97 fewer to 627 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – naproxen 300 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	9/35 (25.7%)	5/17 (29.4%)	RR 0.87 (0.35 to 2.21)	38 fewer per 1000 (from 191 fewer to 356 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – naproxen 550 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	8/30 (26.7%)	4/30 (13.3%)	RR 2.00 (0.67 to 5.94)	133 more per 1000 (from 44 fewer to 659 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – naproxen 600 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	9/35 (25.7%)	5/18 (27.8%)	RR 0.93 (0.36 to 2.36)	19 fewer per 1000 (from 178 fewer to 378 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAIDs	Placebo	Relative (95% CI)	Absolute (95% CI)		

Maternal adverse effects – ketorolac 5 mg

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	6/30 (20.0%)	2/10 (20.0%)	RR 1.00 (0.24 to 4.18)	0 fewer per 1000 (from 152 fewer to 636 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	--------------	--------------	----------------------------------	---	------------------	----------

Maternal adverse effects – ketorolac 10 mg

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	7/30 (23.3%)	2/10 (20.0%)	RR 1.17 (0.29 to 4.73)	34 more per 1000 (from 142 fewer to 746 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	--------------	--------------	----------------------------------	---	------------------	----------

Maternal adverse effects – fenoprofen 200 mg

1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,f}	none	6/12 (50.0%)	5/12 (41.7%)	RR 1.20 (0.50 to 2.88)	83 more per 1000 (from 208 fewer to 783 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	--------------	--------------	----------------------------------	---	------------------	----------

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women.

e. Wide confidence interval including the line of no effect.

f. Less than 300 women and 30 events.

g. Less than 30 events.

Comparison 3: Opioids compared with placebo

Source: Deussen AR, Ashwood P, Martis R, Stewart F, LE G. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Opioids	Placebo	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
5	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	89/165 (53.9%)	53/134 (39.6%)	RR 1.26 (0.99 to 1.61)	103 more per 1000 (from 4 fewer to 241 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – codeine 60 mg vs placebo												
5	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	69/134 (51.5%)	43/118 (36.4%)	RR 1.33 (1.01 to 1.76)	120 more per 1000 (from 4 more to 277 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – codeine 120 mg vs placebo												
1	randomized trials	serious ^e	not serious	serious ^b	very serious ^{c,d}	none	20/31 (64.5%)	10/16 (62.5%)	RR 1.03 (0.65 to 1.64)	19 more per 1000 (from 219 fewer to 400 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief												
3	randomized trials	serious ^a	not serious	serious ^b	serious ^d	none	19/170 (11.2%)	23/103 (22.3%)	RR 0.48 (0.28 to 0.82)	116 fewer per 1000 (from 161 fewer to 40 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – codeine 60 mg vs placebo												
3	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,f}	none	10/104 (9.6%)	13/69 (18.8%)	RR 0.49 (0.24 to 1.02)	96 fewer per 1000 (from 143 fewer to 4 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – codeine 120 mg vs placebo												
1	randomized trials	serious ^e	not serious	serious ^b	very serious ^f	none	1/31 (3.2%)	3/16 (18.8%)	RR 0.17 (0.02 to 1.52)	156 fewer per 1000 (from 184 fewer to 98 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Opioids	Placebo	Relative (95% CI)	Absolute (95% CI)		

Need for additional pain relief – nalbuphine 15 mg vs placebo

1	randomized trials	serious ^e	not serious	serious ^b	very serious ^{c,f}	none	8/35 (22.9%)	7/18 (38.9%)	RR 0.59 (0.25 to 1.36)	159 fewer per 1000 (from 292 fewer to 140 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	--------------	--------------	----------------------------------	---	------------------	----------

Maternal adverse effects

3	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	48/109 (44.0%)	21/79 (26.6%)	RR 1.59 (0.99 to 2.55)	157 more per 1000 (from 3 fewer to 412 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	----------------	---------------	----------------------------------	--	------------------	----------

Maternal adverse effects – codeine 60 mg vs placebo

3	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	21/78 (26.9%)	18/63 (28.6%)	RR 0.95 (0.54 to 1.67)	14 fewer per 1000 (from 131 fewer to 191 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	---------------	---------------	----------------------------------	--	------------------	----------

Maternal adverse effects – codeine 120 mg vs placebo

1	randomized trials	serious ^e	not serious	serious ^b	very serious ^d	none	27/31 (87.1%)	3/16 (18.8%)	RR 4.65 (1.66 to 13.00)	684 more per 1000 (from 124 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	---------------------------	------	---------------	--------------	-----------------------------------	--	------------------	----------

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by studies “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women.

e. The pooled effect provided by study “B”.

f. Less than 300 women and less than 30 events.

EB table A.2.3b: Pharmacological relief of pain due to uterine cramping/involution (pharmacological interventions compared with other pharmacological interventions)

Comparison 1: Lower dose of an oral analgesic compared with a higher dose of the same analgesic

Comparison 1a: Naproxen (lower dose compared with a higher dose)

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Naproxen lower dose	Naproxen higher dose	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – naproxen 300 mg vs naproxen 600 mg												
1	randomized trials	serious ^a	not serious	serious ^c	very serious ^{b,d}	none	27/35 (77.1%)	30/35 (85.7%)	RR 0.90 (0.72 to 1.13)	86 fewer per 1000 (from 240 fewer to 111 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – naproxen 300 mg vs naproxen 600 mg												
1	randomized trials	serious ^a	not serious	serious ^c	very serious ^{d,e}	none	9/35 (25.7%)	9/35 (25.7%)	RR 1.00 (0.45 to 2.22)	0 fewer per 1000 (from 141 fewer to 314 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Less than 300 women.

c. Exclusion: breastfeeding women.

d. Wide confidence interval crossing the line of no effect.

e. Less than 300 women and 30 events.

Comparison 1b: Ketorolac (lower dose compared with a higher dose)

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Ketorolac lower dose	Ketorolac higher dose	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – ketorolac 5 mg vs ketorolac 10 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^c	none	26/30 (86.7%)	29/30 (96.7%)	RR 0.90 (0.77 to 1.05)	97 fewer per 1000 (from 222 fewer to 48 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – ketorolac 5 mg vs ketorolac 10 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	6/30 (20.0%)	7/30 (23.3%)	RR 0.86 (0.33 to 2.25)	33 fewer per 1000 (from 156 fewer to 292 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – ketorolac 5 mg vs ketorolac 10 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	6/30 (20.0%)	7/30 (23.3%)	RR 0.86 (0.33 to 2.25)	33 fewer per 1000 (from 156 fewer to 292 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Less than 300 women.

d. Wide confidence interval crossing the line of no effect.

e. Less than 300 women and less than 30 events.

Comparison 1c: Codeine (lower dose compared with a higher dose)

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Codeine lower dose	Codeine higher dose	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – codeine 60 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	22/32 (68.8%)	20/31 (64.5%)	RR 1.07 (0.75 to 1.51)	45 more per 1000 (from 161 fewer to 329 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – codeine 60 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	1/32 (3.1%)	1/31 (3.2%)	RR 0.97 (0.06 to 14.82)	1 fewer per 1000 (from 30 fewer to 446 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – codeine 60 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^e	none	10/32 (31.3%)	27/31 (87.1%)	RR 0.36 (0.21 to 0.61)	557 fewer per 1000 (from 688 fewer to 340 fewer)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. Pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women.

e. Less than 300 women and 30 events.

Comparison 2: An oral analgesic compared with an alternative oral analgesic of the same class

Comparison 2a: Aspirin compared with naproxen

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Aspirin	Naproxen	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – aspirin 650 mg vs naproxen 275 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^c	none	28/30 (93.3%)	27/30 (90.0%)	RR 1.04 (0.89 to 1.21)	36 more per 1000 (from 99 fewer to 189 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – aspirin 650 mg vs naproxen 275 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	4/30 (13.3%)	5/30 (16.7%)	RR 0.80 (0.24 to 2.69)	33 fewer per 1000 (from 127 fewer to 282 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Less than 300 women.

d. Wide confidence interval crossing the line of no effect.

e. Less than 300 women and less than 30 events.

Comparison 2b: Aspirin compared with flurbiprofen

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Aspirin	Flurbiprofen	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – aspirin 650 mg vs flurbiprofen 50 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	24/34 (70.6%)	26/30 (86.7%)	RR 0.81 (0.63 to 1.05)	165 fewer per 1000 (from 321 fewer to 43 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – aspirin 650 mg vs flurbiprofen 50 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	2/34 (5.9%)	0/30 (0.0%)	RR 4.43 (0.22 to 88.74)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – aspirin 650 mg vs flurbiprofen 50 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	8/34 (23.5%)	6/30 (20.0%)	RR 1.18 (0.46 to 3.01)	36 more per 1000 (from 108 fewer to 402 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women.

e. Less than 300 women and less than 30 events.

Comparison 2c: Aspirin compared with ketorolac

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Aspirin	Ketorolac	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^c	none	26/30 (86.7%)	55/60 (91.7%)	RR 0.95 (0.81 to 1.11)	46 fewer per 1000 (from 174 fewer to 101 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – aspirin 650 mg vs ketorolac 5 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	13/15 (86.7%)	26/30 (86.7%)	RR 1.00 (0.78 to 1.28)	0 fewer per 1000 (from 191 fewer to 243 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – aspirin 650 mg vs ketorolac 10 mg												
1	randomized trials	serious ^a	not serious	serious ^b	serious ^c	none	13/15 (86.7%)	29/30 (96.7%)	RR 0.90 (0.73 to 1.11)	97 fewer per 1000 (from 261 fewer to 106 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	1/30 (3.3%)	2/60 (3.3%)	RR 1.18 (0.16 to 8.52)	6 more per 1000 (from 28 fewer to 251 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – aspirin 650 mg vs ketorolac 5 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	1/15 (6.7%)	1/30 (3.3%)	RR 2.00 (0.13 to 29.81)	33 more per 1000 (from 29 fewer to 960 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – aspirin 650 mg vs ketorolac 10 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	0/15 (0.0%)	1/30 (3.3%)	RR 0.65 (0.03 to 14.97)	12 fewer per 1000 (from 32 fewer to 466 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Aspirin	Ketorolac	Relative (95% CI)	Absolute (95% CI)		

Maternal adverse effects

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	11/30 (36.7%)	13/60 (21.7%)	RR 1.69 (0.86 to 3.31)	150 more per 1000 (from 30 fewer to 501 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	---------------	---------------	----------------------------------	---	------------------	----------

Maternal adverse effects – aspirin 650 mg vs ketorolac 5 mg

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	5/15 (33.3%)	6/30 (20.0%)	RR 1.67 (0.61 to 4.59)	134 more per 1000 (from 78 fewer to 718 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	--------------	--------------	----------------------------------	---	------------------	----------

Maternal adverse effects – aspirin 650 mg vs ketorolac 10 mg

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{d,e}	none	6/15 (40.0%)	7/30 (23.3%)	RR 1.71 (0.70 to 4.20)	166 more per 1000 (from 70 fewer to 747 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	--------------	--------------	----------------------------------	---	------------------	----------

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Less than 300 women.

d. Wide confidence interval crossing the line of no effect.

e. Less than 300 women and less than 30 events.

Comparison 2d: Codeine compared with nalbuphine

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Codeine	Nalbuphine	Relative (95% CI)	Absolute (95% CI)		
Need for additional pain relief – codeine 60 mg vs nalbuphine 15 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	5/37 (13.5%)	8/35 (22.9%)	RR 0.59 (0.21 to 1.64)	94 fewer per 1000 (from 181 fewer to 146 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women and less than 30 events.

Comparison 3: An oral analgesic compared with an alternative oral analgesic from a different class

Comparison 3a: Paracetamol compared with NSAIDs

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Paracetamol	NSAID	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – paracetamol 650 mg vs aspirin 650 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	15/22 (68.2%)	20/26 (76.9%)	RR 0.89 (0.62 to 1.26)	85 fewer per 1000 (from 292 fewer to 200 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects												
2	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	12/58 (20.7%)	13/54 (24.1%)	RR 0.89 (0.29 to 2.78)	26 fewer per 1000 (from 171 fewer to 429 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – paracetamol 650 mg vs aspirin 650 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	10/22 (45.5%)	9/26 (34.6%)	RR 1.31 (0.65 to 2.64)	107 more per 1000 (from 121 fewer to 568 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – paracetamol 1000 mg vs naproxen 500 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	2/36 (5.6%)	4/28 (14.3%)	RR 0.39 (0.08 to 1.97)	87 fewer per 1000 (from 131 fewer to 139 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women.

e. Less than 300 women and less than 30 events.

Comparison 3b: NSAIDs compared with opioids

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Opioid	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
5	randomized trials	serious ^a	not serious	serious ^b	not serious	none	266/395 (67.3%)	89/165 (53.9%)	RR 1.33 (1.13 to 1.57)	178 more per 1000 (from 70 more to 307 more)	⊕⊕○○ LOW	CRITICAL
Adequate pain relief as reported by the woman – aspirin 650 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	12/17 (70.6%)	11/16 (68.8%)	RR 1.03 (0.65 to 1.61)	21 more per 1000 (from 241 fewer to 419 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – aspirin 650 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,d}	none	12/17 (70.6%)	10/16 (62.5%)	RR 1.13 (0.69 to 1.84)	81 more per 1000 (from 194 fewer to 525 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 12.5 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	12/27 (44.4%)	2/5 (40.0%)	RR 1.11 (0.35 to 3.52)	44 more per 1000 (from 260 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 25 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	15/27 (55.6%)	3/5 (60.0%)	RR 0.93 (0.42 to 2.04)	42 fewer per 1000 (from 348 fewer to 624 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 50 mg vs codeine 60 mg												
2	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	33/54 (61.1%)	6/12 (50.0%)	RR 1.24 (0.68 to 2.27)	120 more per 1000 (from 160 fewer to 635 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Opioid	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman – fenoprofen 100 mg vs codeine 60 mg												
2	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	37/56 (66.1%)	6/13 (46.2%)	RR 1.44 (0.77 to 2.66)	203 more per 1000 (from 106 fewer to 766 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 200 mg vs codeine 60 mg												
3	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	42/68 (61.8%)	9/24 (37.5%)	RR 1.42 (0.81 to 2.47)	157 more per 1000 (from 71 fewer to 551 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – fenoprofen 300 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	20/29 (69.0%)	3/8 (37.5%)	RR 1.84 (0.73 to 4.65)	315 more per 1000 (from 101 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – flurbiprofen 50 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	13/15 (86.7%)	11/16 (68.8%)	RR 1.26 (0.86 to 1.85)	179 more per 1000 (from 96 fewer to 584 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – flurbiprofen 50 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	13/15 (86.7%)	10/15 (66.7%)	RR 1.30 (0.86 to 1.96)	200 more per 1000 (from 93 fewer to 640 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – naproxen 300 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	27/35 (77.1%)	9/17 (52.9%)	RR 1.46 (0.90 to 2.36)	244 more per 1000 (from 53 fewer to 720 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – naproxen 600 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,e}	none	30/35 (85.7%)	9/18 (50.0%)	RR 1.71 (1.06 to 2.77)	355 more per 1000 (from 30 more to 885 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Opioid	Relative (95% CI)	Absolute (95% CI)		
Need for additional pain relief												
2	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	2/134 (1.5%)	6/98 (6.1%)	RR 0.37 (0.12 to 1.12)	39 fewer per 1000 (from 54 fewer to 7 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – aspirin 650 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	1/17 (5.9%)	1/16 (6.3%)	RR 0.94 (0.06 to 13.82)	4 fewer per 1000 (from 59 fewer to 801 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – aspirin 650 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	1/17 (5.9%)	0/15 (0.0%)	RR 2.67 (0.12 to 60.93)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – flurbiprofen 50 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^f	none	0/15 (0.0%)	0/16 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – flurbiprofen 50 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	0/15 (0.0%)	1/16 (6.3%)	RR 0.35 (0.02 to 8.08)	41 fewer per 1000 (from 61 fewer to 443 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – naproxen 300 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	0/35 (0.0%)	2/17 (11.8%)	RR 0.10 (0.01 to 1.98)	106 fewer per 1000 (from 116 fewer to 115 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – naproxen 600 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	0/35 (0.0%)	2/18 (11.1%)	RR 0.11 (0.01 to 2.09)	99 fewer per 1000 (from 110 fewer to 121 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Opioid	Relative (95% CI)	Absolute (95% CI)		
Maternal adverse effects												
3	randomized trials	serious ^a	not serious	serious ^b	serious ^e	none	38/146 (26.0%)	48/109 (44.0%)	RR 0.62 (0.43 to 0.89)	167 fewer per 1000 (from 251 fewer to 48 fewer)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – aspirin 650 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	4/17 (23.5%)	5/16 (31.3%)	RR 0.75 (0.24 to 2.32)	78 fewer per 1000 (from 238 fewer to 413 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – aspirin 650 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	4/17 (23.5%)	14/16 (87.5%)	RR 0.27 (0.11 to 0.65)	639 fewer per 1000 (from 779 fewer to 306 fewer)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – fenopufen 200 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,d}	none	6/12 (50.0%)	3/11 (27.3%)	RR 1.83 (0.60 to 5.61)	226 more per 1000 (from 109 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – flurbipufen 50 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	3/15 (20.0%)	5/16 (31.3%)	RR 0.64 (0.18 to 2.22)	112 fewer per 1000 (from 256 fewer to 381 more)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – flurbipufen 50 mg vs codeine 120 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	3/15 (20.0%)	13/15 (86.7%)	RR 0.23 (0.08 to 0.65)	667 fewer per 1000 (from 797 fewer to 303 fewer)	⊕○○○ VERY LOW	CRITICAL
Maternal adverse effects – naproxen 300 mg vs codeine 60 mg												
1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	9/35 (25.7%)	4/17 (23.5%)	RR 1.09 (0.39 to 3.05)	21 more per 1000 (from 144 fewer to 482 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Opioid	Relative (95% CI)	Absolute (95% CI)		

Maternal adverse effects – naproxen 600 mg vs codeine 60 mg

1	randomized trials	serious ^a	not serious	serious ^b	very serious ^{c,d}	none	9/35 (25.7%)	4/18 (22.2%)	RR 1.16 (0.41 to 3.25)	36 more per 1000 (from 131 fewer to 500 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	----------------------	-----------------------------	------	--------------	--------------	----------------------------------	---	------------------	----------

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by studies “B”.

b. Exclusion: breastfeeding women.

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women and less than 30 events.

e. Less than 300 women.

f. Less than 300 women and no events.

Comparison 3c: NSAIDs compared with herbal analgesia

Source: Deussen AR, Ashwood P, Martis R, Stewart F, Grzeskowiak LE. Relief of pain due to uterine cramping/involution after birth. Cochrane Database Syst Rev. 2020;(10):CD004908.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Herbal analgesia	Relative (95% CI)	Absolute (95% CI)		
Adequate pain relief as reported by the woman												
4	randomized trials	serious ^a	not serious	not serious	not serious	none	87/197 (44.2%)	91/197 (46.2%)	RR 0.96 (0.78 to 1.18)	18 fewer per 1000 (from 102 fewer to 83 more)	⊕⊕⊕○ MODERATE	CRITICAL
Adequate pain relief as reported by the woman – mefenamic acid 250 mg vs pimpinella anisum, apium graveolens and crocus sativus 500 mg												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	30/54 (55.6%)	31/54 (57.4%)	RR 0.97 (0.69 to 1.35)	17 fewer per 1000 (from 178 fewer to 201 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – mefenamic acid 250 mg vs melissa officinalis 395 mg												
1	randomized trials	very serious ^d	not serious	not serious	very serious ^{b,e}	none	11/55 (20.0%)	15/55 (27.3%)	RR 0.73 (0.37 to 1.45)	74 fewer per 1000 (from 172 fewer to 123 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – mefenamic acid 250 mg vs fennel 300 mg												
1	randomized trials	very serious ^d	not serious	not serious	very serious ^{b,c}	none	26/43 (60.5%)	26/43 (60.5%)	RR 1.00 (0.71 to 1.41)	0 fewer per 1000 (from 175 fewer to 248 more)	⊕○○○ VERY LOW	CRITICAL
Adequate pain relief as reported by the woman – ibuprofen 400 mg vs fennel essence 20%												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	20/45 (44.4%)	19/45 (42.2%)	RR 1.05 (0.66 to 1.69)	21 more per 1000 (from 144 fewer to 291 more)	⊕○○○ VERY LOW	CRITICAL
Need for additional pain relief – ibuprofen 400 mg vs fennel essence 20%												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	9/45 (20.0%)	9/45 (20.0%)	RR 1.00 (0.44 to 2.29)	0 fewer per 1000 (from 112 fewer to 258 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NSAID	Herbal analgesia	Relative (95% CI)	Absolute (95% CI)		

Pain however measured by the authors – VAS 0–10

1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	54	54	-	MD 0.21 higher (0.13 lower to 0.55 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	----	----	---	--	------------------	----------

Maternal adverse effects – mefenamic acid 250 mg vs pimpinella anisum, apium graveolens and crocus sativus 500 mg

1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	5/54 (9.3%)	1/54 (1.9%)	RR 5.00 (0.60 to 41.39)	74 more per 1000 (from 7 fewer to 748 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	-------------	-------------	-----------------------------------	---	------------------	----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Most of the pooled effect provided by studies “B”.

b. Wide confidence interval crossing the line of no effect.

c. Less than 300 women.

d. The pooled effect provided by study “C”.

e. Less than 300 women and less than 30 events.

EB table A.2.4: Postnatal pelvic floor muscle training (PFMT) for pelvic floor strengthening

Comparison 1: Postnatal PFMT compared with no intervention or usual care for (mixed) prevention or treatment of incontinence

Source: Woodley SJ, Lawrenson P, Boyle R, Cody JD, Mørkved S, Kernohan A, Hay-Smith EJC. Pelvic floor muscle training for preventing and treating urinary and faecal incontinence in antenatal and postnatal women. Cochrane Database Syst Rev.2020;(5):CD007471.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PFMT	No PFMT or usual care	Relative (95% CI)	Absolute (95% CI)		
Urinary incontinence early postnatal period (0–3 months) – PFMT vs no PFMT												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	70/194 (36.1%)	65/127 (51.2%)	RR 0.54 (0.44 to 0.66)	235 fewer per 1000 (from 287 fewer to 174 fewer)	⊕⊕○○ LOW	CRITICAL
Urinary incontinence mid–postnatal period (> 3–6 months) – PFMT vs usual care												
5	randomized trials	very serious ^a	serious ^b	not serious	not serious	none	374/1421 (26.3%)	390/1379 (28.3%)	RR 0.95 (0.75 to 1.19)	14 fewer per 1000 (from 71 fewer to 54 more)	⊕○○○ VERY LOW	CRITICAL
Urinary incontinence late postnatal period (> 6–12 months)												
3	randomized trials	serious ^c	not serious	not serious	serious ^d	none	110/425 (25.9%)	118/401 (29.4%)	RR 0.88 (0.71 to 1.09)	35 fewer per 1000 (from 85 fewer to 26 more)	⊕⊕○○ LOW	CRITICAL
Urinary incontinence late postnatal period (> 6–12 months) – PFMT vs no PFMT												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{d,e,f}	none	6/51 (11.8%)	8/56 (14.3%)	RR 0.82 (0.31 to 2.21)	26 fewer per 1000 (from 99 fewer to 173 more)	⊕○○○ VERY LOW	CRITICAL
Urinary incontinence late postnatal period (> 6–12 months) – PFMT vs usual care												
2	randomized trials	serious ^c	serious ^b	not serious	serious ^d	none	104/374 (27.8%)	110/345 (31.9%)	RR 0.88 (0.71 to 1.10)	38 fewer per 1000 (from 92 fewer to 32 more)	⊕○○○ VERY LOW	CRITICAL
Faecal incontinence early postnatal period (0–3 months) – PFMT vs usual care												
1	randomized trials	very serious ^a	not serious	not serious	serious ^d	none	21/816 (2.6%)	22/793 (2.8%)	RR 0.93 (0.51 to 1.67)	2 fewer per 1000 (from 14 fewer to 19 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PFMT	No PFMT or usual care	Relative (95% CI)	Absolute (95% CI)		

Faecal incontinence late postnatal period (> 6–12 months)

1	randomized trials	very serious ^a	not serious	not serious	very serious ^{d,e,f}	none	2/51 (3.9%)	3/56 (5.4%)	RR 0.73 (0.13 to 4.21)	14 fewer per 1000 (from 47 fewer to 172 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-------------------------------	------	-------------	-------------	----------------------------------	---	------------------	----------

Faecal incontinence late postnatal period (> 6–12 months) – PFMT vs no PFMT

1	randomized trials	very serious ^a	not serious	not serious	very serious ^{d,e,f}	none	2/51 (3.9%)	3/56 (5.4%)	RR 0.73 (0.13 to 4.21)	14 fewer per 1000 (from 47 fewer to 172 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-------------------------------	------	-------------	-------------	----------------------------------	---	------------------	----------

Postnatal quality of life (related to urinary incontinence)

1	randomized trials	serious ^c	not serious	not serious	very serious ^{d,f,g}	none	13	10	-	MD 0.5 higher (5.53 lower to 6.53 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-------------------------------	------	----	----	---	---	------------------	----------

Postnatal quality of life (related to urinary incontinence) – PFMT plus vs PFMT

1	randomized trials	serious ^c	not serious	not serious	very serious ^{d,f,g}	none	13	10	-	MD 0.5 higher (5.53 lower to 6.53 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-------------------------------	------	----	----	---	---	------------------	----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Most of the pooled effect provided by studies “B” or “C” but with a substantial proportion (i.e. > 50%) from studies “C”.

b. Severe, unexplained, heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 < 0.05$).

c. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

d. Wide confidence interval crossing the line of no effect.

e. Less than 300 participants.

f. Few events.

g. Less than 400 participants.

Comparison 2: Postnatal PFMT compared with no intervention or usual care for treatment of incontinence

Source: Woodley SJ, Lawrenson P, Boyle R, Cody JD, Mørkved S, Kernohan A, Hay-Smith EJC. Pelvic floor muscle training for preventing and treating urinary and faecal incontinence in antenatal and postnatal women. Cochrane Database Syst Rev.2020;(5):CD007471.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PFMT	No PFMT or usual care	Relative (95% CI)	Absolute (95% CI)		
Urinary incontinence late postnatal period (> 6–12 months)												
3 ^a	randomized trials	very serious ^b	serious ^c	not serious	serious ^d	none	188/341 (55.1%)	257/355 (72.4%)	RR 0.55 (0.29 to 1.07)	326 fewer per 1000 (from 514 fewer to 51 more)	⊕○○○ VERY LOW	CRITICAL
Urinary incontinence late postnatal period (> 6–12 months) – PFMT vs no PFMT												
1	randomized trials	serious ^e	not serious	not serious	serious ^f	none	12/43 (7.9%)	19/19 (100.0%)	RR 0.29 (0.18 to 0.47)	710 fewer per 1000 (from 820 fewer to 530 fewer)	⊕⊕○○ LOW	CRITICAL
Urinary incontinence late postnatal period (> 6–12 months) – PFMT vs usual care												
2	randomized trials	very serious ^b	not serious	not serious	serious ^d	none	176/298 (59.1%)	238/336 (70.8%)	RR 0.80 (0.61 to 1.06)	142 fewer per 1000 (from 276 fewer to 43 more)	⊕○○○ VERY LOW	CRITICAL
Urinary incontinence long term (> 5–10 years) – PFMT vs usual care												
1	randomized trials	very serious ^b	not serious	not serious	not serious	none	201/263 (76.4%)	201/253 (79.4%)	RR 0.96 (0.88 to 1.05)	32 fewer per 1000 (from 95 fewer to 40 more)	⊕⊕○○ LOW	CRITICAL
Urinary incontinence very long term (> 10 years) – PFMT vs usual care												
1	randomized trials	very serious ^b	not serious	not serious	not serious	none	190/230 (82.6%)	194/241 (80.5%)	RR 1.03 (0.94 to 1.12)	24 more per 1000 (from 48 fewer to 97 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	PFMT	No PFMT or usual care	Relative (95% CI)	Absolute (95% CI)		

Faecal incontinence late postnatal period (> 6–12 months) – PFMT vs usual care

2	randomized trials	very serious ^b	serious ^c	not serious	serious ^d	none	17/292 (5.8%)	45/328 (13.7%)	RR 0.68 (0.24 to 1.94)	44 fewer per 1000 (from 104 fewer to 129 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	----------------------	-------------	----------------------	------	---------------	----------------	----------------------------------	--	------------------	----------

Faecal incontinence long term (> 5–10 years) – PFMT vs usual care

1	randomized trials	very serious ^b	not serious	not serious	serious ^d	none	32/261 (12.3%)	32/248 (12.9%)	RR 0.95 (0.60 to 1.50)	6 fewer per 1000 (from 52 fewer to 65 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	----------------	----------------	----------------------------------	---	------------------	----------

Faecal incontinence very long term (> 10 years) – PFMT vs usual care

1	randomized trials	very serious ^b	not serious	not serious	serious ^d	none	43/228 (18.9%)	35/240 (14.6%)	OR 1.36 (0.84 to 2.22)	43 more per 1000 (from 20 fewer to 129 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	----------------	----------------	----------------------------------	--	------------------	----------

Urinary incontinence-specific quality of life – PFMT vs usual care

1	randomized trials	very serious ^b	not serious	not serious	very serious ^{d,g}	none	9	9	-	MD 1.66 lower (3.51 lower to 0.19 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-----------------------------	------	---	---	---	---	------------------	----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Control group: two trials considered the control group as usual care. The third trial considered the control group as relaxation massage of back and extremities by a physiotherapist, asking women not to exercise the pelvic floor at home.

b. Most of the pooled effect provided by studies “B” or “C” but with a substantial proportion (i.e. > 50%) from studies “C”.

c. Severe, unexplained, heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 < 0.05$).

d. Wide confidence interval crossing the line of no effect.

e. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

f. Less than 300 participants.

g. Less than 400 participants.

EB table A.2.5: Non-pharmacological interventions to treat postpartum breast engorgement

Comparison 1: Cabbage leaf extract cream compared with placebo

Source: Zakarija-Grkovic I, Stewart F. Treatments for breast engorgement during lactation. Cochrane Database Syst Rev. 2020;(9):CD006946.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cabbage leaf extract cream	Placebo	Relative (95% CI)	Absolute (95% CI)		
Breast pain (0–10 VAS; higher score = more pain)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	21	18	-	MD 0.4 higher (0.67 lower to 1.47 higher)	⊕○○○ VERY LOW	CRITICAL
Breast engorgement (measured with 6-point engorgement scale)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	21	18	-	MD 0.2 higher (0.18 lower to 0.58 higher)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; MD: mean difference.

a. The pooled effect provided by study “B”.

b. Wide confidence interval crossing the line of no effect.

c. Small sample size and/or few events.

Comparison 2: Cold cabbage leaves applied directly to the breast compared with usual care

Source: Zakarija-Grkovic I, Stewart F. Treatments for breast engorgement during lactation. Cochrane Database Syst Rev. 2020;(9):CD006946.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cold cabbage leaves	Usual care	Relative (95% CI)	Absolute (95% CI)		
Breast pain (0–10 VAS; higher score = more pain)												
1	randomized trials	not serious	not serious	not serious	serious ^a	none	76	76	-	MD 1.03 lower (1.53 lower to 0.53 lower)	⊕⊕⊕○ MODERATE	CRITICAL
Breast hardness (higher score = more hardness)												
1	randomized trials	not serious	not serious	not serious	serious ^a	none	76	76	-	MD 0.58 lower (0.82 lower to 0.34 lower)	⊕⊕⊕○ MODERATE	CRITICAL
Maternal opinion of treatment — women satisfied or very satisfied												
1	randomized trials	not serious	not serious	not serious	serious ^a	none	75/76 (98.7%)	53/76 (69.7%)	RR 1.42 (1.22 to 1.64)	293 more per 1000 (from 153 more to 446 more)	⊕⊕⊕○ MODERATE	CRITICAL
Cessation of breastfeeding before 6 months												
1	randomized trials	not serious	not serious	not serious	very serious ^{a, b}	none	20/55 (36.4%)	11/53 (20.8%)	RR 1.75 (0.93 to 3.30)	156 more per 1000 (from 15 fewer to 477 more)	⊕⊕○○ LOW	CRITICAL

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Small sample size and/or few events.

b. Wide confidence interval crossing the line of no effect.

Comparison 3: Cold gel packs applied directly to the breast compared with usual care

Source: Zakarija-Grkovic I, Stewart F. Treatments for breast engorgement during lactation. Cochrane Database Syst Rev. 2020;(9):CD006946.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cold gel packs	Usual care	Relative (95% CI)	Absolute (95% CI)		
Breast pain (higher score = more pain)												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	75	76	-	MD 0.4 lower (0.91 lower to 0.11 higher)	⊕⊕○○ LOW	CRITICAL
Breast hardness (higher score = more hardness)												
1	randomized trials	not serious	not serious	not serious	serious ^b	none	75	76	-	MD 0.34 lower (0.6 lower to 0.08 lower)	⊕⊕⊕○ MODERATE	CRITICAL
Maternal opinion of treatment – women satisfied or very satisfied												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	61/75 (81.3%)	53/76 (69.7%)	RR 1.17 (0.97 to 1.40)	119 more per 1000 (from 21 fewer to 279 more)	⊕⊕○○ LOW	CRITICAL
Cessation of breastfeeding before 6 months												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	12/56 (21.4%)	11/53 (20.8%)	RR 1.03 (0.50 to 2.14)	6 more per 1000 (from 104 fewer to 237 more)	⊕⊕○○ LOW	CRITICAL

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Wide confidence interval crossing the line of no effect.

b. Small sample size and/or few events.

Comparison 4: Warm herbal compresses compared with usual care (including warm compresses without herbs)

Source: Zakarija-Grkovic I, Stewart F. Treatments for breast engorgement during lactation. Cochrane Database Syst Rev. 2020;(9):CD006946.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Warm herbal compress	Usual care	Relative (95% CI)	Absolute (95% CI)		
Breast pain (higher score = more pain) – herbal compress vs hot compress												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	250	250	-	MD 1.8 lower (2.07 lower to 1.53 lower)	⊕⊕⊕○ MODERATE	CRITICAL
Breast engorgement (higher score = more pain) – hollyhock leaf compress vs warm compress												
1	randomized trials	very serious ^b	not serious	not serious	serious ^d	none	20	20	-	MD 2.82 lower (4.6 lower to 1.04 lower)	⊕○○○ VERY LOW	CRITICAL
Number of women with adverse effects – herbal compress vs hot compress												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{c,d}	none	2/250 (0.8%)	0/250 (0.0%)	RR 5.00 (0.24 to 103.62)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. The pooled effect provided by study "B".

b. The pooled effect provided by study "C".

c. Wide confidence interval crossing the line of no effect.

d. Small sample size and/or few events.

EB table A.2.6: Pharmacological interventions to treat postpartum breast engorgement

Comparison 1: Subcutaneous oxytocin compared with placebo

Source: Zakarija-Grkovic I, Stewart F. Treatments for breast engorgement during lactation. Cochrane Database Syst Rev. 2020;(9):CD006946.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Subcutaneous oxytocin	Placebo	Relative (95% CI)	Absolute (95% CI)		
Breast engorgement (symptoms not subsided after 3 days of treatment)												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	5/20 (25.0%)	2/25 (8.0%)	RR 3.13 (0.68 to 14.44)	170 more per 1000 (from 26 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "C"

b. Small sample size and/or few events.

c. Wide confidence interval crossing the line of no effect.

Comparison 2: Proteolytic enzymes compared with placebo

Comparison 2a: Oral protease complex compared with placebo

Source: Zakarija-Grkovic I, Stewart F. Treatments for breast engorgement during lactation. Cochrane Database Syst Rev. 2020;(9):CD006946.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Oral protease complex	Placebo	Relative (95% CI)	Absolute (95% CI)		
Breast pain (no improvement)												
1	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	2/35 (5.7%)	8/24 (33.3%)	RR 0.17 (0.04 to 0.74)	277 fewer per 1000 (from 320 fewer to 87 fewer)	⊕○○○ VERY LOW	CRITICAL
Breast swelling (no improvement)												
1	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	6/35 (17.1%)	12/24 (50%)	RR 0.34 (0.15 to 0.79)	330 fewer per 1000 (from 425 fewer to 105 fewer)	⊕○○○ VERY LOW	CRITICAL
Number of women with adverse effects												
2	randomized trials	very serious ^c	not serious	not serious	very serious ^d	none	Adverse effects were measured and reported in the studies investigating serrapeptase (Kee 1989) and protease (Murata 1965). No women in any of the groups experienced adverse events.				⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "C".

b. Small sample size and/or few events.

c. Most of the pooled effect provided by studies "B" or "C" but with a substantial proportion (i.e. > 50%) from studies "C".

d. No meta-analysis done. No events reported.

Comparison 2b: Oral serrapeptase compared with placebo

Source: Zakarija-Grkovic I, Stewart F. Treatments for breast engorgement during lactation. Cochrane Database Syst Rev. 2020;(9):CD006946.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Oral serrapeptase	Placebo	Relative (95% CI)	Absolute (95% CI)		
Breast pain (no improvement)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	5/35 (14.3%)	9/35 (25.7%)	RR 0.56 (0.21 to 1.49)	113 fewer per 1000 (from 203 fewer to 126 more)	⊕○○○ VERY LOW	CRITICAL
Breast swelling (no improvement)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	9/35 (25.7%)	12/35 (34.3%)	RR 0.75 (0.36 to 1.55)	86 fewer per 1000 (from 219 fewer to 189 more)	⊕○○○ VERY LOW	CRITICAL
Breast engorgement (symptoms not subsided after 3 days of treatment)												
1	randomized trials	serious ^a	not serious	not serious	serious ^b	none	5/35 (14.3%)	14/35 (40.0%)	RR 0.36 (0.14 to 0.88)	256 fewer per 1000 (from 344 fewer to 48 fewer)	⊕⊕○○ LOW	CRITICAL
Number of women with adverse effects												
2	randomized trials	serious ^d	not serious	not serious	very serious ^e	none	Adverse effects were measured and reported in the studies investigating serrapeptase (Kee 1989) and protease (Murata 1965). No women in any of the groups experienced adverse events.				⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "B".

b. Small sample size and/or few events.

c. Wide confidence interval crossing the line of no effect.

d. Most of the pooled effect provided by studies "B" or "C" but with a substantial proportion (i.e. > 50%) from studies "C".

e. No meta-analysis done. No events reported.

A.3 Preventive measures

EB table A.3.1: Non-pharmacological interventions to prevent postpartum mastitis

Comparison 1: Probiotics compared with placebo

Source: Crepinsek MA, Taylor EA, Michener K, Stewart F. Interventions for preventing mastitis after childbirth. Cochrane Database Syst Rev. 2020;(9):CD007239.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Probiotics	Placebo	Relative (95% CI)	Absolute (95% CI)		
Nipple damage within 6 months postpartum												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	4/203 (2.0%)	13/221 (5.9%)	RR 0.33 (0.11 to 1.01)	39 fewer per 1000 (from 52 fewer to 1 more)	⊕○○○ VERY LOW	IMPORTANT
Breast pain												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,d}	none	46/139 (33%)	65/152 (42.7%)	RR 0.77 (0.57 to 1.04)	98 fewer per 1000 (from 184 fewer to 17 more)	⊕○○○ VERY LOW	IMPORTANT
Incidence of mastitis within 6 months postpartum												
1	randomized trials	serious ^a	not serious	not serious	serious ^b	none	30/194 (15.5%)	60/205 (29.3%)	RR 0.58 (0.33 to 1.02)	123 fewer per 1000 (from 196 fewer to 6 more)	⊕⊕○○ LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "B".

b. Wide confidence interval crossing the line of no effect.

c. Less than 30 events.

d. Less than 300 women.

Comparison 2: Hydrothermally processed cereal with anti-secretory factor-inducing properties compared with standard cereal (serving as a placebo)

Source: Crepinsek MA, Taylor EA, Michener K, Stewart F. Interventions for preventing mastitis after childbirth. Cochrane Database Syst Rev. 2020;(9):CD007239.

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Hydrothermally processed cereal with AF factor	Standard cereal	Relative (95% CI)	Absolute (95% CI)		
Incidence of mastitis within 6 months postpartum												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	1/12 (8.3%)	6/17 (35.3%)	RR 0.24 (0.03 to 1.72)	268 fewer per 1000 (from 342 fewer to 254 more)	⊕○○○ VERY LOW	IMPORTANT
Recurrence of mastitis within 12 months postpartum												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	0/1 (0.0%)	4/6 (66.7%)	RR 0.39 (0.03 to 4.57)	407 fewer per 1000 (from 647 fewer to 1000 more)	⊕○○○ VERY LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "B".

b. Wide confidence interval crossing the line of no effect.

c. Few events and few participants.

Comparison 3: Specialist breastfeeding education compared with usual care

Source: Crepinsek MA, Taylor EA, Michener K, Stewart F. Interventions for preventing mastitis after childbirth. Cochrane Database Syst Rev. 2020;(9):CD007239.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Specialist breastfeeding education	Usual care	Relative (95% CI)	Absolute (95% CI)		
Breast pain (sore nipples) – at hospital discharge												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	32/74 (43.2%)	60/137 (43.8%)	RR 0.99 (0.72 to 1.36)	4 fewer per 1000 (from 123 fewer to 158 more)	⊕○○○ VERY LOW	CRITICAL
Breast pain (sore nipples) – at 7 days												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	32/73 (43.8%)	67/137 (48.9%)	RR 0.90 (0.66 to 1.22)	49 fewer per 1000 (from 166 fewer to 108 more)	⊕○○○ VERY LOW	CRITICAL
Breast pain (sore nipples) – at 30 days												
1	randomized trials	serious ^a	not serious	not serious	very serious ^b	none	6/71 (8.5%)	12/132 (9.1%)	RR 0.93 (0.36 to 2.37)	6 fewer per 1000 (from 58 fewer to 125 more)	⊕○○○ VERY LOW	CRITICAL
Breast engorgement – at hospital discharge												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	0/74 (0.0%)	1/137 (0.7%)	RR 0.61 (0.03 to 14.87)	3 fewer per 1000 (from 7 fewer to 101 more)	⊕○○○ VERY LOW	CRITICAL
Breast engorgement – at 7 days												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	26/73 (35.6%)	47/137 (34.3%)	RR 1.04 (0.71 to 1.53)	14 more per 1000 (from 99 fewer to 182 more)	⊕○○○ VERY LOW	CRITICAL
Breast engorgement – at 30 days												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	28/71 (39.4%)	50/132 (37.9%)	RR 1.04 (0.73 to 1.49)	15 more per 1000 (from 102 fewer to 186 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Specialist breastfeeding education	Usual care	Relative (95% CI)	Absolute (95% CI)		

Incidence of mastitis within 6 months postpartum – at hospital discharge

1	randomized trials	serious ^a	not serious	not serious	very serious ^{d,e}	none	0/74 (0.0%)	0/137 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	-------------	--------------	---------------	---	------------------	----------

Incidence of mastitis within 6 months postpartum – at 7 days

1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	2/73 (2.7%)	1/137 (0.7%)	RR 3.75 (0.35 to 40.70)	20 more per 1000 (from 5 fewer to 290 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	-------------	--------------	-----------------------------------	---	------------------	----------

Incidence of mastitis within 6 months postpartum – at 30 days

1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	2/71 (2.8%)	4/132 (3.0%)	RR 0.93 (0.17 to 4.95)	2 fewer per 1000 (from 25 fewer to 120 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	-------------	--------------	----------------------------------	--	------------------	----------

Exclusive breastfeeding – at 7 days

1	randomized trials	serious ^a	not serious	not serious	serious ^c	none	60/73 (82.2%)	109/137 (79.6%)	RR 1.03 (0.90 to 1.18)	24 more per 1000 (from 80 fewer to 143 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	----------------------	------	---------------	-----------------	----------------------------------	--	-------------	----------

Exclusive breastfeeding – at 30 days

1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	38/71 (53.5%)	80/132 (60.6%)	RR 0.88 (0.68 to 1.14)	73 fewer per 1000 (from 194 fewer to 85 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	---------------	----------------	----------------------------------	---	------------------	----------

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Wide confidence interval crossing the line of no effect.

c. Small sample size and/or few events.

d. No events.

e. Small sample size.

Comparison 4: Acupoint massage compared with usual care

Source: Crepinsek MA, Taylor EA, Michener K, Stewart F. Interventions for preventing mastitis after childbirth. Cochrane Database Syst Rev. 2020;(9):CD007239.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Acupoint massage	Usual care	Relative (95% CI)	Absolute (95% CI)		
Breast pain												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	10/200 (5.0%)	80/200 (40.0%)	RR 0.13 (0.07 to 0.23)	348 fewer per 1000 (from 372 fewer to 308 fewer)	⊕⊕⊕○ MODERATE	IMPORTANT
Breast engorgement												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	50/200 (25.0%)	102/200 (51.0%)	RR 0.49 (0.37 to 0.65)	260 fewer per 1000 (from 321 fewer to 179 fewer)	⊕⊕⊕○ MODERATE	IMPORTANT
Incidence of mastitis within 6 months postpartum												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	10/200 (5.0%)	26/200 (13.0%)	RR 0.38 (0.19 to 0.78)	81 fewer per 1000 (from 105 fewer to 29 fewer)	⊕⊕⊕○ MODERATE	IMPORTANT
Women's perception of milk supply – moderate or better												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	176/200 (88.0%)	140/200 (70.0%)	RR 1.26 (1.13 to 1.40)	182 more per 1000 (from 91 more to 280 more)	⊕⊕⊕○ MODERATE	IMPORTANT
Exclusive breastfeeding – at 42 days postpartum												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	152/200 (76.0%)	80/200 (40.0%)	RR 1.90 (1.58 to 2.29)	360 more per 1000 (from 232 more to 516 more)	⊕⊕⊕○ MODERATE	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study “B”.

EB table A.3.2: Pharmacological interventions to prevent postpartum mastitis

Comparison 1: Oral prophylactic antibiotics compared with placebo or usual care

Comparison 1a: Oral antibiotics (flucloxacillin) compared with placebo

Source: Crepinsek MA, Taylor EA, Michener K, Stewart F. Interventions for preventing mastitis after childbirth. Cochrane Database Syst Rev. 2020;(9):CD007239.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Oral antibiotics (flucloxacillin)	Placebo	Relative (95% CI)	Absolute (95% CI)		
Incidence of mastitis within 6 months postpartum												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	0/5	1/5	RR 0.33 (0.02 to 6.55)	-	⊕⊕○○ LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. Wide confidence interval crossing the line of no effect.

b. Less than 300 women and less than 30 events.

Comparison 1b: Oral antibiotics (cloxacillin/erythromycin) compared with usual care (breastfeeding advice)

Source: Crepinsek MA, Taylor EA, Michener K, Stewart F. Interventions for preventing mastitis after childbirth. Cochrane Database Syst Rev. 2020;(9):CD007239.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Oral antibiotics (cloxacillin/erythromycin)	Usual care (breastfeeding advice)	Relative (95% CI)	Absolute (95% CI)		
Incidence of mastitis within 6 months postpartum												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	1/19	7/23	RR 0.17 (0.02 to 1.28)	-	⊕⊕○○ LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. Wide confidence interval crossing the line of no effect.

b. Less than 300 women and less than 30 events.

Comparison 2: Topical prophylactic antibiotics compared with usual care (breastfeeding advice)

Source: Crepinsek MA, Taylor EA, Michener K, Stewart F. Interventions for preventing mastitis after childbirth. Cochrane Database Syst Rev. 2020;(9):CD007239.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Topical prophylactic antibiotics	Usual care (breastfeeding advice)	Relative (95% CI)	Absolute (95% CI)		
Incidence of mastitis within 6 months postpartum – fusidic acid ointment vs breastfeeding advice												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	4/17 (23.5%)	7/23 (30.4%)	RR 0.77 (0.27 to 2.22)	70 fewer per 1000 (from 222 fewer to 371 more)	⊕⊕○○ LOW	IMPORTANT
Incidence of mastitis within 6 months postpartum – mupirocin ointment vs breastfeeding advice												
1	randomized trials	not serious	not serious	not serious	very serious ^{a,b}	none	3/25 (12.0%)	7/23 (30.4%)	RR 0.39 (0.12 to 1.35)	186 fewer per 1000 (from 268 fewer to 107 more)	⊕⊕○○ LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

a. Wide confidence interval crossing the line of no effect.

b. Few events and few participants.

EB table A.3.3: Prevention of postpartum constipation

Comparison: Laxatives compared with placebo

Source: Turawa EB, Musekiwa A, Rohwer AC. Interventions for preventing postpartum constipation. Cochrane Database Syst Rev.2020;(8):CD011625.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Laxative	Placebo	Relative (95% CI)	Absolute (95% CI)		
Number of women with first bowel movement less than 24 hours after birth												
1	randomized trials	very serious ^a	not serious	not serious	not serious	none	142/224 (63.4%)	54/247 (21.9%)	RR 2.90 (2.24 to 3.75)	415 more per 1000 (from 271 more to 601 more)	⊕⊕○○ LOW	CRITICAL
Number of women with first bowel movement on day 1 after birth												
1 ^b	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	69/224 (30.8%)	81/247 (32.8%)	RR 0.94 (0.72 to 1.22)	20 fewer per 1000 (from 92 fewer to 72 more)	⊕○○○ VERY LOW	CRITICAL
Number of women with first bowel movement on day 2 after birth												
1 ^b	randomized trials	very serious ^a	not serious	not serious	not serious	none	9/224 (4.0%)	44/247 (17.8%)	RR 0.23 (0.11 to 0.45)	137 fewer per 1000 (from 159 fewer to 98 fewer)	⊕⊕○○ LOW	CRITICAL
Number of women with first bowel movement on day 3 after birth												
1 ^d	randomized trials	very serious ^a	not serious	not serious	serious ^e	none	0/224 (0.0%)	10/247 (4.0%)	RR 0.05 (0.00 to 0.89)	38 fewer per 1000 (from 4 fewer to --)	⊕○○○ VERY LOW	CRITICAL
Number of women with first bowel movement on day 4 after birth												
1 ^d	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,e}	none	1/224 (0.4%)	5/247 (2.0%)	RR 0.22 (0.23 to 1.87)	16 fewer per 1000 (from 16 fewer to 18 more)	⊕○○○ VERY LOW	CRITICAL
Number of postpartum enemas given												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,f}	none	20/123 (16.3%)	31/121 (25.6%)	RR 0.63 (0.38 to 1.05)	95 fewer per 1000 (from 159 fewer to 13 more)	⊕○○○ VERY LOW	CRITICAL
Side-effects – women with abdominal cramps												
1 ^d	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,e}	none	23/224 (10.3%)	6/247 (2.4%)	RR 4.23 (1.75 to 10.19)	78 more per 1000 (from 18 more to 223 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Laxative	Placebo	Relative (95% CI)	Absolute (95% CI)		

Side-effects on the baby – loose stools

1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,e,f}	none	3/126 (2.4%)	6/155 (3.9%)	RR 0.62 (0.16 to 2.41)	15 fewer per 1000 (from 33 fewer to 55 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-------------------------------	------	--------------	--------------	----------------------------------	--	------------------	----------

Side-effects on the baby – diarrhoea

1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,e,f}	none	2/126 (1.6%)	1/155 (0.6%)	RR 2.46 (0.23 to 26.82)	9 more per 1000 (from 5 fewer to 167 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-------------------------------	------	--------------	--------------	-----------------------------------	--	------------------	----------

CI: confidence interval; RR: risk ratio.

a. The effect provided by study “C”.

b. Excluded from the analysis were trials using drugs no longer indicated in the postpartum women (Diamond 1968 and Mundow 1975).

c. Wide confidence interval crossing the line of no effect.

d. Excluded from the analysis were trials using drugs no longer indicated in the postpartum women (Diamond 1968).

e. Less than 30 events.

f. Less than 300 participants.

A.4 Mental health interventions

EB table A.4.1: Screening for postpartum depression and anxiety

Comparison: Screening for common mental disorders (CMDs: depression, anxiety) in the postpartum period compared with no screening or usual care

Source: Waqas A, Kokab A, Meraj H, Dua T, Chowdhary N, Fatima B, et al. Screening programs for common maternal mental health disorders among perinatal women: report of the systematic review evidence. BMC Psychiatry. 2022;22(1):54. doi:10.1186/s12888-022-03694-9.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Screening for CMDs	No screening or usual care	Relative (95% CI)	Absolute (95% CI)		
Rate of postpartum depression – RCTs												
4	randomized trials	very serious ^a	not serious	not serious	not serious	none	500/1648 (30.3%)	604/1516 (39.8%)	OR 0.53 (0.45 to 0.62)	67 fewer per 1000 (from 79 fewer to 53 fewer)	⊕⊕○○ LOW	CRITICAL
Rate of postpartum depression – quasi-RCTs												
2	observational studies	serious ^b	not serious	not serious	not serious	strong association	76/3359 (0.2%)	73/1651 (4.4%)	OR 0.30 (0.24 to 0.48)	31 fewer per 1000 (from 33 fewer to 22 fewer)	⊕⊕○○ LOW	CRITICAL
Postpartum anxiety rate – RCTs												
1	randomized trials	not serious	not serious	not serious	not serious	none	271	294	-	SMD 0.28 SD fewer (0.44 fewer to 0.11 fewer)	⊕⊕⊕⊕ HIGH	CRITICAL
Postpartum anxiety rate – quasi-RCTs												
1	observational studies	very serious ^a	not serious	not serious	not serious	none	1843	1540	-	SMD 0.17 SD fewer (0.24 fewer to 0.09 fewer)	⊕○○○ VERY LOW	CRITICAL
Quality of life – RCTs												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	1072	996	-	SMD 0.24 SD more (0.11 more to 0.38 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Screening for CMDs	No screening or usual care	Relative (95% CI)	Absolute (95% CI)		

Quality of life – quasi-RCTs

1	observational studies	very serious ^a	not serious	not serious	not serious	none	1843	1246	-	SMD 0.04 SD more (0.12 more to 0.26 more)	⊕○○○ VERY LOW	CRITICAL
---	-----------------------	---------------------------	-------------	-------------	-------------	------	------	------	---	---	------------------	----------

Marital satisfaction – RCTs

2	randomized trials	not serious	not serious	not serious	serious ^c	none	-/553	-/464	OR 0.56 (0.205 to 1.525)	not reported ^d	⊕⊕⊕○ MODERATE	CRITICAL
---	-------------------	-------------	-------------	-------------	----------------------	------	-------	-------	------------------------------------	---------------------------	------------------	----------

Parental stress – RCTs

3	randomized trials	not serious	not serious	not serious	not serious	none	-/758	-/824	OR 0.57 (0.45 to 0.74)	not reported ^d	⊕⊕⊕⊕ HIGH	CRITICAL
---	-------------------	-------------	-------------	-------------	-------------	------	-------	-------	----------------------------------	---------------------------	--------------	----------

Parental stress – quasi-RCTs

1	observational studies	serious ^b	not serious	not serious	serious ^c	none	128	626	-	MD 0.14 SD fewer (0.39 fewer to 0.13 more)	⊕○○○ VERY LOW	CRITICAL
---	-----------------------	----------------------	-------------	-------------	----------------------	------	-----	-----	---	--	------------------	----------

Treatment seeking practices – RCTs

2	randomized trials	not serious	very serious ^e	not serious	not serious	none	231/553 (41.8%)	81/464 (17.5%)	OR 3.45 (2.52 to 4.70)	247 more per 1000 (from 173 more to 324 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	-------------	---------------------------	-------------	-------------	------	--------------------	-------------------	----------------------------------	---	-------------	----------

CI: confidence interval; OR: odds ratio; RCT: randomized controlled trial; RR: risk ratio; SMD: standardized mean difference.

a. Most of the pooled effect provided by trials “B” or “C” but with a substantial proportion (i.e. > 50%) from studies “C”.

b. Most of the pooled effect provided by studies “B” or “C” but without a substantial proportion (i.e. < 50%) from studies “C”.

c. Wide confidence interval crossing the line of no effect.

d. Information on total number of events not available from original trials.

e. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

EB table A.4.2: Prevention of postpartum depression and anxiety

Comparison: Interventions to prevent common mental disorders (CMDs: depression, anxiety) in the postpartum period, delivered at any time, compared with no intervention or usual care

Source: Waqas A, Kokab A, Meraj H, Dua T, Chowdhary N, Fatima B, et al. Prevention of common mental disorders among women in the perinatal period: a critical mixed-methods review and meta-analysis. Global Mental Health (in press).

							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to prevent CMDs	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Rate of postpartum depression – any timepoint												
9	randomized trials	serious ^a	not serious	not serious	not serious	none	78/896 (8.7%)	119/935 (12.7%)	OR 0.61 (0.38 to 0.99)	45 fewer per 1000 (from 75 fewer to 2 fewer)	⊕⊕⊕○ MODERATE	CRITICAL
Depression severity – any timepoint												
38	randomized trials	serious ^a	serious ^b	not serious	not serious	none	10 761	9808	-	SMD 0.29 lower (0.44 lower to 0.15 lower)	⊕⊕○○ LOW	CRITICAL
Rate of anxiety disorder – any timepoint												
4	randomized trials	not serious	serious ^b	not serious	not serious	none	66/470 (14.0%)	124/487 (25.5%)	OR 0.20 (0.04 to 0.89)	192 fewer per 1000 (from 240 fewer to 21 fewer)	⊕⊕⊕○ MODERATE	CRITICAL
Anxiety severity – any timepoint												
9	randomized trials	not serious	serious ^b	not serious	not serious	none	906	890	-	SMD 0.79 lower (1.30 lower to 0.28 lower)	⊕⊕⊕○ MODERATE	CRITICAL
Marital discord – any timepoint												
7	randomized trials	not serious	serious ^b	not serious	not serious	none	787	776	-	SMD 0.33 SD lower (0.54 lower to 0.12 lower)	⊕⊕⊕○ MODERATE	IMPORTANT

							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to prevent CMDs	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Perceived social support – any timepoint												
9	randomized trials	serious ^a	not serious	not serious	serious ^c	none	4589	3827	-	SMD 0.002 higher (0.05 lower to 0.05 higher)	⊕⊕○○ LOW	IMPORTANT
Maternal infant attachment – any timepoint												
6	randomized trials	serious ^a	not serious	not serious	not serious	none	1024	1054	-	SMD 0.11 SD lower (0.20 lower to 0.02 lower)	⊕⊕⊕○ MODERATE	IMPORTANT
Rates of exclusive breastfeeding – any timepoint												
1	randomized trials	serious ^a	not serious	not serious	serious ^c	none	395/1206 (32.8%)	402/1232 (32.6%)	OR 1.02 (0.81 to 1.27)	3 more per 1000 (from 44 fewer to 54 more)	⊕⊕○○ LOW	IMPORTANT
Rates of optimum breastfeeding initiation – any timepoint												
2	randomized trials	serious ^a	not serious	not serious	serious ^c	none	300/595 (50.4%)	302/615 (49.1%)	OR 1.10 (0.90 to 1.33)	23 more per 1000 (from 25 fewer to 71 more)	⊕⊕○○ LOW	IMPORTANT
Paternal stress – any timepoint												
4	randomized trials	not serious	serious ^b	not serious	serious ^c	none	302	290	-	SMD 0.07 SD higher (0.21 lower to 0.34 higher)	⊕⊕○○ LOW	IMPORTANT
Maternal dissatisfaction – any timepoint												
8	randomized trials	serious ^a	serious ^b	not serious	not serious	none	2092	1915	-	SMD 0.36 SD lower (0.60 lower to 0.12 lower)	⊕⊕○○ LOW	IMPORTANT
Mental health treatment seeking – any timepoint												
2	randomized trials	serious ^a	serious ^b	not serious	very serious ^{c, d}	none	10/101 (9.9%)	22/96 (22.9%)	OR 0.69 (0.20 to 2.37)	60 fewer per 1000 (from 174 fewer to 185 more)	⊕○○○ VERY LOW	IMPORTANT

							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to prevent CMDs	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Rate of postpartum depression – antenatal only												
2	randomized trials	very serious ^a	not serious	not serious	very serious ^{c, d, f}	none	3/111 (2.7%)	12/114 (10.5%)	OR 0.25 (0.03 to 1.83)	77 fewer per 1000 (from 101 fewer to 72 more)	⊕○○○ VERY LOW	CRITICAL
Rate of postpartum depression – antenatal and postpartum												
5	randomized trials	serious ^a	not serious	not serious	serious ^c	none	43/284 (15.1%)	68/310 (21.9%)	OR 0.57 (0.27 to 1.18)	82 fewer per 1000 (from 29 more to 768 more)	⊕⊕○○ LOW	CRITICAL
Rate of postpartum depression – postpartum only												
2	randomized trials	not serious	not serious	not serious	serious ^c	none	32/501 (6.4%)	39/511 (7.6%)	OR 0.82 (0.48 to 1.41)	13 fewer per 1000 (from 38 fewer to 28 more)	⊕⊕⊕○ MODERATE	CRITICAL
Depression severity – antenatal only												
9	randomized trials	serious ^a	serious ^b	not serious	not serious	none	1614	1392	-	SMD 0.70 lower (1.17 lower to 0.24 lower)	⊕⊕○○ LOW	CRITICAL
Depression severity – antenatal and postpartum												
14	randomized trials	serious ^a	not serious	not serious	not serious	none	1738	1747	-	MD 0.10 lower (0.20 lower to 0.01 lower)	⊕⊕⊕○ MODERATE	CRITICAL
Depression severity – postpartum only												
15	randomized trials	serious ^a	serious ^b	not serious	serious ^c	none	7409	6669	-	SMD 0.25 lower (0.51 lower to 0.01 higher)	⊕○○○ VERY LOW	CRITICAL
Anxiety severity – antenatal only												
3	randomized trials	very serious ^e	serious ^b	not serious	not serious	none	229	203	-	SMD 1.43 lower (2.22 lower to 0.65 lower)	⊕○○○ VERY LOW	CRITICAL

							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to prevent CMDs	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		

Anxiety severity – antenatal and postpartum

2	randomized trials	not serious	not serious	not serious	very serious ^{c, g}	none	85	86	-	SMD 0.20 lower (0.50 lower to 0.11 higher)	⊕⊕○○ LOW	CRITICAL
---	-------------------	-------------	-------------	-------------	------------------------------	------	----	----	---	--	-------------	----------

Anxiety severity – postpartum only

4	randomized trials	not serious	serious ^b	not serious	not serious	none	592	601	-	SMD 0.45 lower (0.88 lower to 0.02 lower)	⊕⊕⊕○ MODERATE	CRITICAL
---	-------------------	-------------	----------------------	-------------	-------------	------	-----	-----	---	---	------------------	----------

CI: confidence interval; MD: mean difference; OR: odds ratio; SMD: standardized mean difference.

a. Most of the pooled effect provided by studies “B”.

b. Severe, unexplained, heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 < 0.05$).

c. Wide confidence interval crossing the line of no effect.

d. Less than 300 women.

e. Most of pooled effect provided by studies “C”.

f. Less than 30 events.

g. Less than 400 participants.

B. NEWBORN CARE

B.1 Newborn assessment

EB table B.1.2: Universal screening for abnormalities of the eye

Comparison: Universal newborn screening for abnormalities of the eye compared with no screening

Source: Malik ANJ, Evans JR, Gupta S, Mariotti S, Gordon I, Bowman R, et al. Universal newborn eye screening: a systematic review of the literature and review of international guidelines (submitted).

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Universal eye screening	No screening	Relative (95% CI)	Absolute (95% CI)		
Proportion of newborns with congenital cataract referred from maternity wards or well-baby clinics in the first year after birth – maternity ward screening compared with no screening												
1	observational studies	very serious ^a	not serious	not serious	not serious	none	49/328 523 (0.0%)	1/65 915 (0.0%)	RR 9.83 (1.36 to 71.20)	0 fewer per 1000 (from 0 fewer to 1 more)	⊕⊕○○ LOW	CRITICAL
Proportion of newborns with congenital cataract referred from maternity wards or well-baby clinics in the first year after birth – well-baby clinic screening compared with no screening												
1	observational studies	very serious ^a	not serious	not serious	very serious ^{b,c}	none	15/149 432 (0.0%)	1/65 915 (0.0%)	RR 6.62 (0.87 to 50.09)	0 fewer per 1000 (from 0 fewer to 1 more)	⊕○○○ VERY LOW	CRITICAL
Proportion of newborns with congenital cataract referred from any health facility* in the first year after birth – maternity ward screening												
1	observational studies	very serious ^a	not serious	not serious	serious ^b	none	61/328 523 (0.0%)	10/65 915 (0.0%)	RR 1.22 (0.63 to 2.39)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Proportion of newborns with congenital cataract referred from any health facility* in the first year after birth – well-baby clinic screening												
1	observational studies	very serious ^a	not serious	not serious	serious ^b	none	22/149 432 (0.0%)	10/65 915 (0.0%)	RR 0.97 (0.46 to 2.05)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Proportion of newborns with congenital cataract referred within 6 weeks (42 days) of birth – maternity ward screening												
1	observational studies	very serious ^a	not serious	not serious	not serious	none	46/328 325 (0.0%)	2/65 915 (0.0%)	RR 4.61 (1.12 to 19.01)	0 fewer per 1000 (from 0 fewer to 1 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Universal eye screening	No screening	Relative (95% CI)	Absolute (95% CI)		
Proportion of newborns with congenital cataract referred within 6 weeks (42 days) of birth – well-baby clinic screening												
1	observational studies	very serious ^a	not serious	not serious	very serious ^{b,c}	none	9/149 432 (0.0%)	2/65 915 (0.0%)	RR 1.98 (0.43 to 9.19)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Proportion of newborns with congenital cataract operated on within 6 weeks (42 days) of birth – maternity ward screening												
1	observational studies	very serious ^a	not serious	not serious	not serious	none	41/328 523 (0.0%)	1/65 915 (0.0%)	RR 8.23 (1.13 to 59.80)	0 fewer per 1000 (from 0 fewer to 1 more)	⊕⊕○○ LOW	CRITICAL
Proportion of newborns with congenital cataract operated on within 6 weeks (42 days) of birth – well-baby clinic screening												
1	observational studies	very serious ^a	not serious	not serious	very serious ^{b,c}	none	9/149 432 (0.0%)	1/65 915 (0.0%)	RR 3.97 (0.50 to 31.33)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Adverse effects associated with screening (red reflex testing) – clinical conjunctivitis												
1	observational studies	very serious ^a	not serious	not serious	not serious	none	219/9338 (2.3%)	201/9532 (2.1%)	OR 1.22 (1.01 to 1.47)	5 more per 1000 (from 0 fewer to 10 more)	⊕⊕○○ LOW	CRITICAL
Adverse effects associated with screening (red reflex testing) – bacterial conjunctivitis												
1	observational studies	very serious ^a	not serious	not serious	serious ^b	none	40/9338 (0.4%)	33/9532 (0.3%)	OR 1.20 (0.76 to 1.90)	1 more per 1000 (from 1 fewer to 3 more)	⊕○○○ VERY LOW	CRITICAL

* includes maternity wards, well-baby clinics, paediatric clinics and others. Note: these analyses use additional data provided by the authors, as the publications did not give an adequate breakdown of the numbers.

a. The pooled effect provided by study “C”.

b. Wide confidence interval crossing the line of no effect.

c. Less than 30 events.

EB table B.1.3: Universal screening for hearing impairment

Comparison: Universal newborn hearing screening (UNHS) compared with no screening or selective screening

Source: Universal Newborn Hearing Screening (UNHS) review group. Effectiveness of universal newborn hearing screening: a systematic review and meta-analysis (in preparation).

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	UNHS	No screening or selective screening	Relative (95% CI)	Absolute (95% CI)		
In all children born, proportion of screened children who had hearing loss (yield of screening)												
3	observational studies	very serious ^a	not serious	not serious	not serious	none	556/574 797 (0.1%)	433/446 700 (0.1%)	RR 1.01 (0.89 to 1.14)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕⊕○○ LOW	CRITICAL
Proportion identified with permanent bilateral hearing loss (PBHL) before 9 months of age												
1	observational studies	serious ^b	not serious	not serious	serious ^c	none	41/68 714 (0.1%)	16/88 019 (0.0%)	RR 3.28 (1.84 to 5.85)	0 fewer per 1000 (from 0 fewer to 1 more)	⊕⊕○○ LOW	CRITICAL
In children with hearing loss, mean age of identification in months												
2	observational studies	very serious ^d	serious ^e	not serious	serious ^c	none	115	82	-	MD 13.16 lower (26.31 lower to 0.01 lower)	⊕○○○ VERY LOW	CRITICAL
In children with hearing loss, mean receptive language at 3–8 years of age (z score)												
1	observational studies	very serious ^a	not serious	not serious	serious ^c	none	52	49	-	MD 0.61 higher (0.07 higher to 1.13 higher)	⊕○○○ VERY LOW	CRITICAL
In children with hearing loss, mean receptive language at 3–8 years of age (development quotient)												
3	observational studies	very serious ^a	serious ^e	not serious	very serious ^{c,f}	none	174	160	-	MD 7.61 higher (1.16 lower to 16.38 higher)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	UNHS	No screening or selective screening	Relative (95% CI)	Absolute (95% CI)		

In children with hearing loss, mean expressive language at 3–8 years of age (z score)

1	observational studies	very serious ^a	not serious	not serious	very serious ^{c,f}	none	46	41	-	MD 0.39 higher (0.2 lower to 0.97 higher)	⊕○○○ VERY LOW	CRITICAL
---	-----------------------	---------------------------	-------------	-------------	-----------------------------	------	----	----	---	---	------------------	----------

In children with hearing loss, mean expressive language at 3–8 years of age (development quotient)

3	observational studies	very serious ^a	serious ^e	not serious	serious ^c	none	174	160	-	MD 10.01 higher (1.77 higher to 18.25 higher)	⊕○○○ VERY LOW	CRITICAL
---	-----------------------	---------------------------	----------------------	-------------	----------------------	------	-----	-----	---	---	------------------	----------

In children with hearing loss, mean literacy at 5–11 years of age (z score)

1	observational studies	very serious ^a	not serious	not serious	very serious ^{c,f}	none	21	20	-	MD 0.58 higher (0.03 higher to 1.13 higher)	⊕○○○ VERY LOW	CRITICAL
---	-----------------------	---------------------------	-------------	-------------	-----------------------------	------	----	----	---	---	------------------	----------

In children with hearing loss, mean literacy at 13–19 years of age (z score)

1	observational studies	very serious ^a	not serious	not serious	very serious ^{c,f}	none	31	29	-	MD 0.15 higher (0.76 lower to 1.05 higher)	⊕○○○ VERY LOW	CRITICAL
---	-----------------------	---------------------------	-------------	-------------	-----------------------------	------	----	----	---	--	------------------	----------

CI: confidence interval; MD: mean difference; OR: odds ratio; RR: risk ratio.

a. Pooled effect provided by studies “C”.

b. Most of the pooled effect is provided by studies “B”.

c. Small sample size (less than 300 participants in dichotomous outcomes or less than 400 in continuous outcomes).

d. Most of the pooled effect is provided by studies “C”.

e. Severe, unexplained, heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 < 0.05$).

f. Wide confidence interval crossing the line of no effect.

EB table B.1.4a: Universal screening for neonatal hyperbilirubinaemia (TcB)

Comparison: Universal screening for identification of neonatal hyperbilirubinaemia by TcB at discharge compared with clinical screening (visual inspection and/or assessment of risk factors), followed by TcB or total serum bilirubin (TSB) if required

Source: Khurshid F, Rao SPN, Sauve C, Gupta S. Universal screening for hyperbilirubinemia in term healthy newborns at discharge: a systematic review and meta-analysis (submitted).

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Universal TcB	Clinical screening	Relative (95% CI)	Absolute (95% CI)		
Severe hyperbilirubinaemia – RCTs												
1	randomized trials	serious ^a	not serious	not serious	serious ^b	none	3/929 (0.3%)	11/929 (1.2%)	RR 0.27 (0.08 to 0.97)	9 fewer per 1000 (from 11 fewer to 0 fewer)	⊕⊕○○ LOW	CRITICAL
Severe hyperbilirubinaemia – non-RCTs												
1	observational studies	very serious ^c	not serious	not serious	not serious	none	-	-	RR 0.25 (0.12 to 0.52)	0 fewer per 1000 (from 1 fewer to 0 fewer)	⊕⊕○○ LOW	CRITICAL
Jaundice requiring exchange transfusion – RCTs												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,d}	none	0/929 (0.0%)	2/929 (0.2%)	RR 0.20 (0.01 to 4.16)	2 fewer per 1000 (from 2 fewer to 7 more)	⊕○○○ VERY LOW	CRITICAL
Jaundice requiring exchange transfusion – non-RCTs												
1	observational studies	very serious ^c	not serious	not serious	not serious	none	-	-	OR 0.28 (0.19 to 0.42)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕⊕○○ LOW	CRITICAL
Bilirubin induced neurological dysfunction/kernicterus – RCTs												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,d}	none	0/929 (0.0%)	1/929 (0.1%)	RR 0.33 (0.01 to 8.17)	1 fewer per 1000 (from 1 fewer to 8 more)	⊕○○○ VERY LOW	CRITICAL
Readmission for jaundice – RCTs												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	12/929 (1.3%)	48/929 (5.2%)	OR 0.24 (0.13 to 0.46)	39 fewer per 1000 (from 45 fewer to 27 fewer)	⊕⊕⊕○ MODERATE	CRITICAL

Certainty assessment							Nº of patients		Effect		Certainty (GRADE)	Importance
Nº of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Universal TcB	Clinical screening	Relative (95% CI)	Absolute (95% CI)		

Readmission for jaundice – non-RCTs

4	observational studies	very serious ^c	serious ^e	not serious	serious ^d	none	55/8223 (0.7%)	89/8266 (1.1%)	OR 1.01 (0.38 to 2.70)	0 fewer per 1000 (from 7 fewer to 18 more)	⊕○○○ VERY LOW	CRITICAL
---	-----------------------	---------------------------	----------------------	-------------	----------------------	------	----------------	----------------	----------------------------------	--	------------------	----------

CI: confidence interval; OR: odds ratio; RCT: randomized controlled trial; RR: risk ratio.

a. The pooled effect provided by study “B”.

b. Less than 30 events.

c. Most of the pooled effect provided by studies “C”.

d. Wide confidence interval crossing the line of no effect.

e. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

EB table B.1.4b: Universal screening for neonatal hyperbilirubinaemia (TSB)

Comparison: Universal screening of total serum bilirubin (TSB) before discharge compared with clinical screening (visual inspection and/or risk factor assessment)

Source: Khurshid F, Rao SPN, Sauve C, Gupta S. Universal screening for hyperbilirubinemia in term healthy newborns at discharge: a systematic review and meta-analysis (submitted).

Certainty assessment							№ of patients *		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Universal TSB	Clinical screening	Relative (95% CI)	Absolute (95% CI)		
Severe hyperbilirubinaemia												
2	observational studies	very serious ^a	serious ^b	serious ^c	not serious	none	370/52 483 (0.7%)	634/48 798 (1.3%)	OR 0.37 (0.15 to 0.88)	8 fewer per 1000 (from 11 fewer to 2 fewer)	⊕○○○ VERY LOW	CRITICAL
Readmission for jaundice												
2	observational studies	serious ^d	serious ^b	serious ^c	serious ^e	none	226/52 483 (0.4%)	268/48 798 (0.5%)	OR 1.01 (0.62 to 1.67)	0 fewer per 1000 (from 2 fewer to 4 more)	⊕○○○ VERY LOW	CRITICAL
Jaundice requiring exchange transfusion												
2	observational studies	serious ^a	serious ^b	serious ^c	serious ^e	none	4/8549 (0.0%)	13/22 510 (0.1%)	OR 0.53 (0.13 to 2.25)	0 fewer per 1000 (from 1 fewer to 1 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; OR: odds ratio.

* No. of participants not reported by one study (Kuzneiwicz 2009), so numbers shown are only from one study for each outcome

a. Most of pooled effect provided by studies "B" or "C" with > 50% studies "C".

b. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

c. The studies enrolled preterm newborns (≥ 35 weeks) and they did not specify the proportion.

d. Most of pooled effects provided by studies "B" or "C" with < 50% studies "C".

e. Wide confidence interval crossing the line of no effect.

B.2 Preventive measures

EB table B.2.1: Timing of first bath to prevent hypothermia and its sequelae

Comparison 1: Delayed first bath (after 24 hours) compared with early first bath (at or before 24 hours)

Source: Priyadarshi M, Balachander B, Gupta S, Sankar MJ. Timing of bathing in term healthy newborns: a systematic review (submitted).

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Delayed first bath (after 24 hours)	Early first bath (at or before 24 hours)	Relative (95% CI)	Absolute (95% CI)		
Infant mortality												
1	observational studies	very serious ^a	not serious	not serious	not serious	none	68/298 (22.8%)	195/491 (39.7%)	RR 0.46 (0.28 to 0.76)	214 fewer per 1000 (from 286 fewer to 95 fewer)	⊕⊕○○ LOW	CRITICAL
Hypothermia												
1	observational studies	very serious ^a	not serious	not serious	not serious	none	23/330 (7.0%)	43/330 (13.0%)	RR 0.50 (0.28 to 0.88)	65 fewer per 1000 (from 94 fewer to 16 fewer)	⊕⊕○○ LOW	CRITICAL
Exclusive breastfeeding at discharge												
1	observational studies	very serious ^a	not serious	not serious	serious ^b	none	188/330 (57.0%)	205/330 (62.1%)	RR 0.81 (0.58 to 1.12)	118 fewer per 1000 (from 261 fewer to 75 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "C".

b. Wide confidence interval crossing the line of no effect.

Comparison 2: Delayed first bath (after 6 hours; i.e. at or after 9, 12 or 24 hours) compared with early first bath (at or before 6 hours)

Source: Priyadarshi M, Balachander B, Gupta S, Sankar MJ. Timing of bathing in term healthy newborns: a systematic review (submitted).

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Delayed first bath (any time after 6 hours)	Early first bath (at or before 6 hours)	Relative (95% CI)	Absolute (95% CI)		
Neonatal mortality – after 6 hours vs at or before 6 hours												
1	observational studies	very serious ^a	not serious	not serious	serious ^b	none	Not available	Not available	RR 0.71 (0.30 to 1.67)	1 fewer per 1000 (from 2 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Hypothermia – at 9, 12 or 24 hours or more vs at or before 6 hours												
4	observational studies	very serious ^a	not serious	not serious	not serious	none	141/1434 (9.8%)	212/1277 (16.6%)	RR 0.47 (0.36 to 0.61)	88 fewer per 1000 (from 106 fewer to 65 fewer)	⊕⊕○○ LOW	CRITICAL
Hypoglycaemia – at 9, 12 or 24 hours or more vs at or before 6 hours												
3	observational studies	very serious ^a	not serious	not serious	not serious	none	27/1420 (1.9%)	67/1355 (4.9%)	RR 0.39 (0.23 to 0.66)	30 fewer per 1000 (from 38 fewer to 17 fewer)	⊕⊕○○ LOW	CRITICAL
Exclusive breastfeeding – at 9, 12 or 24 hours or more vs at or before 6 hours												
6	observational studies	very serious ^a	not serious	not serious	not serious	none	2554/4018 (63.6%)	1606/2750 (58.4%)	RR 1.20 (1.08 to 1.34)	117 more per 1000 (from 47 more to 199 more)	⊕⊕○○ LOW	CRITICAL

CI: confidence interval; RR: risk ratio.

a. Most of the pooled effect provided by studies “C”.

b. Wide confidence interval crossing the line of no effect.

EB table B.2.2: Use of emollients for the prevention of skin conditions

Comparison: Topical emollients compared with no intervention or skin care without emollients

Source: Priyadarshi M, Balachander B, Gupta S, Sankar MJ. Emollients application in term healthy newborns: a systematic review (submitted).

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Topical emollients	No emollients	Relative (95% CI)	Absolute (95% CI)		
Atopic dermatitis												
2	randomized trials	serious ^a	not serious	not serious	serious ^b	none	89/695 (12.8%)	70/713 (9.8%)	RR 1.29 (0.96 to 1.72)	28 more per 1000 (from 4 fewer to 71 more)	⊕⊕○○ LOW	CRITICAL
Food allergy												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	13/118 (11.0%)	15/115 (13.0%)	RR 0.84 (0.42 to 1.70)	21 fewer per 1000 (from 76 fewer to 91 more)	⊕○○○ VERY LOW	CRITICAL
Allergic sensitization – food allergens												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,d}	none	72/119 (60.5%)	53/115 (46.1%)	RR 1.31 (1.03 to 1.68)	143 more per 1000 (from 14 more to 313 more)	⊕○○○ VERY LOW	CRITICAL
Allergic sensitization – inhalation												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	11/119 (9.2%)	11/115 (9.6%)	RR 0.97 (0.44 to 2.14)	3 fewer per 1000 (from 54 fewer to 109 more)	⊕○○○ VERY LOW	CRITICAL
Skin condition – dryness												
2	randomized trials	very serious ^e	not serious	not serious	very serious ^{b,d}	none	51/153 (33.3%)	62/141 (44.0%)	RR 0.74 (0.55 to 1.00)	114 fewer per 1000 (from 198 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Topical emollients	No emollients	Relative (95% CI)	Absolute (95% CI)		

Skin condition – skin problems

2	randomized trials	very serious ^e	not serious	not serious	serious ^d	none	83/152 (54.6%)	95/140 (67.9%)	RR 0.92 (0.81 to 1.05)	54 fewer per 1000 (from 129 fewer to 34 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	----------------	----------------	----------------------------------	---	------------------	----------

CI: confidence interval; RR: risk ratio.

a. Most of the pooled effect provided by studies “B”.

b. Wide confidence interval crossing the line of no effect.

c. Less than 30 events and less than 300 participants.

d. Less than 300 participants.

e. Most of the pooled effect provided by studies “C”.

EB table B.2.3: Application of chlorhexidine to the umbilical cord stump for the prevention of neonatal infection

Comparison: Routine application of chlorhexidine to the umbilical cord stump compared with dry cord care or usual care

Source: Chlorhexidine Umbilical Review Group. Efficacy and safety of umbilical cord cleansing with chlorhexidine in neonates – an individual participant data (IPD) meta-analysis (in preparation).

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Routine application of chlorhexidine to the umbilical cord stump	Dry cord care or usual care	Relative (95% CI)	Absolute (95% CI)		
Neonatal mortality (ITT analysis)												
5	randomized trials	not serious	not serious	serious ^a	not serious	none	1562/70 491 (2.2%)	1464/65 829 (2.2%)	OR 0.90 (0.78 to 1.04)	2 fewer per 1000 (from 5 fewer to 1 more)	⊕⊕⊕○ MODERATE	CRITICAL
Neonatal mortality (ITT analysis) – mortality rate ≥ 30 per 1000 live births												
3	randomized trials	not serious	serious ^b	serious ^c	serious ^d	none	1091/33 696 (3.2%)	980/27 589 (3.6%)	OR 0.83 (0.68 to 1.03)	6 fewer per 1000 (from 11 fewer to 1 more)	⊕○○○ VERY LOW	CRITICAL
Neonatal mortality (ITT analysis) – mortality rate < 30 per 1000 live births												
2	randomized trials	not serious	not serious	not serious	not serious	none	471/36 522 (1.3%)	484/38 240 (1.3%)	OR 0.99 (0.79 to 1.25)	0 fewer per 1000 (from 3 fewer to 3 more)	⊕⊕⊕⊕ HIGH	CRITICAL
Neonatal mortality (ITT analysis) by place of birth – home												
5	randomized trials	not serious	serious ^b	serious ^e	serious ^d	none	1032/44 621 (2.3%)	845/39 049 (2.2%)	OR 0.86 (0.68 to 1.09)	3 fewer per 1000 (from 7 fewer to 2 more)	⊕○○○ VERY LOW	CRITICAL
Neonatal mortality (ITT analysis) by place of birth – facility												
5	randomized trials	not serious	not serious	serious ^e	not serious	none	432/25 000 (1.7%)	430/25 644 (1.7%)	OR 0.95 (0.81 to 1.10)	1 fewer per 1000 (from 3 fewer to 2 more)	⊕⊕⊕○ MODERATE	CRITICAL
Neonatal mortality (ITT analysis) – non-hygienic applications to umbilical cord stump												
5	randomized trials	not serious	not serious	serious ^a	not serious	none	173/11 294 (1.5%)	338/16 523 (2.0%)	OR 0.63 (0.50 to 0.79)	7 fewer per 1000 (from 10 fewer to 4 fewer)	⊕⊕⊕○ MODERATE	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Routine application of chlorhexidine to the umbilical cord stump	Dry cord care or usual care	Relative (95% CI)	Absolute (95% CI)		

Neonatal mortality (ITT analysis) – no non-hygienic applications to umbilical cord stump

5	randomized trials	not serious	not serious	serious ^a	not serious	none	1562/70 491 (2.2%)	1464/65 829 (2.2%)	OR 0.89 (0.77 to 1.03)	2 fewer per 1000 (from 5 fewer to 1 more)	⊕⊕⊕○ MODERATE	CRITICAL
---	-------------------	-------------	-------------	----------------------	-------------	------	--------------------	--------------------	----------------------------------	---	------------------	----------

Post-24-hour neonatal mortality (ITT analysis)

5	randomized trials	not serious	not serious	serious ^a	not serious	none	994/69 923 (1.4%)	949/65 314 (1.5%)	OR 0.91 (0.82 to 1.02)	1 fewer per 1000 (from 3 fewer to 0 fewer)	⊕⊕⊕○ MODERATE	CRITICAL
---	-------------------	-------------	-------------	----------------------	-------------	------	-------------------	-------------------	----------------------------------	--	------------------	----------

Omphalitis (ITT analysis) – moderate omphalitis

5	randomized trials	not serious	not serious	serious ^a	not serious	none	2263/71 570 (3.2%)	3405/66 372 (5.1%)	OR 0.77 (0.71 to 0.83)	11 fewer per 1000 (from 14 fewer to 9 fewer)	⊕⊕⊕○ MODERATE	CRITICAL
---	-------------------	-------------	-------------	----------------------	-------------	------	--------------------	--------------------	----------------------------------	--	------------------	----------

Omphalitis (ITT analysis) – severe omphalitis

5	randomized trials	not serious	serious ^b	serious ^a	not serious	none	1311/71 570 (1.8%)	2067/66 372 (3.1%)	OR 0.55 (0.39 to 0.76)	14 fewer per 1000 (from 19 fewer to 7 fewer)	⊕⊕○○ LOW	CRITICAL
---	-------------------	-------------	----------------------	----------------------	-------------	------	--------------------	--------------------	----------------------------------	--	-------------	----------

Possible serious bacterial infection (PSBI) (ITT analysis) – any PSBI

5	randomized trials	not serious	serious ^b	serious ^a	not serious	none	6846/71 719 (9.5%)	8057/66 223 (12.2%)	OR 0.91 (0.76 to 1.10)	10 fewer per 1000 (from 26 fewer to 11 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	-------------	----------------------	----------------------	-------------	------	--------------------	---------------------	----------------------------------	--	-------------	----------

PSBI – more specific PSBI

5	randomized trials	not serious	serious ^b	serious ^a	not serious	none	1868/71 719 (2.6%)	2103/66 223 (3.2%)	OR 0.91 (0.75 to 1.11)	3 fewer per 1000 (from 8 fewer to 3 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	-------------	----------------------	----------------------	-------------	------	--------------------	--------------------	----------------------------------	---	-------------	----------

PSBI – more severe PSBI

5	randomized trials	not serious	serious ^b	not serious ^a	not serious	none	2030/76 889 (2.6%)	1941/61 053 (3.2%)	OR 0.93 (0.83 to 1.10)	2 fewer per 1000 (from 5 fewer to 3 more)	⊕⊕⊕○ MODERATE	CRITICAL
---	-------------------	-------------	----------------------	--------------------------	-------------	------	--------------------	--------------------	----------------------------------	---	------------------	----------

CI: confidence interval; ITT: intention to treat; OR: odds ratio.

a. 80% of births occurred at home, 30% of babies were of low birthweight, three trials with infant mortality rate $\geq 30/1000$ (downgraded by one level for the combination of these factors).

b. Statistical heterogeneity ($I^2 \geq 60\%$).

c. 80% of births occurred at home, 30% of babies were of low birthweight (downgraded by one level for the combination of both factors).

d. Wide confidence interval crossing the line of no effect.

e. 30% of babies were of low birthweight, three trials with infant mortality rate of $\geq 30/1000$ (downgraded by one level for the combination of these factors).

EB table B.2.4: Sleeping position for the prevention of sudden infant death syndrome

Comparison: Supine (back) sleep position compared with non-supine (prone or side) sleep position

Source: Priyadarshi M, Balachander B, Sankar MJ. Effect of sleep position in term healthy newborns on neonatal mortality and sudden infant death syndrome (SIDS): a systematic review (submitted).

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Sleeping in a supine position	Sleeping in a non-supine (prone or side) position	Relative (95% CI)	Absolute (95% CI)		
Sudden infant death syndrome (SIDS) in infants < 1 year of age – supine vs non-supine												
26	observational studies	very serious ^a	serious ^b	not serious	not serious	publication bias strongly suspected ^c	4720 cases, 54 612 controls		OR 0.51 (0.42 to 0.61)	48 fewer per 1000 (from 58 fewer to 38 fewer)	⊕○○○ VERY LOW	CRITICAL
Sudden unexpected death in infancy (SUDI) – supine vs non-supine												
1	observational study	very serious ^d	not serious	not serious	not serious	none	126 cases, 258 controls		OR 0.39 (0.23 to 0.65)	219 fewer per 1000 (from 313 fewer to 106 fewer)	⊕⊕○○ LOW	CRITICAL
Unexplained SIDS or severe-ALTE in the neonatal period – supine vs non-supine												
1	observational study	very serious ^a	not serious	not serious	serious ^e	none	29 cases, 90 controls		OR 0.16 (0.03 to 0.82)	232 fewer per 1000 (from 282 fewer to 39 fewer)	⊕○○○ VERY LOW	CRITICAL
Gross motor development at 6 months of age – supine vs prone (odds of being 0.5 SD below mean on the Gross Motor Scale, assessed with DDST at 6 months of age)												
1	observational studies	serious ^g	not serious	not serious	not serious	none	-/1777	-/320	OR 1.67 (1.22 to 2.27)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕⊕⊕○ MODERATE	CRITICAL
Gross motor development at 6 months of age – supine vs side (odds of being 0.5 SD below mean on the Gross Motor Scale, assessed with DDST at 6 months of age)												
1	observational studies	serious ^g	not serious	not serious	not serious	none	-/1777	-/6235	OR 1.02 (0.91 to 1.15)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕⊕⊕○ MODERATE	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Sleeping in a supine position	Sleeping in a non-supine (prone or side) position	Relative (95% CI)	Absolute (95% CI)		

Gross motor development at 18 months of age – supine vs prone (odds of being 0.5 SD below mean on the Gross Motor Scale, assessed with DDST at 18 months of age)

1	observational studies	serious ^g	not serious	not serious	serious ^h	none	-/1611	-/308	OR 1.16 (0.96 to 1.43)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕⊕○○ LOW	CRITICAL
---	-----------------------	----------------------	-------------	-------------	----------------------	------	--------	-------	----------------------------------	--	-------------	----------

Gross motor development at 18 months of age – supine vs side (odds of being 0.5 SD below mean on the Gross Motor Scale, assessed with DDST at 18 months of age)

1	observational studies	serious ^g	not serious	not serious	serious ^h	none	-/1611	-/5892	OR 1.12 (0.86 to 1.45)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕⊕○○ LOW	CRITICAL
---	-----------------------	----------------------	-------------	-------------	----------------------	------	--------	--------	----------------------------------	--	-------------	----------

Hospital admissions related to ALTE within 6 months of age – supine vs non-supine

1	observational study	very serious ^a	not serious	not serious	very serious ^{f,h}	none	1/1745 (0.1%)	5/1984 (0.3%)	OR 0.230 (0.005 to 2.040)	2 fewer per 1000 (from 3 fewer to 3 more)	⊕○○○ VERY LOW	CRITICAL
---	---------------------	---------------------------	-------------	-------------	-----------------------------	------	------------------	------------------	-------------------------------------	---	------------------	----------

Positional plagiocephaly within 28 weeks of age – supine vs non-supine

2	observational studies	very serious ^a	not serious	not serious	not serious	none	185/364 (50.8%)	17/107 (15.9%)	OR 6.53 (3.39 to 12.57)	393 more per 1000 (from 231 more to 545 more)	⊕⊕○○ LOW	CRITICAL
---	-----------------------	---------------------------	-------------	-------------	-------------	------	--------------------	-------------------	-----------------------------------	---	-------------	----------

ALTE: apparently life-threatening event; CI: confidence interval; DDST: Denver Developmental Screening Test; OR: odds ratio.

a. Most of the pooled effect provided by studies “C”.

b. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \geq 0.05$).

c. Evident asymmetry in funnel plot.

d. The included study used unadjusted OR and was considered as having very serious risk of bias.

e. Less than 300 newborns in continuous outcomes or less than 400 newborns in dichotomous outcomes.

f. Less than 30 events.

g. The pooled effect provided by study “B”.

h. Wide confidence interval crossing the line of no effect.

B.3 Nutritional interventions

EB table B.3.1: Neonatal vitamin A supplementation

Comparison: Neonatal vitamin A supplementation compared with placebo or no vitamin A supplementation

Source: Imdad A, Rehman F, Davis E, Ranjit D, Surin GSS, Attia SL, et al. Effects of neonatal nutrition interventions on neonatal mortality and child health and development outcomes: a systematic review. Campbell Syst Rev. 2019;17(1):e1141. doi:10.1002/cl2.1021.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin A	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		
All-cause neonatal mortality												
6	randomized trials	not serious	not serious	not serious	not serious	none	-/63 371	-/63 177	RR 0.99 (0.90 to 1.08)	not estimable ^a	⊕⊕⊕⊕ HIGH	CRITICAL
All-cause infant mortality at 6 months of age												
12	randomized trials	not serious	not serious	not serious	not serious	publication bias strongly suspected ^b	-/77 505	-/77 435	RR 0.98 (0.89 to 1.07)	not estimable ^a	⊕⊕⊕○ MODERATE	CRITICAL
All-cause infant mortality at 12 months of age												
8	randomized trials	not serious	not serious	not serious	not serious	none	-/60 071	-/58 305	RR 1.04 (0.94 to 1.14)	not estimable ^a	⊕⊕⊕⊕ HIGH	CRITICAL
Adverse effects – bulging fontanel												
6	randomized trials	not serious	serious ^c	not serious	not serious	none	-/50 459	-/49 797	RR 1.53 (1.12 to 2.09)	not estimable ^a	⊕⊕⊕○ MODERATE	CRITICAL
Adverse effects – vomiting												
5	randomized trials	not serious	not serious	not serious	not serious	none	-/49 904	-/49 678	RR 1.00 (0.93 to 1.07)	not estimable ^a	⊕⊕⊕⊕ HIGH	CRITICAL

CI: confidence interval; RR: risk ratio.

a. It was not possible to calculate the absolute risks because data on the number of events were not available.

b. Evident asymmetry in the funnel plot.

c. Statistical heterogeneity (I^2 65%).

EB table B.3.2: Vitamin D supplementation for breastfed, term infants

Comparison: Vitamin D supplementation for breastfed, term infants compared with placebo or no supplementation

Source: Tan ML, Abrams SA, Osborn DA. Vitamin D supplementation for term breastfed infants to prevent vitamin D deficiency and improve bone health. Cochrane Database Syst Rev. 2020;(12):CD013046.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin D	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L)												
4	randomized trials	serious ^a	not serious	not serious	serious ^b	none	29/132 (22.0%)	64/142 (45.1%)	RR 0.57 (0.41 to 0.80)	194 fewer per 1000 (from 266 fewer to 90 fewer)	⊕⊕○○ LOW	CRITICAL
Serum 25(OH) vitamin D level at latest time reported to 6 months of age												
6	randomized trials	serious ^a	not serious	not serious	serious ^c	none	159	175	-	MD 22.63 higher (17.05 higher to 28.21 higher)	⊕⊕○○ LOW	CRITICAL
Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L)												
2	randomized trials	serious ^d	not serious	not serious	very serious ^{e,g}	none	5/58 (8.6%)	14/64 (21.9%)	RR 0.41 (0.16 to 1.05)	129 fewer per 1000 (from 184 fewer to 11 more)	⊕○○○ VERY LOW	CRITICAL
Nutritional rickets – biochemical												
2	randomized trials	serious ^d	not serious	not serious	very serious ^{b,h}	none	0/17 (0.0%)	0/17 (0.0%)	not estimable	not estimable	⊕○○○ VERY LOW	CRITICAL
Size at latest time measured – weight												
2	randomized trials	serious ^d	not serious	not serious	very serious ^e	none	71	72	-	MD 123.63 higher (170.02 lower to 417.28 higher)	⊕○○○ VERY LOW	CRITICAL
Size at latest time measured – length												
3	randomized trials	serious ^d	not serious	not serious	very serious ^{d,e}	none	77	79	-	MD 0.73 higher (0.11 lower to 1.57 higher)	⊕○○○ VERY LOW	CRITICAL
Size at latest time measured – head circumference at 6 months of age												
1	randomized trials	serious ^d	not serious	not serious	very serious ^{d,e}	none	52	53	-	MD 0 (0.6 lower to 0.6 higher)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin D	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		

Bone mineral content at the end of intervention

2	randomized trials	serious ^d	serious ^f	not serious	very serious ^{c,e}	none	28	28	-	MD 3.93 higher (2.42 lower to 10.27 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	----------------------	-------------	-----------------------------	------	----	----	---	---	------------------	----------

Adverse effect – hypercalcaemia

1	randomized trials	serious ^d	not serious	not serious	very serious ^{e,g}	none	8/47 (17.0%)	6/51 (11.8%)	RR 1.45 (0.54 to 3.86)	53 more per 1000 (from 54 fewer to 336 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	--------------	--------------	----------------------------------	--	------------------	----------

Adverse effect – others

3	randomized trials	very serious ⁱ	not serious	not serious	very serious ^{e,g}	none	1/25 (4.0%)	0/24 (0.0%)	RR 3.00 (0.14 to 64.26)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-----------------------------	------	-------------	-------------	-----------------------------------	--	------------------	----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Most of the pooled effect provided by studies “B”.

b. Less than 300 babies.

c. Less than 400 babies.

d. The effect provided by studies “B”.

e. Wide confidence interval crossing the line of no effect.

f. Statistical heterogeneity (I^2 94%).

g. Less than 300 participants and less than 30 events.

h. No events.

i. Most of the pooled effect provided by study “C”.

Subgroup analysis by neonatal risk status (high risk or low risk)

Source: Tan ML, Abrams SA, Osborn DA. Vitamin D supplementation for term breastfed infants to prevent vitamin D deficiency and improve bone health. Cochrane Database Syst Rev. 2020;(12):CD013046.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin D	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) – high-risk infants												
3	randomized trials	serious ^a	not serious	not serious	serious ^c	none	25/64 (39.1%)	42/70 (60.0%)	RR 0.65 (0.46 to 0.94)	210 fewer per 1000 (from 324 fewer to 36 fewer)	⊕⊕○○ LOW	CRITICAL
Serum 25(OH) vitamin D level at latest time reported to 6 months of age – high-risk infants												
3	randomized trials	serious ^a	not serious	not serious	serious ^c	none	64	70	-	MD 18.24 higher (9.39 higher to 27.09 higher)	⊕⊕○○ LOW	CRITICAL
Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L) – high-risk infants												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,d,e}	none	5/58 (8.6%)	14/64 (21.9%)	RR 0.41 (0.16 to 1.05)	129 fewer per 1000 (from 184 fewer to 11 more)	⊕○○○ VERY LOW	CRITICAL
Nutritional rickets: biochemical – high risk infants: D2 400 IU/day from birth to 6 months of age; all seasons												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,f}	none	0/9 (0.0%)	0/9 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) – low risk infants												
1	randomized trials	serious ^a	not serious	not serious	serious ^c	none	4/68 (5.9%)	22/72 (30.6%)	RR 0.19 (0.07 to 0.53)	248 fewer per 1000 (from 284 fewer to 144 fewer)	⊕⊕○○ LOW	CRITICAL
Serum 25(OH) vitamin D level at latest time reported to 6 months of age – low risk infants												
3	randomized trials	serious ^a	not serious	not serious	serious ^c	none	95	105	-	MD 25.53 higher (18.34 higher to 32.72 higher)	⊕⊕○○ LOW	CRITICAL
Nutritional rickets: biochemical – low-risk infants: D2 400 IU/day from birth to 6 months of age												
1	randomized trials	very serious ^g	not serious	not serious	very serious ^{b,f}	none	0/8 (0.0%)	0/8 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin D	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		

Bone mineral content at the end of intervention – low-risk infants; D2 400 IU/day from birth to 3 months of age

1	randomized trials	serious ^a	not serious	not serious	serious ^c	none	9	9	-	MD 15 higher (6.68 higher to 23.32 higher)	⊕⊕○○ LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	----------------------	------	---	---	---	--	-------------	----------

Bone mineral content at the end of intervention – low-risk infants; D2 400 IU/day from birth to 6 months of age

1	randomized trials	serious ^a	not serious	not serious	serious ^c	none	19	19	-	MD 11.5 lower (21.32 lower to 1.68 lower)	⊕⊕○○ LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	----------------------	------	----	----	---	---	-------------	----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. The pooled effect provided by studies “B”.

b. Less than 300 babies.

c. Less than 400 babies.

d. Less than 30 events.

e. Wide confidence interval crossing the line of no effect.

f. No events.

g. The pooled effect provided by studies “C”.

Subgroup analyses by neonatal active form (Vitamin D2 or D3), dosage (single oral dose of 50 000 IU or 400 IU daily), time of administration (from birth, from one month age), and duration of supplementation (single, oral 50 000 IU at birth, 1–2 months or > 6 months)

Source: Tan ML, Abrams SA, Osborn DA. Vitamin D supplementation for term breastfed infants to prevent vitamin D deficiency and improve bone health. Cochrane Database Syst Rev. 2020;(12):CD013046.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin D	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) – vitamin D3												
3	randomized trials	serious ^a	not serious	not serious	serious ^b	none	27/126 (21.4%)	60/136 (44.1%)	RR 0.58 (0.40 to 0.82)	185 fewer per 1000 (from 265 fewer to 79 fewer)	⊕⊕○○ LOW	CRITICAL
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) – vitamin D2												
1	randomized trials	very serious ^d	not serious	not serious	very serious ^{b,c,e}	none	2/6 (33.3%)	4/6 (66.7%)	RR 0.50 (0.14 to 1.77)	333 fewer per 1000 (from 573 fewer to 513 more)	⊕○○○ VERY LOW	CRITICAL
Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L) – vitamin D3												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	5/58 (8.6%)	14/64 (21.9%)	not estimable	90 more per 1000 (from 20 fewer to 200 more)	⊕○○○ VERY LOW	CRITICAL
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) by dosage – vitamin D 400 IU/day												
3	randomized trials	serious ^a	not serious	not serious	serious ^b	none	25/121 (20.7%)	58/132 (43.9%)	RR 0.56 (0.39 to 0.81)	193 fewer per 1000 (from 268 fewer to 83 fewer)	⊕⊕○○ LOW	CRITICAL
Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L) by dosage – vitamin D 400 IU/day												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	5/47 (10.6%)	14/54 (25.9%)	not estimable	150 more per 1000 (from 10 more to 300 more)	⊕○○○ VERY LOW	CRITICAL
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) by dosage – single, oral vitamin D 50 000 IU at birth												
1	randomized trials	serious ^a	not serious	not serious	serious ^{b,c,e}	none	4/11 (36.4%)	6/10 (60.0%)	RR 0.61 (0.24 to 1.54)	234 fewer per 1000 (from 456 fewer to 324 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin D	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		
Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L) by dosage – single, oral vitamin D 50 000 IU at birth												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e,f}	none	0/11 (0.0%)	0/10 (0.0%)	not estimable	0 fewer per 1000 (from 170 fewer to 170 more)	⊕○○○ VERY LOW	CRITICAL
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) by timing of commencement – from birth												
3	randomized trials	serious ^a	not serious	not serious	serious ^b	none	25/64 (39.1%)	42/70 (60.0%)	RR 0.65 (0.46 to 0.94)	210 fewer per 1000 (from 324 fewer to 36 fewer)	⊕⊕○○ LOW	CRITICAL
Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L) by timing of commencement – from birth												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	5/58 (8.6%)	14/64 (21.9%)	not estimable	90 more per 1000 (from 20 fewer to 200 more)	⊕○○○ VERY LOW	CRITICAL
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) by timing of commencement – from 1 month of age												
1	randomized trials	serious ^a	not serious	not serious	serious ^{b,c}	none	4/68 (5.9%)	22/72 (30.6%)	RR 0.19 (0.07 to 0.53)	248 fewer per 1000 (from 284 fewer to 144 fewer)	⊕⊕○○ LOW	CRITICAL
Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) by duration of supplementation – single, oral vitamin D 50 000 IU at birth												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,d}	none	4/11 (36.4%)	6/10 (60.0%)	RR 0.61 (0.24 to 1.54)	234 fewer per 1000 (from 456 fewer to 324 more)	⊕○○○ VERY LOW	CRITICAL
Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L) by duration of supplementation – single, oral vitamin D 50 000 IU at birth												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e,f}	none	0/11 (0.0%)	0/10 (0.0%)	not estimable	0 fewer per 1000 (from 170 fewer to 170 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Vitamin D	Placebo or no supplementation	Relative (95% CI)	Absolute (95% CI)		

Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) by duration of supplementation – 1–2 months

1	randomized trials	very serious ^c	not serious	not serious	very serious ^{a,b,d}	none	2/6 (33.3%)	4/6 (66.7%)	RR 0.50 (0.14 to 1.77)	333 fewer per 1000 (from 573 fewer to 513 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-------------------------------	------	-------------	-------------	----------------------------------	---	------------------	----------

Vitamin D insufficiency (25(OH) vitamin D < 50 nmol/L) by duration of supplementation – > 6 months

2	randomized trials	serious ^a	not serious	not serious	serious ^c	none	23/115 (20.0%)	54/126 (42.9%)	RR 0.57 (0.39 to 0.83)	184 fewer per 1000 (from 261 fewer to 73 fewer)	⊕⊕○○ LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	----------------------	------	-------------------	----------------	----------------------------------	---	-------------	----------

Vitamin D deficiency (25(OH) vitamin D < 30 nmol/L) by duration of supplementation – > 6 months

1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	5/47 (10.6%)	14/54 (25.9%)	not estimable	150 more per 1000 (from 10 more to 300 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-------------------------------	------	--------------	---------------	---------------	--	------------------	----------

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by studies "B".

b. Less than 300 babies.

c. Less than 30 events.

d. The pooled effect provided by studies "C".

e. Wide confidence interval crossing the line of no effect.

f. No events.

B.4 Infant growth and development

EB table B.4.1: Whole-body massage

Comparison: Whole-body massage compared with no massage

Source: Priyadarshi M, Kumar V, Balachander B, Gupta S, Sankar MJ. Effect of whole-body massage on growth and neurodevelopment in term healthy newborns: a systematic review (submitted).

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Whole-body massage	No massage	Relative (95% CI)	Absolute (95% CI)		
Weight (g) – end of intervention period												
16	randomized trials	very serious ^a	serious ^b	not serious	not serious	none	1072	1076	-	MD 343.43 higher (260.73 higher to 426.12 higher)	⊕○○○ VERY LOW	CRITICAL
Weight (g) – follow-up at 8–12 months												
2	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	74	83	-	MD 455.07 higher (86.33 higher to 823.8 higher)	⊕○○○ VERY LOW	CRITICAL
Length (cm) – end of intervention period												
8	randomized trials	very serious ^a	not serious	not serious	not serious	none	630	630	-	MD 1.53 higher (1.37 higher to 1.70 higher)	⊕⊕○○ LOW	CRITICAL
Length (cm) – follow-up at 12 months												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,d}	none	54	62	-	MD 0.71 higher (0.15 lower to 1.57 higher)	⊕○○○ VERY LOW	CRITICAL
Head circumference (cm) – end of intervention period												
6	randomized trials	very serious ^a	serious ^b	not serious	not serious	none	500	500	-	MD 0.85 higher (0.57 higher to 1.14 higher)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Whole-body massage	No massage	Relative (95% CI)	Absolute (95% CI)		
Head circumference (cm) – follow-up at 6 months												
1	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	55	60	-	MD 1.31 higher (0.55 higher to 2.07 higher)	⊕○○○ VERY LOW	CRITICAL
Bilirubin levels at 4 days												
4	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	168	177	-	MD 31.75 lower (40.05 lower to 23.46 lower)	⊕○○○ VERY LOW	CRITICAL
Sleep duration over 24-hour period (hours/day) – end of intervention												
3	randomized trials	very serious ^a	serious ^b	not serious	not serious	none	266	268	-	MD 0.62 higher (0.12 higher to 1.12 higher)	⊕○○○ VERY LOW	CRITICAL
Sleep duration over 24-hour period (hours/day) – follow-up at 6 months												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,d}	none	61	63	-	MD 0.08 higher (0.48 lower to 0.64 higher)	⊕○○○ VERY LOW	CRITICAL
Psychomotor Development Indices (PDI) meta-analysis post-intervention – end of intervention period												
3	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	234	154	-	SMD 0.39 higher (0.6 higher to 0.18 higher)	⊕○○○ VERY LOW	CRITICAL
Psychomotor Development Indices (PDI) meta-analysis post-intervention – follow-up at 24 months												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,d}	none	20	21	-	SMD 7.52 higher (1.49 lower to 16.53 higher)	⊕○○○ VERY LOW	CRITICAL
Mental Development Indices (MDI) meta-analysis post-intervention – end of intervention period												
3	randomized trials	very serious ^a	serious ^b	not serious	very serious ^{c,d}	none	234	154	-	SMD 0.29 higher (0.18 lower to 0.77higher)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Whole-body massage	No massage	Relative (95% CI)	Absolute (95% CI)		
Mental Development Indices (MDI) meta-analysis – follow-up at 24 months												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,d}	none	20	21	-	SMD 8.59 higher (1.62 lower to 18.80 higher)	⊕○○○ VERY LOW	CRITICAL
Gross motor development at end of intervention (Gesell development quotient/Capital Institute mental checklist)												
2	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	117	120	-	SMD 0.44 higher (0.18 higher to 0.7 higher)	⊕○○○ VERY LOW	CRITICAL
Fine motor development at end of intervention (Gesell development quotient/Capital Institute mental checklist)												
2	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	117	120	-	SMD 0.61 higher (0.35 higher to 0.87 higher)	⊕○○○ VERY LOW	CRITICAL
Language at end of intervention (Gesell development quotient/Capital Institute mental checklist)												
2	randomized trials	very serious ^a	serious ^b	not serious	very serious ^{c,d}	none	117	120	-	SMD 0.82 higher (0.03 lower to 1.67 higher)	⊕○○○ VERY LOW	CRITICAL
Personal-social behaviour at end of intervention (Gesell development quotient/Capital Institute mental checklist)												
2	randomized trials	very serious ^a	serious ^b	not serious	serious ^c	none	117	120	-	SMD 0.9 higher (0.18 higher to 1.61 higher)	⊕○○○ VERY LOW	CRITICAL
Gross motor development at 12 months (Gesell development quotient)												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{c,d}	none	54	62	-	MD 2.85 higher (2.48 lower to 8.18 higher)	⊕○○○ VERY LOW	CRITICAL
Fine motor development at 12 months (Gesell development quotient)												
1	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	54	62	-	MD 8.12 higher (4.57 higher to 11.67 higher)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Whole-body massage	No massage	Relative (95% CI)	Absolute (95% CI)		

Language at 12 months (Gesell development quotient)

1	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	54	62	-	MD 7.9 higher (4.1 higher to 11.7 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	----	----	---	---	------------------	----------

Personal-social behaviour at 12 months (Gesell development quotient)

1	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	54	62	-	MD 6.19 higher (2.55 higher to 9.83 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	----	----	---	---	------------------	----------

Maternal Attachment Inventory score

1	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	57	60	-	MD 5.77 higher (0.95 higher to 10.59 higher)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	----	----	---	--	------------------	----------

Crying or fussing time – end of intervention

3	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	136	135	-	MD 0.36 lower (0.16 lower to 0.56 lower)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	-----	-----	---	--	------------------	----------

Crying or fussing time – follow-up at 6 months

1	randomized trials	very serious ^a	not serious	not serious	serious ^c	none	61	63	-	MD 0.15 lower (0.01 lower to 0.29 lower)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	----------------------	------	----	----	---	--	------------------	----------

CI: confidence interval; MD: mean difference; SMD: standardized mean difference.

a. The pooled effect provided by studies C.

b. Severe, unexplained, heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 < 0.05$).

c. Less than 400 participants.

d. Wide confidence interval crossing the line of no effect.

C. HEALTH SYSTEMS AND HEALTH PROMOTION INTERVENTIONS

EB table C.1: Schedules for postnatal care contacts

Comparison 1: Schedules involving four postnatal home visits (3, 7, 28 and 42 days after birth) compared with one postnatal home visit (at about 42 days after birth)

Source: Yonemoto N, Nagai S, Mori R. Schedules for home visits in the early postpartum period. Cochrane Database Syst Rev. 2021;(7):CD009326.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Schedules involving four postnatal home visits	Schedules involving one postnatal home visit	Relative (95% CI)	Absolute (95% CI)		
Maternal health problems (as identified by a doctor)												
1	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	22/178 (12.4%)	24/174 (13.8%)	RR 0.90 (0.52 to 1.54)	14 fewer per 1000 (from 66 fewer to 74 more)	⊕○○○ VERY LOW	CRITICAL
Neonatal mortality												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	2/208 (1.0%)	4/200 (2.0%)	RR 0.48 (0.09 to 2.60)	10 fewer per 1000 (from 18 fewer to 32 more)	⊕○○○ VERY LOW	CRITICAL
Infant respiratory tract infection within 42 days												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	4/178 (2.2%)	10/174 (5.7%)	RR 0.39 (0.12 to 1.22)	35 fewer per 1000 (from 51 fewer to 13 more)	⊕○○○ VERY LOW	CRITICAL
Infant referral to paediatrician up to 42 days												
1	randomized trials	very serious ^a	not serious	not serious	not serious	none	30/178 (16.9%)	71/174 (40.8%)	RR 0.41 (0.28 to 0.60)	241 fewer per 1000 (from 294 fewer to 163 fewer)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Schedules involving four postnatal home visits	Schedules involving one postnatal home visit	Relative (95% CI)	Absolute (95% CI)		

Exclusive breastfeeding (last assessment up to 6 weeks)

1	randomized trials	very serious ^a	not serious	not serious	not serious	none	169/178 (94.9%)	146/174 (83.9%)	RR 1.13 (1.05 to 1.22)	109 more per 1000 (from 42 more to 185 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	---------------------------	-------------	-------------	-------------	------	-----------------	-----------------	----------------------------------	--	-------------	----------

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "C".

b. Wide confidence interval crossing the line of no effect.

c. Less than 300 women and/or less than 30 events.

Comparison 2: Schedules involving two postnatal visits (3–5 and 10–14 days after birth) compared with one outpatient visit (10–14 days after birth)

Source: Yonemoto N, Nagai S, Mori R. Schedules for home visits in the early postpartum period. Cochrane Database Syst Rev. 2021;(7):CD009326.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Schedules involving two postnatal visits	Schedules involving one outpatient visit	Relative (95% CI)	Absolute (95% CI)		
Discontinuation of breastfeeding (up to 30 days)												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,c}	none	18/95 (18.9%)	22/90 (24.4%)	RR 0.78 (0.45 to 1.35)	54 fewer per 1000 (from 134 fewer to 86 more)	⊕○○○ VERY LOW	CRITICAL
Any breastfeeding (last assessment up to 6 months)												
1	randomized trials	serious ^d	not serious	not serious	not serious	none	367/509 (72.1%)	326/491 (66.4%)	RR 1.09 (1.00 to 1.18)	60 more per 1000 (from 0 fewer to 120 more)	⊕⊕⊕○ MODERATE	CRITICAL

CI: confidence interval; RR: risk ratio.

a. The pooled effect provided by study "C".

b. Wide confidence interval crossing the line of no effect.

c. Less than 300 women and/or less than 30 events.

d. The pooled effect provided by study "B".

EB table C.2: Length of stay in health facilities after birth

Comparison 1: Early discharge following vaginal birth compared with usual discharge

Source: Jones E, Stewart F, Taylor B, Davis PG, Brown SJ. Early postnatal discharge from hospital for healthy mothers and term infants. Cochrane Database Syst Rev. 2021;(6):CD002958.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		
Postpartum depression within 6 months												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	5/263 (1.9%)	13/271 (4.8%)	RR 0.43 (0.15 to 1.19)	27 fewer per 1000 (from 41 fewer to 9 more)	⊕○○○ VERY LOW	CRITICAL
Maternal readmission within 6 weeks												
6	randomized trials	serious ^d	not serious	serious ^e	serious ^b	none	37/2213 (1.7%)	9/715 (1.3%)	RR 1.32 (0.58 to 3.02)	4 more per 1000 (from 5 fewer to 25 more)	⊕○○○ VERY LOW	CRITICAL
Women's satisfaction with postnatal care (continuous data)												
2	randomized trials	serious ^a	not serious	not serious	serious ^f	none	171	135	-	SMD 0.74 higher (0.5 higher to 0.98 higher)	⊕⊕○○ LOW	CRITICAL
Number of women who perceive their hospital stay to be too short												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,g}	none	2/41 (4.9%)	1/41 (2.4%)	RR 2.00 (0.19 to 21.21)	24 more per 1000 (from 20 fewer to 493 more)	⊕○○○ VERY LOW	CRITICAL
Number of women who perceive their hospital stay to be too long												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,g}	none	5/41 (12.2%)	9/41 (22.0%)	RR 0.56 (0.20 to 1.52)	97 fewer per 1000 (from 176 fewer to 114 more)	⊕○○○ VERY LOW	CRITICAL
Infant mortality within 28 days												
1	randomized trials	very serious ^h	not serious	serious ⁱ	very serious ^{b,c}	none	3/1667 (0.2%)	1/217 (0.5%)	RR 0.39 (0.04 to 3.74)	3 fewer per 1000 (from 4 fewer to 13 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		
Infant mortality within 1 year												
2	randomized trials	very serious ^h	not serious	serious ⁱ	very serious ^{b,g}	none	4/1716 (0.2%)	2/270 (0.7%)	RR 0.45 (0.07 to 2.77)	4 fewer per 1000 (from 7 fewer to 13 more)	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 7 days												
1	randomized trials	serious ^d	not serious	not serious	very serious ^{b,g}	none	1/50 (2.0%)	0/54 (0.0%)	RR 3.24 (0.13 to 77.63)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 28 days – mode of birth subgroups												
5	randomized trials	serious ^d	not serious	serious ^e	serious ^b	none	26/2160 (1.2%)	8/694 (1.2%)	RR 1.30 (0.55 to 3.09)	3 more per 1000 (from 5 fewer to 24 more)	⊕○○○ VERY LOW	CRITICAL
Women breastfeeding (exclusively or partially) at 6 weeks postpartum												
6	randomized trials	serious ^d	serious ^j	serious ^e	serious ^b	none	641/2388 (26.8%)	315/724 (43.5%)	RR 1.15 (0.90 to 1.47)	65 more per 1000 (from 44 fewer to 204 more)	⊕○○○ VERY LOW	CRITICAL
Women breastfeeding (exclusively or partially) at 12 weeks postpartum												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	141/213 (66.2%)	119/217 (54.8%)	RR 1.21 (1.03 to 1.41)	115 more per 1000 (from 16 more to 225 more)	⊕⊕⊕○ MODERATE	CRITICAL

CI: confidence interval; RR: risk ratio; SMD: standardized mean difference.

a. The pooled effect provided by studies “B”.

b. Wide confidence interval crossing the line of no effect.

c. Less than 30 events.

d. Most of the pooled effect provided by studies “B” or “C” with ≤ 50% of studies “B”.

e. Time of discharge from two studies (Hellman 1962 and Smith-Hanrahan 1995) was reported as over 72 hours.

f. Less than 400 women.

g. Less than 300 women and less than 30 events.

h. Most of the pooled effect provided by studies “B” or “C” with > 50% of studies “C”.

i. Time of discharge from one of the trials (Hellman 1962) was reported as over 72 hours.

j. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

Comparison 2: Early discharge following caesarean birth compared with usual discharge

Source: Jones E, Stewart F, Taylor B, Davis PG, Brown SJ. Early postnatal discharge from hospital for healthy mothers and term infants. Cochrane Database Syst Rev. 2021;(6):CD002958.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early	standard discharge	Relative (95% CI)	Absolute (95% CI)		
Maternal mortality												
2	randomized trials	serious ^a	not serious	not serious	very serious ^b	none	No maternal deaths within one year after childbirth among the 1545 women allocated to early discharge and 1653 women allocated to standard discharge				⊕○○○ VERY LOW	CRITICAL
Women reporting health problems in the first 6 weeks postpartum												
1	randomized trials	serious ^c	not serious	not serious	serious ^d	none	5/50 (10.0%)	60/150 (40.0%)	RR 0.25 (0.11 to 0.59)	300 fewer per 1000 (from 356 fewer to 164 fewer)	⊕⊕○○ LOW	CRITICAL
Postpartum depression within 6 months												
2	randomized trials	serious ^a	not serious	not serious	serious ^e	none	1172/1665 (70.4%)	917/1675 (54.7%)	RR 1.08 (0.44 to 2.64)	44 more per 1000 (from 307 fewer to 898 more)	⊕⊕○○ LOW	CRITICAL
Maternal readmission within 6 weeks												
4	randomized trials	serious ^a	not serious	not serious	serious ^e	none	62/1798 (3.4%)	59/1807 (3.3%)	RR 1.05 (0.74 to 1.49)	2 more per 1000 (from 8 fewer to 16 more)	⊕⊕○○ LOW	CRITICAL
Women who had extra contacts with healthcare professionals due to maternal health issues within 6 weeks												
2	randomized trials	not serious	not serious	not serious	serious ^e	none	22/231 (9.5%)	31/233 (13.3%)	RR 0.72 (0.43 to 1.20)	37 fewer per 1000 (from 76 fewer to 27 more)	⊕⊕⊕○ MODERATE	CRITICAL
Infant mortality within 28 days												
1	randomized trials	serious ^c	not serious	not serious	very serious ^f	none	0/1495 (0.0%)	0/1503 (0.0%)	Not estimable	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 7 days												
1	randomized trials	serious ^c	not serious	not serious	very serious ^{e,g}	none	6/72 (8.3%)	6/71 (8.5%)	RR 0.99 (0.33 to 2.91)	1 fewer per 1000 (from 57 fewer to 161 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early	standard discharge	Relative (95% CI)	Absolute (95% CI)		

Infants readmitted for neonatal morbidity within 28 days

4	randomized trials	serious ^a	not serious	not serious	not serious	none	163/1798 (9.1%)	104/1807 (5.8%)	RR 1.57 (1.24 to 1.99)	33 more per 1000 (from 14 more to 57 more)	⊕⊕⊕○ MODERATE	CRITICAL
---	-------------------	----------------------	-------------	-------------	-------------	------	-----------------	-----------------	----------------------------------	--	------------------	----------

Women breastfeeding (exclusively or partially) at 6 weeks postpartum

2	randomized trials	serious ^a	serious ^h	not serious	not serious	none	1091/1665 (65.5%)	1172/1675 (70.0%)	RR 0.99 (0.83 to 1.18)	7 fewer per 1000 (from 119 fewer to 126 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	----------------------	----------------------	-------------	-------------	------	-------------------	-------------------	----------------------------------	---	-------------	----------

CI: confidence interval; RR: risk ratio.

a. Most of the pooled effect provided by studies “B” or “C” with $\leq 50\%$ of studies “B”.

b. Not pooled.

c. The pooled effect provided by studies “B”.

d. Less than 300 women.

e. Wide confidence interval crossing the line of no effect.

f. No events.

g. Less than 30 events and less than 300 women.

h. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

Ad-hoc analyses by time of discharge and mode of birth

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		
Vaginal birth – Policy of discharge within 24 hours compared with any time later												
Women with probable postpartum depression within 6 months												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	2/213 (0.9%)	8/217 (3.7%)	RR 0.25 (0.05 to 1.19)	28 fewer per 1000 (from 35 fewer to 7 more)	⊕○○○ VERY LOW	CRITICAL
Women readmitted within 6 weeks												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	4/257 (1.6%)	5/261 (1.9%)	RR 0.82 (0.22 to 2.99)	3 fewer per 1000 (from 15 fewer to 38 more)	⊕○○○ VERY LOW	CRITICAL
Women satisfied with postnatal care – dichotomous data												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	170/172 (98.8%)	113/125 (90.4%)	RR 1.09 (1.03 to 1.16)	81 more per 1000 (from 27 more to 145 more)	⊕⊕⊕○ MODERATE	CRITICAL
Women satisfied with postnatal care – continuous data												
1	randomized trials	serious ^a	not serious	not serious	serious ^d	none	44	19	-	SMD 1.1 SD higher (0.53 higher to 1.68 higher)	⊕⊕○○ LOW	CRITICAL
Women who perceive their hospital stay to be too short)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	2/41 (4.9%)	1/41 (2.4%)	RR 2.00 (0.19 to 21.21)	24 more per 1000 (from 20 fewer to 493 more)	⊕○○○ VERY LOW	CRITICAL
Women perceive their hospital stay to be too long												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	5/41 (12.2%)	9/41 (22.0%)	RR 0.56 (0.20 to 1.52)	97 fewer per 1000 (from 176 fewer to 114 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		
Infants readmitted for neonatal morbidity within 28 days												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	5/257 (1.9%)	5/261 (1.9%)	RR 1.01 (0.31 to 3.28)	0 fewer per 1000 (from 13 fewer to 44 more)	⊕○○○ VERY LOW	CRITICAL
Women breastfeeding (exclusively or partially) at 6 weeks postpartum												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	190/213 (89.2%)	182/217 (83.9%)	RR 1.06 (0.99 to 1.15)	50 more per 1000 (from 8 fewer to 126 more)	⊕⊕⊕○ MODERATE	CRITICAL
Women breastfeeding (exclusively or partially) at 12 weeks postpartum												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	141/213 (66.2%)	119/217 (54.8%)	RR 1.21 (1.03 to 1.41)	115 more per 1000 (from 16 more to 225 more)	⊕⊕⊕○ MODERATE	CRITICAL
Women breastfeeding (exclusively or partially) at 6 months postpartum												
1	randomized trials	serious ^a	not serious	not serious	serious ^f	none	94/213 (44.1%)	76/217 (35.0%)	RR 1.26 (1.00 to 1.60)	91 more per 1000 (from 0 fewer to 210 more)	⊕⊕○○ LOW	CRITICAL
Vaginal birth – Policy of discharge within 48 hours compared with any time later*												
Women with probable postpartum depression within 6 months												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	3/50 (6.0%)	5/54 (9.3%)	RR 0.65 (0.16 to 2.57)	32 fewer per 1000 (from 78 fewer to 145 more)	⊕○○○ VERY LOW	CRITICAL
Women readmitted within 6 weeks												
4	randomized trials	very serious ^g	not serious	not serious	serious ^b	none	33/1956 (1.7%)	4/454 (0.9%)	RR 1.72 (0.58 to 5.12)	6 more per 1000 (from 4 fewer to 36 more)	⊕○○○ VERY LOW	CRITICAL
Women reporting infant feeding problems												
1	randomized trials	very serious ^h	not serious	not serious	serious ^b	none	207/1683 (12.3%)	25/266 (9.4%)	RR 1.31 (0.88 to 1.94)	29 more per 1000 (from 11 fewer to 88 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		
Women satisfied with postnatal care – dichotomous data												
2	randomized trials	very serious ^h	serious ⁱ	not serious	serious ^b	none	1568/1991 (78.8%)	294/370 (79.5%)	RR 1.41 (0.56 to 3.59)	326 more per 1000 (from 350 fewer to 1000 more)	⊕○○○ VERY LOW	CRITICAL
Women satisfied with postnatal care – continuous data												
2	randomized trials	serious ^a	not serious	not serious	serious ^d	none	127	116	-	SMD 0.66 SD higher (0.4 higher to 0.93 higher)	⊕⊕○○ LOW	CRITICAL
Infant mortality within 28 days												
1	randomized trials	very serious ^h	not serious	not serious	very serious ^{b,c}	none	3/1667 (0.2%)	1/217 (0.5%)	RR 0.39 (0.04 to 3.74)	3 fewer per 1000 (from 4 fewer to 13 more)	⊕○○○ VERY LOW	CRITICAL
Infant mortality within one year												
2	randomized trials	very serious ^g	not serious	not serious	very serious ^{b,c}	none	4/1716 (0.2%)	2/270 (0.7%)	RR 0.45 (0.07 to 2.77)	4 fewer per 1000 (from 7 fewer to 13 more)	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 7 days												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	1/50 (2.0%)	0/54 (0.0%)	RR 3.24 (0.13 to 77.63)	0 fewer per 1000 (from 0 fewer to 0 fewer)	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 28 days												
3	randomized trials	very serious ^g	not serious	not serious	very serious ^{b,c}	none	21/1903 (1.1%)	3/433 (0.7%)	RR 1.67 (0.46 to 5.99)	5 more per 1000 (from 4 fewer to 35 more)	⊕○○○ VERY LOW	CRITICAL
Extra contacts with health professionals regarding infant health issues within 4 weeks of birth												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	12/78 (15.4%)	17/97 (17.5%)	RR 0.88 (0.45 to 1.73)	21 fewer per 1000 (from 96 fewer to 128 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		
Women breastfeeding (exclusively or partially) at 6 weeks postpartum												
5	randomized trials	serious ^j	serious ⁱ	not serious	serious ^b	none	451/2175 (20.7%)	133/507 (26.2%)	RR 1.19 (0.80 to 1.78)	50 more per 1000 (from 52 fewer to 205 more)	⊕○○○ VERY LOW	CRITICAL
Women breastfeeding (exclusively or partially) at 6 months postpartum												
1	randomized trials	serious ^a	not serious	not serious	very serious ^k	none	0/49 (0.0%)	0/59 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Caesarean birth – Policy of discharge within 24 hours compared with any time later												
Women with probable postpartum depression within 6 months (within 24 hours – caesarean birth)												
2	randomized trials	serious ^a	not serious	not serious	not serious	none	1172/1665 (70.4%)	917/1675 (54.7%)	RR 1.28 (1.22 to 1.35)	153 more per 1000 (from 120 more to 192 more)	⊕⊕⊕○ MODERATE	CRITICAL
Women readmitted within 6 weeks (within 24 hours – caesarean birth)												
2	randomized trials	serious ^j	not serious	not serious	serious ^b	none	57/1665 (3.4%)	52/1675 (3.1%)	RR 1.10 (0.76 to 1.59)	3 more per 1000 (from 7 fewer to 18 more)	⊕⊕○○ LOW	CRITICAL
Women who had extra contacts with health professionals regarding maternal health issues within 6 weeks (within 24 hours – caesarean birth)												
1	randomized trials	not serious	not serious	not serious	serious ^b	none	16/170 (9.4%)	18/172 (10.5%)	RR 0.90 (0.47 to 1.70)	10 fewer per 1000 (from 55 fewer to 73 more)	⊕⊕⊕○ MODERATE	CRITICAL
Infant mortality within 28 days (within 24 hours – caesarean birth)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^k	none	0/1495 (0.0%)	0/1503 (0.0%)	not estimable	-	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 28 days (within 24 hours – caesarean birth)												
2	randomized trials	serious ^j	not serious	not serious	not serious	none	155/1665 (9.3%)	92/1675 (5.5%)	RR 1.69 (1.32 to 2.17)	38 more per 1000 (from 18 more to 64 more)	⊕⊕⊕○ MODERATE	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		
Number of contacts with health professionals regarding infant health issues within 4 weeks of birth (within 24 hours – caesarean birth)												
1	randomized trials	not serious	not serious	not serious	very serious ^{b,c}	none	30/170 (17.6%)	32/172 (18.6%)	RR 0.95 (0.60 to 1.49)	9 fewer per 1000 (from 74 fewer to 91 more)	⊕⊕○○ LOW	CRITICAL
Women breastfeeding (exclusively or partially) at 6 weeks postpartum (within 24 hours – caesarean birth)												
2	randomized trials	serious ^j	serious ⁱ	not serious	not serious	none	1091/1665 (65.5%)	1172/1675 (70.0%)	RR 0.94 (0.89 to 0.98)	42 fewer per 1000 (from 77 fewer to 14 fewer)	⊕⊕○○ LOW	CRITICAL
Caesarean birth – Policy of discharge within 72 hours compared with any time later*												
Women reporting health problems in the first 6 weeks postpartum (within 72 hours – caesarean birth)												
1	randomized trials	serious ^a	not serious	not serious	serious ^e	none	5/50 (10.0%)	60/150 (40.0%)	RR 0.25 (0.11 to 0.59)	300 fewer per 1000 (from 356 fewer to 164 fewer)	⊕⊕○○ LOW	CRITICAL
Women readmitted within 6 weeks (within or after 72 hours – caesarean birth)												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	5/133 (3.8%)	7/132 (5.3%)	RR 0.73 (0.25 to 2.13)	14 fewer per 1000 (from 40 fewer to 60 more)	⊕○○○ VERY LOW	CRITICAL
Women reporting extra contacts with health professionals regarding maternal health issues within 6 weeks of birth (within 72 hours – caesarean birth)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	6/61 (9.8%)	13/61 (21.3%)	RR 0.46 (0.19 to 1.14)	115 fewer per 1000 (from 173 fewer to 30 more)	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 7 days (within 72 hours – caesarean birth)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	6/72 (8.3%)	6/71 (8.5%)	RR 0.99 (0.33 to 2.91)	1 fewer per 1000 (from 57 fewer to 161 more)	⊕○○○ VERY LOW	CRITICAL
Infants readmitted for neonatal morbidity within 28 days (within 72 hours – caesarean birth)												
2	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c,e}	none	8/133 (6.0%)	12/132 (9.1%)	RR 0.66 (0.28 to 1.57)	31 fewer per 1000 (from 65 fewer to 52 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Early discharge	Usual discharge	Relative (95% CI)	Absolute (95% CI)		

Number of contacts with health professionals regarding infant health issues within 4 weeks of birth (within 72 hours – caesarean birth)

1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,e}	none	25/61 (41.0%)	31/61 (50.8%)	RR 0.81 (0.55 to 1.19)	97 fewer per 1000 (from 229 fewer to 97 more)	⊕○○○ VERY LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	-----------------------------	------	---------------	---------------	----------------------------------	---	------------------	----------

CI: confidence interval; RR: risk ratio; SMD: standardized mean difference.

*Comparison corresponds to subgroup > 24 hours in the Cochrane review. All trials after vaginal birth had a discharge policy of within 48 hours in the intervention arm; all trials after caesarean birth had a discharge policy of within 72 hours in the intervention arm.

a. The pooled effect provided by studies “B”.

b. Wide confidence interval crossing the line of no effect.

c. Less than 30 events.

d. Less than 400 participants.

e. Less than 300 participants.

f. Wide confidence interval touching the line of no effect.

g. Most of the pooled effect provided by studies “C”.

h. Pooled effects provided by studies “C”.

i. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

j. Most of the pooled effect provided by studies “B”.

k. No events.

EB table C.4: Approaches to strengthen preparation for discharge from the facility to home after birth

Comparison 1: Written education booklets for women compared with control leaflets

Source: Smith HJJ, Portela AG, Harvey C. Discharge preparation and readiness after birth: a scoping review of global policies, guidelines and literature. BMC Pregnancy Childbirth (in press).

Certainty assessment							№ of participants		Effect	Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Written education booklet	Control leaflet			
Postpartum visits to a health professional after discharge (6–20 weeks postpartum)											
1	randomized trials	not serious	not serious	not serious	serious ^a	none	187	191	Proportion: 30% higher (<i>P</i> < 0.001)	⊕⊕⊕○ MODERATE	PRIORITY
Satisfaction with care (6–20 weeks postpartum)											
1	randomized trials	not serious	not serious	not serious	serious ^a	none	187	191	Proportion: 18.3% higher (<i>P</i> < 0.001)	⊕⊕⊕○ MODERATE	PRIORITY

a. One study, small sample size.

Comparison 2: Discharge education by a designated nurse compared with usual care

Source: Smith HJJ, Portela AG, Harvey C. Discharge preparation and readiness after birth: a scoping review of global policies, guidelines and literature. BMC Pregnancy Childbirth (in press).

Certainty assessment							№ of participants		Effect	Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Discharge education by a designated nurse	Usual care			

Discharge preparedness (assessed prior to discharge)

1	non-randomized evaluation	very serious ^a	not serious	not serious	serious ^b	none	30	30	<u>Personal status</u> Median score: 0.3 higher $P = 0.437$ <u>Knowledge</u> Median score: 2.396 higher $P < 0.001$ <u>Coping ability</u> Median score: 1.8 higher $P < 0.001$ <u>Expected support</u> Median score: 1.308 higher $P < 0.005$	⊕○○○ VERY LOW	IMPORTANT
---	---------------------------	---------------------------	-------------	-------------	----------------------	------	----	----	--	------------------	-----------

a. Study did not use randomization.

b. One study, small sample size.

EB table C.5a: Home visits for postnatal care contacts compared with usual care

Comparison: Home visits for postnatal care contacts compared with usual care (evidence source 1)

Source: Yonemoto N, Nagai S, Mori R. Schedules for home visits in the early postpartum period. Cochrane Database Syst Rev.2021;(7):CD009326.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Postnatal care home visits	Usual care (without home visits)	Relative (95% CI)	Absolute (95% CI)		
Severe maternal morbidity												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	206/579 (35.6%)	109/297 (36.7%)	RR 0.97 (0.80 to 1.17)	11 fewer per 1000 (from 73 fewer to 62 more)	⊕⊕○○ LOW	CRITICAL
Secondary postpartum haemorrhage												
2	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	40/577 (6.9%)	26/296 (8.8%)	RR 0.78 (0.49 to 1.26)	19 fewer per 1000 (from 45 fewer to 23 more)	⊕○○○ VERY LOW	CRITICAL
Abdominal pain up to 42 days postpartum												
2	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	152/574 (26.5%)	74/295 (25.1%)	RR 1.06 (0.83 to 1.34)	15 more per 1000 (from 43 fewer to 85 more)	⊕○○○ VERY LOW	CRITICAL
Back pain up to 42 days postpartum												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	268/577 (46.4%)	143/294 (48.6%)	RR 0.96 (0.83 to 1.11)	19 fewer per 1000 (from 83 fewer to 54 more)	⊕⊕○○ LOW	CRITICAL
Maternal fever up to 42 days postpartum												
2	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	104/579 (18.0%)	41/297 (13.8%)	RR 1.30 (0.93 to 1.82)	41 more per 1000 (from 10 fewer to 113 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Postnatal care home visits	Usual care (without home visits)	Relative (95% CI)	Absolute (95% CI)		
Urinary tract complications up to 42 days postpartum												
2	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	102/579 (17.6%)	63/297 (21.2%)	RR 0.83 (0.63 to 1.10)	36 fewer per 1000 (from 78 fewer to 21 more)	⊕○○○ VERY LOW	CRITICAL
Dyspareunia												
2	randomized trials	very serious ^a	serious ^c	not serious	serious ^b	none	138/574 (24.0%)	60/295 (20.3%)	RR 1.18 (0.90 to 1.55)	37 more per 1000 (from 20 fewer to 112 more)	⊕○○○ VERY LOW	CRITICAL
Maternal satisfaction with postnatal care												
2	randomized trials	very serious ^a	serious ^c	not serious	not serious	none	459/570 (80.5%)	246/292 (84.2%)	RR 0.96 (0.90 to 1.02)	34 fewer per 1000 (from 84 fewer to 17 more)	⊕○○○ VERY LOW	CRITICAL
Unscheduled visits to hospital												
2	randomized trials	very serious ^a	serious ^d	not serious	serious ^b	none	25/500 (5.0%)	18/248 (7.3%)	RR 0.69 (0.38 to 1.24)	23 fewer per 1000 (from 45 fewer to 17 more)	⊕○○○ VERY LOW	CRITICAL
Maternal contraceptive use												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	224/565 (39.6%)	118/291 (40.5%)	RR 0.98 (0.82 to 1.16)	8 fewer per 1000 (from 73 fewer to 65 more)	⊕⊕○○ LOW	CRITICAL
Infant jaundice												
2	randomized trials	very serious ^a	not serious	not serious	serious ^b	none	199/568 (35.0%)	99/293 (33.8%)	RR 1.04 (0.85 to 1.26)	14 more per 1000 (from 51 fewer to 88 more)	⊕○○○ VERY LOW	CRITICAL
Infant respiratory tract infection within 42 days of birth												
2	randomized trials	very serious ^a	serious ^c	not serious	not serious	none	312/572 (54.5%)	158/293 (53.9%)	RR 1.01 (0.89 to 1.15)	5 more per 1000 (from 59 fewer to 81 more)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Postnatal care home visits	Usual care (without home visits)	Relative (95% CI)	Absolute (95% CI)		
Infant diarrhoea within 42 days of birth												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	256/568 (45.1%)	155/293 (52.9%)	RR 0.85 (0.74 to 0.98)	79 fewer per 1000 (from 138 fewer to 11 fewer)	⊕⊕○○ LOW	CRITICAL
Infant immunization												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	550/572 (96.2%)	289/296 (97.6%)	RR 0.99 (0.96 to 1.01)	10 fewer per 1000 (from 39 fewer to 10 more)	⊕⊕○○ LOW	CRITICAL
Exclusive breastfeeding (last assessment up to 6 weeks of age)												
1	randomized trials	very serious ^a	not serious	not serious	very serious ^{b,d}	none	18/30 (60.0%)	10/30 (33.3%)	RR 1.80 (1.00 to 3.23)	267 more per 1000 (from 0 fewer to 743 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding (last assessment up to 6 months of age)												
3	randomized trials	very serious ^a	not serious	not serious	not serious	none	159/528 (30.1%)	59/288 (20.5%)	RR 1.50 (1.15 to 1.94)	102 more per 1000 (from 31 more to 193 more)	⊕⊕○○ LOW	CRITICAL
Any breastfeeding (last assessment up to 6 months of age)												
2	randomized trials	very serious ^a	not serious	not serious	not serious	none	531/544 (97.6%)	269/278 (96.8%)	RR 1.01 (0.99 to 1.04)	10 more per 1000 (from 10 fewer to 39 more)	⊕⊕○○ LOW	CRITICAL
Mean duration of any breastfeeding (months)												
1	randomized trials	very serious ^a	not serious	not serious	serious ^e	none	27	27	-	MD 3 higher (2.33 higher to 3.67 higher)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Most of the pooled effect provided by studies “C”.

b. Wide confidence interval crossing the line of no effect.

c. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

d. Less than 300 women and/or less than 30 events.

e. Less than 400 women.

Comparison: Home visits for postnatal care contacts compared with usual care (evidence source 2)

Source: Tiruneh GT, Shiferaw CB, Worku A. Effectiveness and cost-effectiveness of home-based postpartum care on neonatal mortality and exclusive breastfeeding practice in low-and-middle-income countries: a systematic review and meta-analysis. BMC Pregnancy Childbirth. 2019;19(1):507. doi:10.1186/s12884-019-2651-6.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Postnatal care home visits	Usual care (without home visits)	Relative (95% CI)	Absolute (95% CI)		
Neonatal mortality												
9	randomized trials	not serious	serious ^a	not serious	not serious	none	32/46 269 (3.2%)	42/46 814 (4.2%)	RR 0.76 (0.62 to 0.92)	10 fewer per 1000 (from 8 fewer to 12 fewer)	⊕⊕⊕○ MODERATE	CRITICAL

CI: confidence interval; RR: risk ratio.

a. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

EB table C.5b: Home visits for postnatal care contacts compared with routine outpatient postnatal care

Comparison: Home visits for postnatal care contacts compared with routine outpatient postnatal care

Source: Source: Yonemoto N, Nagai S, Mori R. Schedules for home visits in the early postpartum period. Cochrane Database Syst Rev. 2021;(7):CD009326.

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Postnatal care home visits	Outpatient postnatal care	Relative (95% CI)	Absolute (95% CI)		
Postnatal depression (last assessment up to 42 days postpartum)												
2	randomized trials	serious ^a	not serious	not serious	serious ^b	none	229/1088 (21.0%)	209/1089 (19.2%)	RR 1.10 (0.93 to 1.30)	19 more per 1000 (from 13 fewer to 58 more)	⊕⊕○○ LOW	CRITICAL
Postpartum depression at 60 days (Edinburgh Postnatal Depression Scale)												
1	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	6/92 (6.5%)	14/184 (7.6%)	RR 0.86 (0.34 to 2.16)	11 fewer per 1000 (from 50 fewer to 88 more)	⊕○○○ VERY LOW	CRITICAL
Mean maternal anxiety score (last assessment up to 42 days postpartum)												
1	randomized trials	serious ^a	not serious	not serious	serious ^b	none	259	254	-	MD 0.3 higher (1.08 lower to 1.68 higher)	⊕⊕○○ LOW	CRITICAL
Maternal depression and anxiety (Hospital Anxiety and Depression Scale)												
1	randomized trials	very serious ^d	not serious	not serious	very serious ^{b,c}	none	2/213 (0.9%)	8/217 (3.7%)	RR 0.25 (0.05 to 1.19)	28 fewer per 1000 (from 35 fewer to 7 more)	⊕○○○ VERY LOW	CRITICAL
Maternal satisfaction with postnatal care												
3	randomized trials	serious ^a	serious ^e	not serious	not serious	none	825/1185 (69.6%)	644/1183 (54.4%)	RR 1.36 (1.14 to 1.62)	196 more per 1000 (from 76 more to 338 more)	⊕⊕○○ LOW	CRITICAL
Mean satisfaction score with postnatal care												
1	randomized trials	serious ^a	not serious	not serious	serious ^b	none	259	254	-	MD 0.1 lower (0.88 lower to 0.68 higher)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of patients		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Postnatal care home visits	Outpatient postnatal care	Relative (95% CI)	Absolute (95% CI)		
Emergency maternal health care visits												
3	randomized trials	serious ^a	not serious	not serious	serious ^b	none	177/1626 (10.9%)	170/1616 (10.5%)	RR 1.04 (0.82 to 1.33)	4 more per 1000 (from 19 fewer to 35 more)	⊕⊕○○ LOW	CRITICAL
Maternal hospital readmissions												
3	randomized trials	serious ^a	not serious	not serious	very serious ^{b,c}	none	8/1347 (0.6%)	6/1343 (0.4%)	RR 1.33 (0.46 to 3.81)	1 more per 1000 (from 2 fewer to 13 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding (last assessment up to 6 weeks)												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	183/259 (70.7%)	171/254 (67.3%)	RR 1.05 (0.93 to 1.18)	34 more per 1000 (from 47 fewer to 121 more)	⊕⊕○○ LOW	CRITICAL
Any breastfeeding (last assessment up to 6 months)												
1	randomized trials	serious ^a	not serious	not serious	not serious	none	367/509 (72.1%)	326/491 (66.4%)	RR 1.09 (1.00 to 1.18)	60 more per 1000 (from 0 fewer to 120 more)	⊕⊕⊕○ MODERATE	CRITICAL
Discontinuation breastfeeding (up to 30 days)												
1	randomized trials	very serious ^d	not serious	not serious	very serious ^{b,c}	none	18/95 (18.9%)	22/90 (24.4%)	RR 0.78 (0.45 to 1.35)	54 fewer per 1000 (from 134 fewer to 86 more)	⊕○○○ VERY LOW	CRITICAL
Discontinued breastfeeding (up to 6 weeks)												
2	randomized trials	serious ^a	not serious	not serious	not serious	none	180/1088 (16.5%)	193/1089 (17.7%)	RR 0.93 (0.78 to 1.12)	12 fewer per 1000 (from 39 fewer to 21 more)	⊕⊕⊕○ MODERATE	CRITICAL
Infant emergency health care visits (health care utilization)												
3	randomized trials	serious ^a	not serious	not serious	serious ^b	none	185/1633 (11.3%)	160/1624 (9.9%)	RR 1.15 (0.95 to 1.38)	15 more per 1000 (from 5 fewer to 37 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							No of patients		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Postnatal care home visits	Outpatient postnatal care	Relative (95% CI)	Absolute (95% CI)		

Infant hospital readmissions

3	randomized trials	serious ^a	not serious	not serious	serious ^b	none	30/1347 (2.2%)	25/1343 (1.9%)	RR 1.20 (0.71 to 2.02)	4 more per 1000 (from 5 fewer to 19 more)	⊕⊕○○ LOW	CRITICAL
---	-------------------	----------------------	-------------	-------------	----------------------	------	----------------	----------------	----------------------------------	---	-------------	----------

CI: confidence interval; MD: mean difference; RR: risk ratio.

a. Most of the pooled effect provided by studies “B”.

b. Wide confidence interval crossing the line of no effect.

c. Less than 300 women and/or less than 30 events.

d. The pooled effect provided by study “C”.

e. Statistical heterogeneity ($I^2 \geq 60\%$ or $\text{Chi}^2 \leq 0.05$).

EB table C.9: Involvement of men in postnatal care and maternal and newborn health

Note: Study author names are provided in each of the GRADE tables to distinguish studies in cases where meta-analyses were not possible.

Comparison 1: Couples education compared with no intervention or usual care

Source: Baguiya A, Portela A, Moyvisan A, Gerlach N, Gopal P, Sauvé C, et al. Effectiveness of male involvement intervention on maternal and newborn health outcomes (in preparation).

Certainty assessment							No of participants		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Couples education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Postnatal visits for women – at least one within 2 weeks of childbirth												
1 (Mullany et al., 2007)	randomized trial	serious ^a	not serious	not serious	serious ^b	none	81/133 (60.9%)	60/128 (46.9%)	RR 1.29 (1.04 to 1.60)	136 more per 1000 (from 19 more to 281 more)	⊕⊕○○ LOW	CRITICAL
Postnatal visits for women – two or more within 6 weeks of childbirth												
1 (Daniele et al., 2018)	randomized trial	not serious	not serious	not serious	not serious	none	342/560 (61.1%)	265/541 (49.0%)	RR 1.23 (1.11 to 1.37)	113 more per 1000 (from 54 more to 181 more)	⊕⊕⊕⊕ HIGH	CRITICAL
Family planning – timely initiation of a modern contraceptive method												
1 (Daniele et al., 2018)	randomized trial	not serious	not serious	not serious	not serious	none	249/329 (75.7%)	188/281 (66.9%)	RR 1.11 (1.00 to 1.24)	74 more per 1000 (from 0 fewer to 161 more)	⊕⊕⊕⊕ HIGH	CRITICAL
Family planning – use of any contraceptive method at 3 months after childbirth												
1 (Daniele et al., 2018)	randomized trial	not serious	not serious	not serious	not serious	none	315/553 (57.0%)	262/532 (49.2%)	RR 1.16 (1.04 to 1.30)	79 more per 1000 (from 20 more to 148 more)	⊕⊕⊕⊕ HIGH	CRITICAL
Family planning – use of a modern contraceptive method at 6 months after childbirth												
1 (Kunene et al., 2004)	(cluster) randomized trial	serious ^a	not serious	not serious	not serious	none	466/526 (88.6%)	352/395 (89.1%)	RR 1.01 (0.90 to 1.12)	9 fewer per 1000 (from 45 fewer to 27 more)	⊕⊕⊕○ MODERATE	CRITICAL

Certainty assessment							No of participants		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Couples education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Breastfeeding initiation within 1 hour of childbirth												
1 (Kunene et al., 2004)	(cluster) randomized trial	serious ^a	not serious	not serious	serious ^c	none	107/630 (17.0%)	95/592 (16.0%)	RR 1.06 (0.82 to 1.36)	10 more per 1000 (from 29 fewer to 58 more)	⊕⊕○○ LOW	CRITICAL
Exclusive breastfeeding up to 3 months of age												
4 (Abbass-Dick et al., 2015; Daniele et al., 2018; Turan et al., 2018 Sakkaki, 2013) ^d	randomized trials	not serious	not serious	not serious	serious ^c	none	70/104 (67.3%) 232/535 (43.4%) 48/53 (90.6%) 23/34 (67.6%)	63/105 (60.0%) 161/511 (31.5%) 40/52 (76.9%) 12/33 (36.4%)	RR 1.12 (0.91 to 1.38) RR 1.35 (1.15 to 1.59) RR 1.18 (0.99 to 1.41) RR 1.86 (1.12 to 3.09)	72 more per 1000 (from 54 fewer to 228 more) 110 more per 1000 (from 47 more to 186 more) 138 more per 1000 (from 8 fewer to 315 more) 313 more per 1000 (from 44 more to 760 more)	⊕⊕⊕○ MODERATE	CRITICAL
Breastfeeding at 6 months of age												
1 (Kunene et al., 2004)	(cluster) randomized trial	serious ^a	not serious	not serious	not serious	none	496/671 (73.9%)	458/627 (73.0%)	RR 1.01 (0.87 to 1.19)	7 more per 1000 (from 37 fewer to 58 more)	⊕⊕⊕○ MODERATE	CRITICAL

Certainty assessment							No of participants		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Couples education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		

Breastfeeding at 6 months of age

3 (Su, 2016; Susin, 2008; Lovera et al., 2010) ^e	quasi-experimental pre-post with a control, non-randomized controlled trial, and analytic cohort	serious ^f	serious ^g	not serious	very serious ^{b, c}	none	14/35 (40.0%)	6/34 (17.6%)	OR 3.11 (1.02 to 9.45)	223 more per 1000 (from 3 more to 493 more)	⊕○○○ VERY LOW	CRITICAL
							90/180 (50.0%)	87/187 (46.4%)	OR 1.15 (0.76 to 1.73)	35 more per 1000 (from 67 fewer to 136 more)		
							19/101 (18.8%)	20/99 (20.2%)	OR 0.92 (0.46 to 1.84)	13 fewer per 1000 (from 98 fewer to 116 more)		

Breastfeeding initiation before discharge

1 (Su, 2016)	quasi-experimental with a control	serious ^f	not serious	not serious	very serious ^{b, c}	none	14/36 (38.9%)	12/36 (33.3%)	OR 1.27 (0.49 to 3.34)	55 more per 1000 (from 137 fewer to 393 more)	⊕○○○ VERY LOW	CRITICAL
--------------	-----------------------------------	----------------------	-------------	-------------	------------------------------	------	---------------	---------------	----------------------------------	---	------------------	----------

Exclusive breastfeeding up to 4–6 weeks

2 (Abbass-Dick et al., 2015; Sakkaki, 2013) ^h	randomized trials	very serious ⁱ	not serious	not serious	very serious ^{b, c}	none	28/34 (82.4%)	19/33 (57.6%)	RR 1.43 (1.03 to 1.99)	248 more per 1000 (from 17 more to 570 more)	⊕○○○ VERY LOW	CRITICAL
							75/104 (72.1%)	62/102 (60.8%)	RR 1.19 (0.98 to 1.44)	115 more per 1000 (from 12 fewer to 267 more)		

Exclusive breastfeeding at 1 month of age

1 (Su, 2016)	quasi-experimental pre-post with a control	serious ^f	not serious	not serious	very serious ^{b, c}	none	22/36 (61.1%)	21/34 (61.8%)	OR 0.97 (0.37 to 2.55)	7 fewer per 1000 (from 244 fewer to 187 more)	⊕○○○ VERY LOW	CRITICAL
--------------	--	----------------------	-------------	-------------	------------------------------	------	---------------	---------------	----------------------------------	---	------------------	----------

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Couples education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Exclusive breastfeeding at 2 months of age												
1 (Sakkaki, 2013)	randomized trial	very serious ⁱ	not serious	not serious	serious ^b	none	25/34 (73.5%)	14/33 (42.4%)	RR 1.73 (1.11 to 2.71)	310 more per 1000 (from 47 more to 752 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding at 4 months of age												
1 (Sakkaki, 2013)	randomized trial	very serious ⁱ	not serious	not serious	serious ^b	none	18/34 (52.9%)	8/33 (24.2%)	RR 2.18 (1.11 to 4.32)	286 more per 1000 (from 27 more to 805 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding at 4 months of age												
1 (Su, 2016)	quasi-experimental pre-post with a control	serious ^f	not serious	not serious	serious ^b	none	18/35 (51.4%)	9/34 (26.5%)	OR 2.94 (1.07 to 8.07)	249 more per 1000 (from 13 more to 479 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding at 6 months of age												
1 (Sakkaki, 2013)	randomized trial	very serious ⁱ	not serious	not serious	serious ^b	none	15/34 (44.1%)	6/33 (18.2%)	RR 2.43 (1.07 to 5.49)	260 more per 1000 (from 13 more to 816 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding discontinuation in the first 6 months after childbirth												
1 (Susin, 2008)	non-randomized controlled trial	serious ^f	not serious	not serious	not serious	none	180	187	HR 0.80 (0.65 to 0.98)	-	⊕○○○ VERY LOW	CRITICAL
Co-parenting at 6 weeks after childbirth (as perceived by mothers)												
1 (Abbass-Dick et al., 2015)	randomized trial	not serious	not serious	not serious	very serious ^{b, c}	none	91	98	-	SMD 0.17 higher (0.12 lower to 0.45 higher)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Couples education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Co-parenting at 12 weeks after childbirth (as perceived by mothers)												
1 (Abbass-Dick et al., 2015)	randomized trial	not serious	not serious	not serious	very serious ^{b, c}	none	100	96	-	SMD 0.18 higher (0.10 lower to 0.46 higher)	⊕⊕○○ LOW	CRITICAL
Quality of father–child interaction at 6 months after childbirth												
1 (Doherty et al., 2006)	randomized trial	serious ^j	not serious	not serious	serious ^b	none	95	70	-	SMD 0.46 higher (0.15 higher to 0.77 higher)	⊕⊕○○ LOW	CRITICAL
Father involvement at 4 weeks after childbirth												
1 (Bagheri et al., 2015)	randomized trial	very serious ^{a, j}	not serious	not serious	serious ^b	none	50	50	-	SMD 1.83 higher (1.36 higher to 2.30 higher)	⊕○○○ VERY LOW	CRITICAL
Father involvement at 8 weeks after childbirth												
1 (Bagheri et al., 2015)	randomized trial	very serious ^{a, j}	not serious	not serious	serious ^b	none	50	50	-	SMD 0.96 higher (0.55 higher to 1.38 higher)	⊕○○○ VERY LOW	CRITICAL
Paternal responsibility at 6 months after childbirth												
1 (Doherty et al., 2006)	randomized trial	serious ^j	not serious	not serious	very serious ^{b, c}	none	95	70	-	SMD 0.19 higher (0.12 lower to 0.50 higher)	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of participants		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Couples education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		

Paternal engaged interaction at 6 months after childbirth

1 (Doherty et al., 2006)	randomized trial	serious ^j	not serious	not serious	very serious ^{b, c}	none	95	70	-	SMD 0.05 lower (0.36 lower to 0.26 higher) SMD 0.21 higher (0.10 lower to 0.52 higher)*	⊕○○○ VERY LOW	CRITICAL
--------------------------	------------------	----------------------	-------------	-------------	------------------------------	------	----	----	---	--	------------------	----------

Paternal parallel interaction at 6 months after childbirth

1 (Doherty et al., 2006)	randomized trial	serious ^j	not serious	not serious	very serious ^{b, c}	none	95	70	-	SMD 0.08 higher (0.23 lower to 0.39 higher) SMD 0.39 higher (0.08 higher to 0.70 higher)*	⊕○○○ VERY LOW	CRITICAL
--------------------------	------------------	----------------------	-------------	-------------	------------------------------	------	----	----	---	--	------------------	----------

Total accessibility at 6 months after childbirth

1 (Doherty et al., 2006)	randomized trial	serious ^j	not serious	not serious	very serious ^{b, c}	none	95	70	-	SMD 0.19 lower (0.50 lower to 0.12 higher) SMD 0.41 higher (0.10 higher to 0.72 higher)*	⊕○○○ VERY LOW	CRITICAL
--------------------------	------------------	----------------------	-------------	-------------	------------------------------	------	----	----	---	---	------------------	----------

CI: confidence interval; HR: hazard ratio; OR: odds ratio; RR: risk ratio; SMD: standardized mean difference.

* The first estimate applies to involvement on a day at home and the second to involvement on a work day, as measured separately in the study.

a. Concerns with missing data.

b. Limited sample size and/or limited number of events.

c. Wide confidence interval crossing the line of no effect.

d. Data not meta-analysed due to heterogeneity in the interventions.

e. Data not meta-analysed due to variation in study designs and timing of the interventions.

f. Lack of appropriate accounting for confounders.

g. Inconsistent direction of effect in the body of evidence.

h. Data not meta-analysed because of differences in the study populations (nulliparous women regardless of the mode of birth in one study and caesarean birth only in the other).

i. Inappropriate randomization.

j. Lack of blinding (subjective self-reported outcome).

Comparison 2: Couples education compared with women's education alone

Source: Baguiya A, Portela A, Moyvisan A, Gerlach N, Gopal P, Sauvé C, et al. Effectiveness of male involvement intervention on maternal and newborn health outcomes (in preparation).

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Couples education	Women’s education	Relative (95% CI)	Absolute (95% CI)		
Postnatal visits for women – at least one within 2 weeks of childbirth												
1 (Mullany et al., 2007)	randomized trial	serious ^a	not serious	not serious	serious ^b	none	81/133 (60.9%)	61/125 (48.8%)	RR 1.25 (1.01 to 1.54)	122 more per 1000 (from 5 more to 264 more)	⊕⊕○○ LOW	CRITICAL
Exclusive breastfeeding at 4 months after childbirth												
1 (Susin, 2008)	non-randomized controlled trial	serious ^c	not serious	not serious	serious ^d	none	32/193 16.6%	11/201 5.5%	RR 3.02 (0.90 to 3.24)	111 more per 1000 (from 5 fewer to 123 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding at 6 months after childbirth												
1 (Susin, 2008)	non-randomized controlled trial	serious ^c	not serious	not serious	serious ^d	none	90/180 (50.0%)	108.5/180 (60.3%)	OR 0.66 (0.43 to 1.01)	102 fewer per 1000 (from 208 fewer to 2 more)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; OR: odds ratio; RR: risk ratio.

a. Concerns with missing data.

b. Limited sample size and/or limited number of events.

c. Lack of appropriate accounting for confounders.

d. Wide confidence interval crossing the line of no effect.

Comparison 3. Men's education compared with no intervention or usual care

Source: Baguiya A, Portela A, Moyvisan A, Gerlach N, Gopal P, Sauvé C, et al. Effectiveness of male involvement intervention on maternal and newborn health outcomes (in preparation).

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Men's education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Postnatal visits for women – at least one within 7 days of childbirth												
1 (Hazra et al, 2018)	quasi-experimental pre-post with a control	not serious	not serious	not serious	serious ^a	none	68	79	OR 3.02 <i>P</i> < 0.05	-	⊕○○○ VERY LOW	CRITICAL
Maternal morbidity – general psychosocial problems at 3 weeks after the intervention												
1 (Nosrati et al., 2017)	randomized trial	serious ^b	not serious	not serious	very serious ^{a, c}	none	30	30	-	SMD 0.24 lower (0.75 lower to 0.27 higher)	⊕○○○ VERY LOW	CRITICAL
Maternal morbidity – general psychosocial problems at 6 weeks after the intervention												
1 (Nosrati et al., 2017)	randomized trial	serious ^b	not serious	not serious	serious ^a	none	30	30	-	SMD 0.96 lower (1.50 lower to 0.43 lower)	⊕⊕○○ LOW	CRITICAL
Care practices for newborns – delaying bathing by at least 2 days												
1 (Hazra et al., 2018)	quasi-experimental pre-post with a control	not serious	not serious	not serious	serious ^a	none	68	79	OR 1.93 <i>P</i> < 0.05	-	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding at 6 weeks after childbirth												
1 (Maycock et al., 2013)	randomized trial	serious ^d	not serious	not serious	serious ^c	none	164/353 (46.5%)	133/298 (44.6%)	aOR 1.09 (0.79 to 1.51)	21 more per 1000 (from 57 fewer to 103 more)	⊕⊕○○ LOW	CRITICAL
Breastfeeding until 6 months after childbirth												
1 (Raeisi et al., 2014)	randomized trial	serious ^d	not serious	not serious	serious ^a	none	47/50 (94%)	38/50 (76%)	RR 1.24 (1.04 to 1.47)	182 more (from 30 more to 357 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Men’s education	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Paternal involvement at 4 weeks after childbirth												
1 (Bagheri et al., 2015)	randomized trial	very serious ^{b, d}	not serious	not serious	serious ^a	none	50	50	-	SMD 1.48 higher (1.04 higher to 1.93 higher)	⊕○○○ VERY LOW	CRITICAL
Paternal involvement at 8 weeks after childbirth												
1 (Bagheri et al., 2015)	randomized trial	very serious ^{b, d}	not serious	not serious	serious ^a	none	50	50	-	SMD 0.92 higher (0.51 higher to 1.34 higher)	⊕○○○ VERY LOW	CRITICAL
Paternal responsiveness at 6 months after the intervention												
1 (Mihelic et al., 2018)	randomized trial	very serious ^{b, d}	not serious	not serious	very serious ^{a, c}	none	57	55	-	SMD 0.12 lower (0.49 lower to 0.24 higher)	⊕○○○ VERY LOW	CRITICAL
Paternal bonding difficulties at 6 months after the intervention												
1 (Mihelic et al., 2018)	randomized trial	very serious ^{b, d}	not serious	not serious	very serious ^{a, c}	none	57	55	-	SMD 0.02 higher (0.35 lower to 0.39 higher]	⊕○○○ VERY LOW	CRITICAL

aOR: adjusted odds ratio; CI: confidence interval; OR: odds ratio; RR: risk ratio; SMD: standardized mean difference.

a. Limited sample size and/or limited number of events.

b. Lack of blinding (subjective self-reported outcome).

c. Wide confidence interval crossing the line of no effect.

d. Concerns with missing data.

Comparison 4. Father as a labour companion compared with no companion

Source: Baguiya A, Portela A, Moyvisan A, Gerlach N, Gopal P, Sauvé C, et al. Effectiveness of male involvement intervention on maternal and newborn health outcomes (in preparation).

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Labour companion (father)	No companion	Relative (95% CI)	Absolute (95% CI)		
Maternal morbidity – depressive symptoms 6–8 weeks after childbirth												
1 (Sapkota et al., 2013)	Non-randomized controlled trial	serious ^a	not serious	not serious	very serious ^{b, c}	none	77	79	-	SMD 0.28 lower (0.60 lower to 0.04 higher)	⊕○○○ VERY LOW	CRITICAL
Maternal morbidity – anxiety 6–8 weeks after childbirth												
1 (Sapkota et al., 2013)	non-randomized controlled trial	serious ^a	not serious	not serious	serious ^b	none	77	79	-	SMD 0.40 lower (0.71 lower to 0.08 lower)	⊕○○○ VERY LOW	CRITICAL
Father–infant bonding on the first day after childbirth												
1 (Brandao, 2012)	quasi-experimental	serious ^d	not serious	not serious	very serious ^{b, c}	none	45	28	-	SMD 0.11 lower (0.58 lower to 0.36 higher)	⊕○○○ VERY LOW	CRITICAL
Father–infant bonding in the first month after childbirth												
1 (Brandao, 2012)	quasi-experimental	serious ^d	not serious	not serious	serious ^b	none	45	28	-	SMD 0.87 SD higher (0.37 higher to 1.36 higher)	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; SMD: standardized mean difference.

a. Concerns with missing data.

b. Limited sample size and/or limited number of events.

c. Wide confidence interval crossing the line of no effect.

d. Lack of appropriate accounting for confounders.

Comparison 5. Father as a labour companion compared with a female friend as a labour companion

Source: Baguiya A, Portela A, Moyvisan A, Gerlach N, Gopal P, Sauvé C, et al. Effectiveness of male involvement intervention on maternal and newborn health outcomes (in preparation).

Certainty assessment							No of participants		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Labour companion (father)	Labour companion (female friend)	Relative (95% CI)	Absolute (95% CI)		

Maternal morbidity – depressive symptoms 6–8 weeks after childbirth

1 (Sapkota et al., 2013)	non-randomized controlled trial	serious ^a	not serious	not serious	very serious ^{b, c}	none	77	75	-	SMD 0.21 lower (0.53 lower to 0.11 higher)	⊕○○○ VERY LOW	CRITICAL
--------------------------	---------------------------------	----------------------	-------------	-------------	------------------------------	------	----	----	---	--	------------------	----------

Maternal morbidity – anxiety 6–8 weeks after childbirth

1 (Sapkota et al., 2013)	non-randomized controlled trial	serious ^a	not serious	not serious	very serious ^{b, c}	none	77	75	-	SMD 0.14 lower (0.46 lower to 0.03 higher)	⊕○○○ VERY LOW	CRITICAL
--------------------------	---------------------------------	----------------------	-------------	-------------	------------------------------	------	----	----	---	--	------------------	----------

CI: confidence interval; SMD: standardized mean difference.

a. Bias due to missing data.

b. Limited sample size and/or limited number of events.

c. Wide confidence interval crossing the line of no effect.

Comparison 6. Multi-component interventions compared with no intervention or usual care

Source: Baguiya A, Portela A, Moyvisan A, Gerlach N, Gopal P, Sauvé C, et al. Effectiveness of male involvement intervention on maternal and newborn health outcomes (in preparation).

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Multi-component	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Postnatal visits for women – any postnatal care from a skilled professional within 2 days of childbirth												
1 (Rahman et al., 2019)	pre-post with a control and propensity score matching	serious ^a	not serious	not serious	serious ^b	none	Baseline: 39/235 (16.4%) Endline: 46/217 (21.2%)	Baseline: 53/235 (22.4%) Endline: 47/217 (21.7%)	DID 0.05% <i>P</i> = 0.333	-	⊕○○○ VERY LOW	CRITICAL
Breastfeeding initiation within the first hour of childbirth												
1 (Baqui et al., 2008)	(cluster) randomized trial	not serious	not serious	not serious	not serious	none	1426/1760 (81%)	963/1689 (57%)	RR 1.42 (1.35 to 1.49)	239 more per 1000 (from 200 more to 279 more)	⊕⊕⊕⊕ HIGH	CRITICAL
Breastfeeding initiation within the first hour of childbirth												
1 (Bich et al., 2016)	pre-post with a control	serious ^c	not serious	not serious	not serious	none	194/239 (81.2%)	91/230 (39.6%)	OR 7.64 (4.81 to 12.12)	438 more per 1000 (from 363 more to 494 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding at 2 months after childbirth												
1 (Kohan et al., 2019)	randomized trial	not serious	not serious	not serious	serious ^d	none	33/35 (94.3%)	23/35 (65.7%)	RR 1.43 (1.11 to 1.85)	283 more per 1000 (from 72 more to 559 more)	⊕⊕⊕○ MODERATE	CRITICAL
Exclusive breastfeeding at 4 months after childbirth												
1 (Kohan et al., 2019)	randomized trial	not serious	not serious	not serious	very serious ^{d, e}	none	23/35 (65.7%)	30/35 (85.7%)	RR 0.77 (0.58 to 1.01)	197 fewer per 1000 (from 360 fewer to 9 more)	⊕⊕○○ LOW	CRITICAL

Certainty assessment							№ of participants		Effect		Certainty (GRADE)	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Multi-component	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		
Exclusive breastfeeding at 4 months after childbirth												
2 (Bich et al., 2014; Bich et al., 2019) ^f	pre-post with a control	serious ^c	not serious	not serious	not serious	none	49/238 (20.6%)	26/230 (11.3%)	OR 2.36 (1.35 to 4.14)	118 more per 1000 (from 34 more to 232 more)	⊕○○○ VERY LOW	CRITICAL
							67/359 (18.7%)	16/397 (4.0%)	OR 7.46 (3.95 to 14.11)	198 more per 1000 (from 102 more to 332 more)		
Early initiation of exclusive breastfeeding												
1 (Bich et al., 2019)	pre-post with a control	serious ^c	not serious	not serious	not serious	none	179/368 (48.6%)	144/403 (35.7%)	OR 1.69 (1.19 to 2.41)	127 more per 1000 (from 41 more to 215 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding at 1 month after childbirth												
1 (Bich et al., 2019)	pre-post with a control	serious ^c	not serious	not serious	not serious	none	128/368 (34.8%)	23/403 (5.7%)	OR 10.15 (6.06 to 17.02)	323 more per 1000 (from 211 more to 450 more)	⊕○○○ VERY LOW	CRITICAL
Exclusive breastfeeding until 6 months after childbirth												
2 (Bich et al., 2014; Bich et al., 2019) ^f	pre-post with a control	serious ^c	not serious	not serious	very serious ^{b, e}	none	16/238 (6.7%)	2/230 (0.9%)	OR 6.29 (1.35 to 29.29)	43 more per 1000 (from 3 more to 196 more)	⊕○○○ VERY LOW	CRITICAL
							7.5/362 (2.1%)	0.5/397 (0.1%)	OR 16.78 (0.96 to 294.8)	19 more per 1000 (from 0 fewer to 270 more)		
Exclusive breastfeeding cessation at 6 months after childbirth												
1 (Bich et al., 2019)	pre-post with a control	serious ^c	not serious	not serious	not serious	none	361	396	HR 0.69 (0.59 to 0.81)	-	⊕○○○ VERY LOW	CRITICAL

Certainty assessment							No of participants		Effect		Certainty (GRADE)	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Multi-component	No intervention or usual care	Relative (95% CI)	Absolute (95% CI)		

Father-infant play at 1 month after childbirth

1 (Rempel et al., 2017)	pre-post with a control	serious ^a	not serious	not serious	not serious	none	350	382	-	SMD 0.34 higher (0.19 higher to 0.49 higher)	⊕○○○ VERY LOW	CRITICAL
-------------------------	-------------------------	----------------------	-------------	-------------	-------------	------	-----	-----	---	--	------------------	----------

Father care-taking of infant at 1 month after childbirth

1 (Rempel et al., 2017)	pre-post with a control	serious ^a	not serious	not serious	serious ^e	none	350	382	-	SMD 0.06 higher (0.09 lower to 0.20 higher)	⊕○○○ VERY LOW	CRITICAL
-------------------------	-------------------------	----------------------	-------------	-------------	----------------------	------	-----	-----	---	---	------------------	----------

Father affection towards infant at 1 month after childbirth

1 (Rempel et al., 2017)	pre-post with a control	serious ^a	not serious	not serious	not serious	none	350	382	-	SMD 0.39 higher (0.25 higher to 0.54 higher)	⊕○○○ VERY LOW	CRITICAL
-------------------------	-------------------------	----------------------	-------------	-------------	-------------	------	-----	-----	---	--	------------------	----------

Father-infant attachment at 1 month after childbirth

1 (Rempel et al., 2017)	pre-post with a control	serious ^a	not serious	not serious	not serious	none	350	382	-	SMD 0.59 higher (0.44 higher to 0.73 higher)	⊕○○○ VERY LOW	CRITICAL
-------------------------	-------------------------	----------------------	-------------	-------------	-------------	------	-----	-----	---	--	------------------	----------

CI: confidence interval; DID: difference-in-difference; HR: hazard ratio; OR: odds ratio; RR: risk ratio; SMD: standardized mean difference.

a. Lack of appropriate accounting for confounders.

b. Insufficient data reported to enable assessment of imprecision.

c. Concerns with selecting participants into the study.

d. Limited sample size and/or limited number of events.

e. Wide confidence interval crossing the line of no effect.

f. Data not meta-analysed due to heterogeneity in the studies.

For more information, please contact:

World Health Organization

Avenue Appia 20
CH-1211 Geneva 27
Switzerland

Department of Sexual and Reproductive Health and Research

Email: srhmph@who.int
Website: [www.who.int/teams/sexual-and-reproductive-health-and-research-\(srh\)](http://www.who.int/teams/sexual-and-reproductive-health-and-research-(srh))

Department of Maternal, Newborn, Child and Adolescent Health and Ageing

Email: mncah@who.int
Website: www.who.int/teams/maternal-newborn-child-adolescent-health-and-ageing/

Department of Mental Health and Substance Use

Email: mhgap-info@who.int
Website: www.who.int/teams/mental-health-and-substance-use

Department of Nutrition and Food Safety

Email: nutrition@who.int
Website: www.who.int/teams/nutrition-and-food-safety/overview

